Chapter 1

Zorich Mathematical Analysis

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1.1 Logical Symbolism

$\boxed{\neg A}$

A	0	1
$\neg A$	1	0

$\boxed{A \wedge B}$

$A \backslash B$	0	1
0	0	0
1	0	1

$\boxed{A \vee B}$

$A \backslash B$	0	1
0	0	1
1	1	1

$\boxed{A \Rightarrow B}$

$A \backslash B$	0	1
0	1	1
1	0	1

Problem (1)

Check whether all of these tables agree with your concept of the corresponding logical operation. (In particular, pay attnetion to the fact that if A is false, then the implication $A \implies B$ is always true.)

$\neg A$: If A is true, then $\neg A$ is false, and vice versa.

$A \wedge B$: If A and B are true, then $A \wedge B$ is true. If A is true, B is false, then $A \wedge B$ is false, if A is false, B is true, $A \wedge B$ is false. Finally, if A and B are false, $A \wedge B$ is false.

$A \vee B$: If A and B are false, $A \vee B$ is false. If at least one of A and B is true, $A \vee B$ is true.

$A \implies B$: If A and B are false, the statement is true. If A is true but B is false, the statement is false. If A is false but B is true, the statement is true. If A is true and B is true, the statement is true.

Problem (2)

Show that the following simple, but very useful relations, which are widely used in mathematical reasoning, are true:

(a) $\neg(A \wedge B) \iff \neg A \vee \neg B$;

(b) $\neg(A \vee B) \iff \neg A \wedge \neg B$;

(c) $(A \implies B) \iff (\neg B \implies \neg A)$;

(d) $(A \implies B) \iff (\neg A \wedge B)$l

(e) $\neg(A \implies B) \iff A \wedge \neg B$.

(a) If $A \wedge B$ is true, then A and B are true. $\neg(A \wedge B)$ is false, and since both $\neg A$ and $\neg B$ are false, so is $\neg A \vee \neg B$.

If $A \wedge B$ is false, then at least one of A or B is false, so at least one of $\neg A$ and $\neg B$ is true, so $\neg A \vee \neg B$ is true. $\neg(A \wedge B)$ is true, so they are equivalent.

(b) If $\neg(A \vee B)$ is false, then $A \vee B$ is true, at least one of A and B is true, so at least one of $\neg A$ and $\neg B$ is false, so $\neg A \wedge \neg B$ is false.

If $\neg(A \vee B)$ is true, then $A \vee B$ is false, so both of A and B are false, so both $\neg A$ and $\neg B$ are true, so $\neg A \wedge \neg B$ is true.

(c) A statement and its contrapositive are equivalent.

(d) $(A \implies B) \iff (-\neg A \vee B)$

(e) $\neg(A \implies B) \iff A \wedge \neg B$

1.2 Sets and Elementary Operations on Them

In Exercises 1, 2, and 3 the letters A, B, and C denote subsets of a set M.

Problem (1)

Verify the following relations.

(a) $(A \subset C) \wedge (B \subset C) \iff ((A \cup B) \subset C)$;

(b) $(C \subset A) \wedge (C \subset B) \iff (C \subset (A \cap B))$;

(c) $C_M(C_M A) = A$;

(d) $(A \subset C_M B) \iff (B \subset C_M A)$;

(e) $(A \subset B) \iff (C_M A \supset C_M B)$;

(a) If both A and B are subsets of C, then $A \cup B$ is a subset of C. If one of A or B is not a subset of C, say A, then there exists $x \in A$ such that $x \in A \cup B$ and $x \notin C$. Thus $A \cup B$ is not a subset of C¿

(b) If C is a subset of both A and B, then C is a subset of $A \cap B$. If C is not a subset of one of A and B, then C is not a subset of $A \cap B$.

(c) $C_M A$ is the complement of A. $C_M(C_M A)$ is the complement of the complement, or simply A.

(d) If A is in the complement of B, this means that A and B share no common elements. Since this relationship is symmetric, then B is a subset of the complement of A.

(e) If A is a subset of B, this means that there are elements not in A but in B, so $C_M B \subset C_M A$.

Problem (2)

Prove the following statements.

(a) $A \cup (B \cup C) = (A \cup B) \cup C =: A \cup B \cup C$;

(b) $A \cap (B \cap C) = (A \cap B) \cap C =: A \cap B \cap C$;

(c) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;

(d) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

(a) Note that all elements of $A \cup (B \cup C)$ and $(A \cup B) \cup C$ are elements of $A \cup B \cup C$, as desired.

(b) All elements of $A \cap (B \cap C)$ and $(A \cap B) \cap C$ are elements of $A \cap B \cap C$.

(c) Using a venn diagram, if $x \in A \cap (B \cup C)$, then $x \in A$ and $x \in B \cup C$. Thus $x \in B$ or C, so $x \in A \cap B$ or $A \cap C$. Therefore $x \in (A \cap B) \cup (A \cap C)$.

(d) Let $x \in A \cup (B \cap C)$. Then $x \in A$ or $x \in B \cap C$.

- If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$, so $x \in (A \cup B) \cap (A \cup C)$.
- If $x \in B \cap C$, then $x \in B$ and $x \in C$, so $x \in A \cup B$ and $x \in A \cup C$. Therefore $x \in (A \cup B) \cap (A \cup C)$.

Problem (3)

Verify the connection (duality) between the operations of union and intersection.

(a) $C_M(A \cup B) = C_M A \cap C_M B$;

(b) $C_M(A \cap B) = C_M A \cup C_M B$.

(a) If $x \in C_M(A \cup B)$, then $x \notin A \cup B$, so $x \notin A$ and $x \notin B$. Hence $x \in C_M A$ and $x \in C_M B$, so $x \in C_M A \cap C_M B$.

If $x \in C_M A \cap C_M B$, then $x \in C_M A$ and $x \in C_M B$, so $x \notin A$ and $x \notin B$. Then $x \notin A \cup B$, so $x \in C_M(A \cup B)$.

(b) If $x \in C_M(A \cap B)$, then $x \notin A \cap B$ so $x \notin A$ or $x \notin B$. Then $x \in C_M A$ or $x \in C_M B$, so $x \in C_M A \cup C_M B$.

Problem (4)

Give geometric representations of the following Cartesian products.

(a) The product of two line segments (a rectangle).

(b) The product of two lines (a plane).

(c) The product of a line and a circle (an infinite cylindrical surface).

(d) The product of a line and a disk (an infinite solid cylinder).

(e) The product of two circles (a torus).

(f) The product of a circle and a disk (a solid torus).

(a) The product of two line segments is a rectangle

(b) the product of two lines is a plane

(c) the product of a line and a circle is an infinitely long cylindrical surface

(d) the product of a line and a disk is an infinitely long solid cylinder

(e) the product of two circles is a torus

(f) the product of a circle and a disk is a solid torus

Problem (5)

The set $\Delta = \{(x_1, x_2) \in X^2 \mid x_1 = x_2\}$ is called the *diagonal* of the Cartesian square X^2 of the set X. Give geometric representations of the diagonals of the sets obtained in parts (a), (b), and (e) of Exercise 4.

(a) the diagonal of the rectangle (b) the set (x, x) (e) the center circle of the torus

Problem (6)

Show that

(a) $(X \times Y = \varnothing) \iff (X = \varnothing) \vee (Y = \varnothing)$, and if $X \times Y \neq \varnothing$, then

(b) $(A \times B \subset X \times Y) \iff (A \subset X) \wedge (B \subset Y)$,

(c) $(X \times Y) \cup (Z \times Y) = (X \cup Z) \times Y$,

(d) $(X \times Y) \cap (X' \times Y') = (X \cap X') \times (Y \cap Y')$.

Here $\varnothing$ denotes the empty set, that is, the set having no elements.

(a) If $X \times Y = \varnothing$, then $X = \varnothing$ or $Y = \varnothing$, so $(X = \varnothing) \vee (Y = \varnothing)$.

(b) Suppose $A \times B = \{(a, b) \mid a \in A, b \in B\}$, and the same with $X \times Y$. Then if $A \times B \subset X \times Y$, then all $a \in A$ also have $a \in X$, so $A \subset X$, and similarly $B \subset Y$.

(c) $\{(x,y) \mid x \in X, y \in Y\}$ and $\{(z,y) \mid z \in Z, y \in Y\}$. The second number isn't changed but the first numbers becomes the set of all numbers in $X \cup Z$, so $(X \times Y) \cup (Z \times Y) = (X \cup Z) \times Y$.

(d) If $x \in X \cap x' \in X'$, we get $x \in X \cap X'$, etc.

Problem (7)

By comparing the relations of Exercise 3 with relations (a) and (b) from Exercise 2 of Sect. 1.1, establish a correspondence between the logical operators $\neg$, $\wedge$, $\vee$ and the operations C, $\cap$, and $\cup$ on sets.

$C \approx \neg$, $\cup \approx \vee$, $\cap \approx \wedge$

1.3 Functions

Problem (1)

The *composition* $\mathcal{R}_2 \circ \mathcal{R}_1$ of the relations $\mathcal{R}_1$ and $\mathcal{R}_2$ is defined as follows:

$$\mathcal{R}_2 \circ \mathcal{R}_1 := \{(x,z) \mid \exists y (x\mathcal{R}_\infty y \wedge y\mathcal{R}_2 z)\}.$$

In particular, if $\mathcal{R}_1 \subset X \times Y$ and $\mathcal{R}_2 \subset Y \times Z$, then $\mathcal{R} = \mathcal{R}_2 \circ \mathcal{R}_1 \subset X \times Z$, and

$$x\mathcal{R}z := \exists y\,((y \in Y) \wedge (x\mathcal{R}_1 y) \wedge (y\mathcal{R}_2 z)).$$

(a) Let Δ_X be the diagonal of X^2 and Δ_Y the diagonal of Y^2. Show that if the relations $\mathcal{R}_1 \subset X \times Y$ and $\mathcal{R}_2 \subset Y \times X$ are such that $(\mathcal{R}_2 \circ \mathcal{R}_1 = \Delta X) \wedge (\mathcal{R}_1 \circ \mathcal{R}_2 = \Delta_Y)$, then both relations are functional and define mutually inverse mappings of X and Y.

(b) Let $\mathcal{R} \subset X^2$. Show that the condition of transitivity of the relation $\mathcal{R}$ is equivalent to the condition $\mathcal{R} \circ \mathcal{R} \subset \mathcal{R}$.

(c) The relation $\mathcal{R}' \subset Y \times X$ is called the *transpose* of the relation $R \subset X \times Y$ if $(y\mathcal{R}'x) \iff (x\mathcal{R}y)$.

Show that a relation $\mathcal{R} \subset X^2$ is antisymmetric if and only if $\mathcal{R} \cap \mathcal{R}' \subset \Delta_X$.

(d) Verify that any two elements of X are connected (in some order) by the relation $\mathcal{R} \subset X^2$ if and only if $\mathcal{R} \cup \mathcal{R}' = X^2$.

(a) If $\mathcal{R}_1 \subset X \times Y$ and $\mathcal{R}_2 \subset Y \times X$ are such that $(\mathcal{R}_2 \circ \mathcal{R}_1 = \Delta_X) \wedge (\mathcal{R}_1 \circ \mathcal{R}_2 = \Delta_Y)$, then both relations are functional. This is because $\mathcal{R}_2 \circ \mathcal{R}_1$ equals Δ_X

(b) Transitivity: $(a\mathcal{R}b) \wedge (b\mathcal{R}c) \implies a\mathcal{R}c$. This is equivalent to the condition that if 2 functions are applied, we get a function, so basically $\mathcal{R} \circ \mathcal{R} \subset \mathcal{R}$.

(c) Since $\mathcal{R} \subset X^2$, then it is antisymmetric if $\mathcal{R} \cup \mathcal{R}' \subset \Delta_X$, as $(x,y) \in \Delta$ if and only if $x = y$.

(d) Two elements of X are connected in some order by the relation $\mathcal{R} \subset X^2$ if and only if $\mathcal{R} \cup \mathcal{R}' = X^2$.

Problem (2)

Let $f : X \mapsto Y$ be a mapping. The pre-image $f^{-1}(y) \subset X$ of the element $y \in Y$ is called the *fiber* over y.

1. Find the fibers for the following mappings:

$$\mathrm{pr}_1 : X_1 \times X_2 \mapsto X_1, \qquad \mathrm{pr}_2 : X_1 \times X_2 \mapsto X_2.$$

2. An element $x_1 \in X$ will be considered to be connected with an element $x_2 \in X$ by the relation $\mathcal{R} \subset X^2$, and we shall write $x_1 \mathcal{R} x_2$ if $f(x_1) = f(x_2)$, that is x_1 and x_2 both lie in the same fiber.

   Verify that $\mathcal{R}$ is an equivalence relation.

3. Show that the fibers of a mapping $f : X \mapsto Y$ do not intersect one another and that the union of all the fibers is the whole set X.

4. Verify that any equivalence relation between elements of a set makes it possible to represent the set as a union of mutually disjoint equivalence classes of elements.

(a) Since this maps from $X_1 \times X_2$ to X_1 and $X_1 \times X_2$ to X_2, then the fibers are from X_1 to $X_1 \times X_2$ and X_2 to $X_1 \times X_2$.

(b) Since $f(x_1) = f(x_1)$, then $x_1 \mathcal{R} x_1$, so $\mathcal{R}$ is reflexive. Since $f(x_1) = f(x_2)$ implies $f(x_2) = f(x_1)$, then $x_1 \mathcal{R} x_2$ and $x_2 \mathcal{R} x_1$ are equivalent, so this is symmetric. Since $f(x_1) = f(x_2)$ and $f(x_2) = f(x_3)$ implies $f(x_1) = f(x_3)$, then $(x_1 \mathcal{R} x_2) \wedge (x_2 \mathcal{R} x_3) = x_1 \mathcal{R} x_3$, so this is transitive. Therefore $\mathcal{R}$ is an equivalence relation.

(c) Note that a function, has, by definition, the same output for the same input. Therefore, if $f(x) = y_1$, and $f(x) = y_2$, we must have $y_1 = y_2$. This way, the fibers corresponding to two values will not have any common members. Since f maps all of the members of X to Y, then for every $x \in X$, there must be some element $y \in Y$ such that $f(x) = y$. Then x is the fiber of y, so the union of all x is the set X.

(d) If we have an equivalence relation, we can split the set into smaller classes of elements that are equivalent to eachother. These classes are disjoint due to the transitive property, because if an element is equivalent to another element, then they must be in the same class.

Problem (3)

Let $f : X \mapsto Y$ be a mapping from X into Y. Show that if A and B of X, then

(a) $(A \subset B) \implies (f(A) \subset f(B)) \neq (A \subset B)$.

(b) $(A \neq \varnothing) \implies (f(A) \neq \varnothing)$,

(c) $f(A \cap B) \subset f(A) \cap f(B)$,

(d) $f(A \cup B) = f(A) \cup f(B)$;

if A' and B' are subsets of Y, then

(e) $(A' \subset B') \implies (f^{-1}(A') \subset f^{-1}(B'))$,

(f) $f^{-1}(A' \cap B') = f^{-1}(A') \cap f^{-1}(B')$,

(g) $f^{-1}(A' \cup B') = f^{-1}(A') \cup f^{-1}(B')$;

if $Y \supset A' \supset B'$, then

(h) $f^{-1}(A' \backslash B') = f^{-1}(A') \backslash f^{-1}(B')$,

(i) $f^{-1}(C_Y A') = C_X f^{-1}(A')$;

and for any $A \subset X$ and $B' \subset Y$

(j) $f^{-1}(f(A)) \supset A$,

(k) $f(f^{-1}(B')) \subset B'$.

(a) If $A \subset B$, then $f(A) \subset f(B)$, but this does not necessarily imply that $A \subset B$. For example, if we let $f(x) = x^2$ and set $A = \{-1, 0, 1\}$ and $B = \{0, 1, 2\}$. Then $f(A) \subset f(B)$ but $A \not\subset B$.

(b) If $(A \neq \varnothing)$, then this means that $f(A) \neq \varnothing$ because otherwise this would require A to be the empty set.

(c) Let $x \in f(A \cap B)$. Since $A \cap B$ is a subset of A and B, then $f(A \cap B)$ is a subset of $f(A)$ and $f(B)$, so $f(A \cap B) \subset f(A) \cap f(B)$.

(d) Since A and B are subsets of $A \cup B$, then $f(A)$ and $f(B)$ are subsets of $f(A \cup B)$, so $f(A \cup B) = f(A) \cup f(B)$.

(e) Since $A' \subset B'$ and these are outputs, then the subset of X that map to A' and the subset of X that maps to B' is a subset of each other, so $f^{-1}(A') \subset f^{-1}(B')$.

(f) Consider two sets A and B. Then $f^{-1}(A') \cap f^{-1}(B') = A \cap B$ and $f^{-1}(A' \cap B') = A \cap B$, so $f^{-1}(A' \cap B') = f^{-1}(A') \cap f^{-1}(B')$.

(g) Note that $f^{-1}(A' \cup B') = A \cup B$ and $f^{-1}(A') = A$, $f^{-1}(B') = B$, so $f^{-1}(A' \cup B') = A \cup B = f^{-1}(A') \cup f^{-1}(B')$.

(h) $f^{-1}(A' \backslash B') = A \backslash B$. $f^{-1}(A') = A$, $f^{-1}(B) = B$, so $f^{-1}(A') \backslash f^{-1}(B') = A \backslash B$. Therefore $f^{-1}(A' \backslash B') = A \backslash B = f^{-1}(A') \backslash f^{-1}(B')$.

(i) Suppose $X \supset A \supset B$. Then $f^{-1}(C_Y A') = C_X A$ and since $f^{-1}(A') = A$, then $f^{-1}(C_Y A') = C_X f^{-1}(A')$.

(j) Suppose $x \in A$. Then $f(x) \in f(A)$, so $x \in f^{-1}(f(A))$. Therefore $x^{-1}(f(A)) \supset A$.

(k) Suppose $y \in f(f^{-1}(B'))$. Then $f^{-1}(y) \in f^{-1}(B')$ and $y \in B$, so $f(f^{-1}(B')) \subset B'$.

Problem (4)

Show that the mapping $f : X \mapsto Y$ is

(a) surjective if and only if $f(f^{-1}(B')) = B'$ for every set $B' \subset Y$;

(b) bijective if and only if

$$\left(f^{-1}\left(f(A)\right) = A\right) \wedge \left(f\left(f^{-1}\left(B'\right)\right) = B'\right)$$

for every set $A \subset X$ and every set $B' \subset Y$.

(a) This is surjective if every element in Y is mapped to. Therefore $f^{-1}(B') = B$, so $f(B) = B'$ (since it's surjective).

(b) Bijective if it is both injective and surjective. It is injective if every unique output corresponds to a unique input. $f(A) = A'$, and since this is injective, $f^{-1}(A') = A$. Putting these two together gives us bijective.

Problem (5)

Verify that the following statements about a mapping $f : X \mapsto Y$ are equivalent:

(a) f is injective;

(b) $f^{-1}(f(A)) = A$ for every $A \subset X$;

(c) $f(A \cap B) = f(A) \cap f(B)$ for any two subsets A and B of X;

(d) $f(A) \cap f(B) = \varnothing \iff A \cap B = \varnothing$;

(e) $f(A \backslash B) = f(A) \backslash f(B)$ whenever $X \supset A \supset B$.

The first two are clearly equivalent by the previous problem.

(c) $f(A \cap B) = f(A) \cap f(B)$ is equivalent to being injective because no two disjoint set of values can map to the same output set, hence the $\cap$.

(d) is equivalent as well because by the previous problem, $f(A \cap B) = f(A) \cap f(B)$, so $f(A \cap B) = \varnothing$, so $A \cap B = \varnothing$.

(e) is because if $y \in f(A) \backslash f(B)$, then $y \in f(A)$ but $y \notin f(B)$. Therefore, if this is injective, then there is only one x satisfying $f(x) = y$. Then $x \in A$, but $x \notin B$, so $x \in A \backslash B$. Therefore $y \in f(A \backslash B)$.

Problem (6) (a) If the mappings $f : X \mapsto Y$ and $g : Y \mapsto X$ are such that $g \circ f = e_X$, where e_X is the identity mapping on X, then g is called a *left inverse* of f and f a *right inverse* of g. SHow that, in contrast to the uniqueness of the inverse mapping, there may exist many one-sided inverse mappings.

Consider, for example, the mappings $f : X \mapsto Y$ and $g : Y \mapsto X$, where X is a one-element set and Y a two-element set, or the mappings of sequences given by

$$(x_1, \dots, x_n, \dots) \xmapsto{f_a} (a, x_1, \dots, x_n, \dots),$$
$$(y_2, \dots, y_n, \dots) \xleftarrow{g} (y_1, y_2, \dots, y_n, \dots).$$

(b) Let $f : X \mapsto Y$ and $g : Y \mapsto Z$ be bijective mappings. Show that the mapping $g \circ f : X \mapsto Z$ is bijective and that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

(c) Show that the equality

$$(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C))$$

holds for any mappings $f : X \mapsto Y$ and $g : Y \mapsto Z$ and any set $C \subset Z$.

(d) Verify that the mappings $F : X \times Y \mapsto Y \times X$ defined by the correspondence $(x, y) \mapsto (y, x)$ is bijective. Describe the connection between the graphs of mutually inverse mappings $f : X \mapsto Y$ and $f^{-1} : Y \mapsto X$.

(a) There are many one-sided inverse mappings, as we only need to worry about the outside canceling out the inside, and we don't need to worry about any special cases that the inside might have on the outside. Some more examples are $f(x) = x^2$ and $g(x) = \sqrt{x}$. Then $f(g(x)) = x$ but $g(f(x)) = |x|$. Not the same.

(b) Since f is surjective, then $f(X) = Y$, and the range of f completely equals Y. Since g is also surjective, then the range of $g(f(X)) = g(Y)$ completely equals Z. This means that $g \circ f$ is surjective.

Since both f and g are injective, every distinct input in X maps to a distinct output in Y (via f), and every distinct input in Y maps to a distinct output in Z (via g). So since the domain of f maps distinctly to the range of f (and every output in Y is distinct), then all of the distinct outputs in Y input to distinct outputs in Z. Now $g \circ f$ is distinct inputs to X to distinct outputs to Z.

(c) Since $g \circ f$ maps from X to Z, then $(g \circ f)^{-1}$ maps from Z to X. g maps from Y to Z, so g^{-1} maps from Y to Z. f^{-1} maps from X to Y, so $f^{-1}(g^{-1})$ also maps from Z to X. Since f and g are bijective, then so are f^{-1} and g^{-1} and $(g \circ f)^{-1}$, so the equality holds.

(d) Since $f : X \mapsto Y$ is bijective, then $f^{-1} : Y \mapsto X$ is also bijective. f^{-1} maps the outputs of f to the inputs of f. So f^{-1} could be described as reflecting f over the identity mapping on X, e_X.

Problem (7) (a) Show that for any mapping $f : X \mapsto Y$ the mapping $F : X \mapsto X \times Y$ defined by the correspondence $x \overset{F}{\longmapsto} (x, f(x))$ is injective.

(b) Suppose a particle is moving at uniform speed on a circle Y; let X be the time axis and $x \overset{F}{\longmapsto}$ the correspondence between the time $x \in X$ and the position $y = f(x) \in Y$ of the particle. Describe the graph of the function $f : X \mapsto Y$ in $X \times Y$.

(a) Note that the output $(x, f(x))$ is distinct for every distinct x because the first number in the pair is x, which is distinct for every distinct x. Therefore F is injective.

(b) It depends on the speed and method of describing the position, but the graph should be periodic with period equal to the time required for one full cycle.

Problem (8) (a) For each of the examples 1-12 considered in Sect. 1.3 determine whether the mapping defined in the examples is surjective, injective, or whether it belongs to none of these classes.

(b) Ohm's law $I = V/R$ connects the current I in the conductor with the potential difference V at the ends of hte counductor and the resistance R of the conductor. Given sets X and Y for which some mapping $O : X \mapsto Y$ corresponds to Ohm's law. What set is the relation corresponding to Ohm's law a subset of?

(c) Find the mappings G^{-1} and L^{-1} inverse to the Galilean and Lorentz transformations.

(a)
1. This formula is bijective because different radii map to different volumes, and all positive volumes can be achieved. Same with circumference.
2. This formula is neither injective nor surjective, because all inertial systems map to the same value, and there is only one value.
3. This is injective but not surjective, assuming v is constant.
4. This is surjective but not injective because we can change all of the values of x_1 to make X_1, but changing x_2 will change the input but not the output.
5. This is both injective and surjective. For any $B \in \mathcal{P}(M)$, we can obtain the input by taking the complement, so it's surjective, and it's injective because the input is unique.
6. This is not injective because there are only 2 possible outputs so many of them will map to the same output, and it's not surjective because 0 and 1 obviously do not encompass all of $\mathbb{R}$.
7. We don't know for sure if this is injective or surjective, because $f(x_0)$ could be the same for different f, and it does not necessarily encompass all of Y.
8. This is not injective because there are multiple ways to get the same length. This is also not surjective because it is impossible to get from one point to another with 0 or negative length.

9. This is injective because every unique input function will be shifted to the left by a, which will result in another unique output function. This is also surjective because we can obtain all of the functions in $M(\mathbb{R};\mathbb{R})$ by shifting the functions to the left by a.
10. The motion of a particle's function is not necessarily injective nor surjective because a particle could visit a point in space twice, and it does not necessarily visit every single point in space.
11. This is not injective because two configurations could have the same potential energy. This is not surjective because the configuration couldn't have infinite potential energy.
12. This is not injective because there is only one value for the mechanical energy, so it's not surjective either.

(b) We can have $X = \mathbb{R}^2$ and $Y = \mathbb{R}$, where $V \times R \subset \mathbb{R}^2$ and $I \subset \mathbb{R}$.

(c) $G^{-1} : \mathbb{R}^2 \mapsto \mathbb{R}^2$, $L^{-1} : \mathbb{R}^2 \mapsto \mathbb{R}^2$.

Problem (9) (a) A set $S \subset X$ is *stable* with respect to a mapping $f : X \mapsto X$ if $f(S) \subset S$. Describe the sets that are stable with respect to a shift of the plane by a given vector lying in the plane.

(b) A set $I \subset X$ is *invariant* with respect to a mapping $f : X \mapsto X$ if $f(I) = I$. Describe the sets that are invariant with respect to rotation of the plane about a fixed poin.

(c) A point $p \in X$ is a *fixed point* of a mapping $f : X \mapsto X$ if $f(p) = p$. Verify that any composition of a shift, rotation, and a similarity transformation of the plane has a fixed point, provided the coefficient of the similarity transformation is less than 1.

(d) Regarding the Galilean and Lorentz transformations as mappings of the plane into itself for which the point with coordinates (x, t) maps to the point with coordinates (x', t'), find the invariant sets of these transformations.

(a) Because the vector is lying in the plane and we can assume that the plane is infinitely large, then as long as the vector lies in the plane, then the plane will only go along the vector so it's stable.

(b) This occurs when the point is on the plane, and the plane rotates around the line through the point perpendicular to the plane.

(c) Let this be a function mapping $f : \mathbb{R}^3 \mapsto \mathbb{R}^3$. There must be a solution to $f(x, y, z) = (x, y, z)$, so since this function represents the composition of a shift, a rotation, and a similarity transformation, then there must be a fixed point.

(d) This occurs when the velocity or time is 0, so when the inertial states are equal.

Problem (10)

Consider the steady flow of a fluid (that is, the velocity at each point of the flow does not change over time). In time t a particle at point x of the flow will move to some new point $f_t(x)$ of space. The mapping $x \mapsto f_t(x)$ that arises thereby on the points of space occupied by the flow depends on time and is called the *mapping after time t*. Show that $f_{t_2} \circ f_{t_1} = f_{t_1} \circ f_{t_2} = f_{t_1+t_2}$ and $f_t \circ f_{-t} = e_X$.

This is saying that if after t_1 time, x will move to a point $f_{t_1}(x)$, and after another t_2 time, x will move to a point $f_{t_2}(f_{t_1}(x))$. However, since t_1 and t_2 are continuous, we can just look at this like a whole $t_1 + t_2$ time, so $f_{t_2} \circ f_{t_1} = f_{t_2+t_1}$. Similarly for $f_{t_1} \circ f_{t_2}$, so we're done. Also $f_t \circ f_{-t} = f_{t-t} = f_0 = e_X$, because after 0 time nothing changes.

1.4 Supplementary Material

Problem (1) (a) Prove the equipotence of the closed interval $\{x \in \mathbb{R} \mid 0 \le x \le 1\}$ and the open interval $\{x \in \mathbb{R} \mid 0 < x < 1\}$ of the real line $\mathbb{R}$ both using the Schroder-Bernstein theorem and by direct exhibition of a suitable bijection.

(b) Analyze the following proof of the Schroder-Bernstein theorem:

$$(\mathrm{card}X \le \mathrm{card}Y) \wedge (\mathrm{card}Y \le \mathrm{card}X) \implies (\mathrm{card}X = \mathrm{card}Y).$$

Proof. It suffices to prove that if the sets X, Y, and Z are such that $X \supset Y \supset Z$ and $\mathrm{card}X = \mathrm{card}Z$, then $\mathrm{card}X = \mathrm{card}Y$. Let $f : X \mapsto Z$ be a bijection. A bijection $g : X \mapsto Y$ can be defined, for example, as follows:

$$g(x) = \begin{cases} f(x), & \text{if } x \in f^n(X) \backslash f^n(Y) \text{ for some } n \in \mathbb{N}, \\ x & \text{otherwise.} \end{cases}$$

Here $f^n = f \circ \cdots \circ f$ is the nth iteration of the mapping f and $\mathbb{N}$ is the set of natural numbers. ☐

(a) Let $A = \{x \in \mathbb{R} \mid 0 \le x \le 1\}$ and $B = \{x \in \mathbb{R} \mid 0 < x < 1\}$. Note that $\mathrm{card}B \le \mathrm{card}A$ because we can simply map each element x in B to the element x in A. Also, $\mathrm{card}A \le \mathrm{card}B$ because we can map each nonzero element x in A to the element $\frac{x}{2}$ in B, and we can map the 0 in A to the 0.6 in B. Therefore, by the Schroder-Bernstein theorem, $\mathrm{card}A = \mathrm{card}B$, so A and B are equipotent.

(b) Okay, done.

Problem (2) (a) Starting from the definition of a pair, verify that the definition of the direct product $X \times Y$ of sets X and Y given in Sect. 1.4.2 is unambiguous, that is, the set $\mathcal{P}(\mathcal{P}(X) \cup \mathcal{P}(Y))$ contains all ordered pairs (x, y) in which $x \in X$ and $y \in Y$.

(b) Show that the mappings $f : X \mapsto Y$ from one given set X into another given set Y themselves form a set $M(X, Y)$.

(c) Verify that if $\mathcal{R}$ is a set of ordered pairs (that is, a relation), then the first elements of the pairs belonging to $\mathcal{R}$ (like the second elements) form a set.

(a) We have to prove that there is a bijection between $X \times Y$ and the set of all ordered pairs (x, y) in which $x \in X$ and $y \in Y$. Note that the cardinality of the set of (x, y) for $x \in X$, $y \in Y$ is less than or equal to the cardinality of $X \times Y$ because we can simply map the set to the pairs (x, y) in $X \times Y$. Similarly, the cardinality of $X \times Y$ is less than or equal to the cardinality of the set of pairs (x, y).

(b) We know that by definition, $M(X, Y)$ is the set of all functions that map from the set X to the set Y, so isn't that obvious?

(c) A set can consist of anything. Therefore, we can group all of the first elements together, then delete off the redundant terms to get a set.

Problem (3) (a) Using the axioms of extensionability, pairing, separation, union, and infinity, verify that the following statements hold for the elements of the set $\mathbb{N}_0$ of natural numbers in the sense of von Neumann:

1^0 $x = y \implies x^+ = y^+$;

2^0 $(\forall x \in \mathbb{N}_0)\ (x^+ \neq \varnothing)$;

3^0 $x^+ = y^+ \implies x = y$;

4^0 $(\forall x \in \mathbb{N}_0)\ (x \neq \varnothing \implies (\exists y \in \mathbb{N}_0)\ (x = y^+))$.

(b) Using the fact that $\mathbb{N}_0$ is an inductive set, show that the following statements hold for any of its elements x and y (which in turn are themselves sets):

1^0 $\text{card}x \leq \text{card}x^+$;

2^0 $\text{card}\varnothing < \text{card}x^+$;

3^0 $\text{card}x < \text{card}y \iff \text{card}x^+ < \text{card}y^+$;

4^0 $\text{card}x < \text{card}x^+$;

5^0 $\text{card}x < \text{card}y \implies \text{card}x^+ \leq \text{card}y$;

6^0 $x = y \iff \text{card}x = \text{card}y$;

7^0 $(x \subset y) \vee (x \supset y)$.

(c) Show that in any subset X of $\mathbb{N}_0$ there exists a (minimal) element x_m such that $(\forall x \in X)(\text{card}x_m \leq \text{card}x)$. (If you have difficulty doing so, come back to this problem after reading Chap. 2.)

(a) If $x = y$, then $x^+ = y^+$, and conversely, if $x^+ = y^+$, then $x = y$. For all $x \in \mathbb{N}_0$, $x^+ \neq \varnothing$. If, for all $x \in \mathbb{N}_0$, $x \neq \varnothing$, then there exists $y \in \mathbb{N}_0$ such that $x = y^+$.

(b) Note that card$x$ < cardx^+, so therefore card$x \le$ cardx^+ If card$x$ < card$y$, then cardx^+ < cardy^+, and vice versa. Also, card$\varnothing = 0 <$ cardx^+, and card$x$ < card$y$ implies that card$x^+ \le$ card$y$ and vice versa. If $x = y$, then card$x$ = card$y$, and vice versa. ALso, either $x \subset y$ or $x \supset y$, we cannot have $x = y$.

(c) In any subset X of $\mathbb{N}_0$ there exists a minimal element x_m such that for all $x \in X$, card$x_m \le cardx$. Any set has a greatest lower bound, which in this case would be x_m.

Problem (4)

We shall deal only with sets. Since a set consisting of different elements may itself be an element of another set, logicians usually denote all sets by uppercase letters. In the present exercise, it is very convenient to do so.

(a) Verify that the statement

$$\forall x \exists y \forall z\,(z \in y \iff \exists w(z \in w \wedge w \in x))$$

expresses the axiom of union, according to which y is the union of the sets belonging to x.

(b) State which axioms of set theory are represented by the following statements:

$$\forall x\ \forall y\ \forall z\,((z \in x \iff z \in y) \iff x = y)\,,$$
$$\forall x\ \forall y\ \exists z\ \forall v\,(v \in z \iff (v = x \vee v = y))\,,$$
$$\forall x\ \exists y\ \forall z(z \in y \iff \forall u\ (u \in z \implies u \in x))$$
$$\exists x\ (\forall y\ (\neg\exists z\ (z \in y) \implies y \in x) \wedge$$
$$\wedge\, \forall w\,(w \in x \implies \forall u\,(\forall v\,(v \in u \iff (v = w \vee v \in w)) \implies u \in x))).$$

(c) Verify that the formula

$$\forall z\,(z \in f \implies (\exists x_1\ \exists y_1\,(x_1 \in x \wedge y_1 \in y \wedge z = (x_1, y_1)))) \wedge$$
$$\wedge\, \forall x_1\ (x_1 \in x \implies \exists y_1\ \exists z_1\,(y_1 \in y \wedge z = (x_1, y_1) \wedge z \in f)) \wedge$$
$$\wedge\, \forall x_1\ \forall y_1\ \forall y_2(\exists z_1 \exists z_2\,(z_1 \in f \wedge z_2 \in f \wedge z_1 = (x_1, y_1) \wedge z_2 = (x_1, x_2)) \implies$$
$$\implies y_1 = y_2)$$

imposes three successive restrictions on the set f : f is a subset of $x \times y$; the projection of f on x is equal to x; to each element x_1 of x there corresponds exactly one y_1 in y such that $(x_1, y_1) \in f$.

Thus what we have here is a definition of a mapping $f : x \mapsto y$.

This example shows yet again that the formal expression of a statement is by no means always the shortest and most transparent in comparison with its expression in ordinary language. Taking this circumstance into account, we shall henceforth use logical symbolism only to the extent that it seems useful to us to achieve greater compactness or clarity of exposition.

(a) This states that for all x and z, there exists y such that $z \in y$ is equivalent to there existing w such that $z \in w$ and $w \in x$. The axiom of union states that for any set

M whose elements are sets, there exists a set whose elements are the elements of the elements of M. This is equivalent to the statement because z are the elements of the elements of M, y is the set with elements of elements, and w is the elements of M.

(b) First statement: for all x, y, and z, $z \in x$ implying $z \in y$ means that $x = y$. This is the axiom of extensionality.

Second statement: For all x, y, and v, there exists z such that $v \in z$ is equivalent to $v = x$ and $v = y$. This is the pairing axiom.

Third statement: For all x and z, there exists y such that $z \in y$ is equivalent to the statement that for all u, $u \in z$ implies that $u \in x$. Axiom of separation

Fourth statement: There exists x such that for all y, there does not exist z such that $z \in y$ implies that $y \in x$ and for all w, $w \in x$ implies that for all u, for all v, $v \in u$ implies that $v = w$ or $v \in w$, implies that $u \in x$. This is the axiom of choice.

(c) f is a subset of $x \times y$ (first requirement), the projection of f on x is equal to x (second requirement), and to each element x_1 of x there corresponds exactly one y_1 in y such that $(x_1, y_1) \in f$ (third requirement). This mathematical notation is barely legible.

Problem (5)

Let $f : X \mapsto Y$ be a mapping. Write the logical negation of each of the following statements:

(a) f is surjective;

(b) f is injective;

(c) f is bijective.

(a) f is not surjective, so when f maps the elements of X, it misses some of the elements in Y

(b) f is not injective, so some of the elements in X map to the same element in Y.

(c) f is not bijective, meaning that there is at least one element, either in X or Y, that doesn't map to or from exactly one other element.

Problem (6)

Let X and Y be sets and $f \subset X \times Y$. Write what it means to say that the set f is not a function.

If f is not a function, then this means that there exist multiple mappings from the same value of X to Y.

Chapter 3

Zorich Mathematical Analysis

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3.1 The Limit of a Sequence

1 Suppose a number $x \in \mathbb{R}$'s base q representation has periodically repeating digits. Separate the repeating part from the non-repeating part. For example, if we have $x = 0.1232323\ldots$, we can separate this into $a = 0.1$ and $y = 0.0\overline{23}$. Then we can multiply x by q^n to get $x \cdot q^n - (x - a)$ is equal to a rational number. Therefore we can solve for x, which is a rational number, and then convert the numerators and denominators into standard base.

2 The heights are h, qh, q^2h, etc. Note that if the initial or final velocity is v, then the final or initial velocity is 0, so $v^2 = 2gy$. This means that $v = \sqrt{2gy}$, so $t = \sqrt{\frac{2y}{g}}$. Thus the time in total is

$$\sqrt{\frac{2h}{g}} + \sqrt{\frac{2qh}{g}} + \sqrt{\frac{2q^2h}{g}} + \cdots = \sqrt{\frac{2h}{g}}(1 + \sqrt{q} + \sqrt{q^2} + \cdots) = \frac{\sqrt{\frac{2h}{g}}}{1 - \sqrt{q}}.$$

The total distance traveled is $h + qh + q^2h + \cdots = \frac{h}{1-q}$.

3 We mark all the points on a cicle obtained from a fixed point by rotations of the circle through angles of n radians, where $n \in \mathbb{Z}$ ranges over all integers. This way, every single point on the circle will eventually be marked because we need to find the possible values of $n \pmod{2\pi}$ span the entirety of the range $[0, 2\pi)$.

4

(a) Using the hint, we have $\frac{m}{n} = q_1 + \frac{r_1}{n} = q_1 + \frac{1}{n/r_1}$. We also have $\frac{n}{r_1} = q_2 + \frac{r_2}{r_1} = q_2 + \frac{1}{r_1/r_2}$. This gives

$$\frac{m}{n} = q_1 + \frac{1}{q_2 + \frac{1}{r_1/r_2}}.$$

Then $r_1/r_2 = q_3 + \frac{r_3}{r_2} = q_3 + \frac{1}{r_2/r_3}$, so we get

$$\frac{m}{n} = q_1 + \frac{1}{q_2 + \frac{1}{q_3 + \frac{1}{r_2/r_3}}}.$$

Because the m and n are relatively prime, then there will be n equations, so there will be n of those nested fractions.

(b) We prove the first half of the inequalities first, namely that $R_1 < R_3 < \cdots < R_{2k-1}$. Consider any odd n. Then the nested fraction ends in something like $\frac{1}{q_{n-1} + \frac{1}{q_n}}$.

Then R_{n+2} gives an ending like

$$\frac{1}{q_{n-1}+\frac{1}{q_n+\frac{1}{q_{n+1}+\frac{1}{q_{n+2}}}}}.$$

Since the denominator of $\frac{1}{q_n}$ increases, then $\frac{1}{q_n}$ decreases, so the denominator of $\frac{1}{q_{n-1}}$ decreases, and hence $\frac{1}{q_{n-1}}$ increases. By repeating this, and utilizing the fact that n is odd, we arrive that $R_{n+2} > R_n$, increases for odd n. In a similar fashion, $R_{n+2} < R_n$ for even n, and since $\frac{m}{n}$ is larger than R_{2k-1} but smaller than R_{2k}, we're done.

(c) We prove this by induction. Note that $R_1 = q_1 = \frac{P_1}{Q_1}$. Suppose that for some k, $P_k = P_{k-1}q_k + P_{k-2}$ and $Q_k = Q_{k-1}q_k + Q_{k-2}$. Observe that we can obtain R_{k+1} from R_k by replacing the q_k in R_k with $q_k + \frac{1}{q_{k+1}} = \frac{q_kq_{k+1}+1}{q_{k+1}}$. We get

$$\begin{aligned}
\frac{P_{k+1}}{Q_{k+1}} &= \frac{P_{k-1}\left(\frac{q_kq_{k+1}+1}{q_{k+1}}\right)+P_{k-2}}{Q_{k-1}\cdot\left(\frac{q_kq_{k+1}+1}{q_{k+1}}\right)+Q_{k-2}}\\
&= \frac{P_{k-1}(q_kq_{k+1}+1)+q_{k+1}P_{k-2}}{Q_{k-1}(q_kq_{k+1}+1)+q_{k+1}Q_{k-2}}\\
&= \frac{P_{k-1}q_kq_{k+1}+P_{k-1}+q_{k+1}P_{k-2}}{Q_{k-1}q_kq_{k+1}+Q_{k-1}+q_{k+1}Q_{k-2}}\\
&= \frac{(P_{k-1}q_k+P_{k-2})q_{k+1}+P_{k-1}}{(Q_{k-1}q_k+Q_{k-2})q_{k+1}+P_{k-1}}\\
&= \frac{P_kq_{k+1}+P_{k-1}}{Q_kq_{k+1}+Q_{k-1}}.
\end{aligned}$$

(d) Observe that

$$R_k - R_{k-1} = \frac{P_k}{Q_k} - \frac{P_{k-1}}{Q_{k-1}} = \frac{P_kQ_{k-1}-P_{k-1}Q_k}{Q_{k-1}Q_k}.$$

Then

$$P_kQ_{k-1}-P_{k-1}Q_k = (P_{k-1}q_k+P_{k-2})Q_{k-1}-P_{k-1}(Q_{k-1}q_k+Q_{k-2}) = P_{k-2}Q_{k-1}-Q_{k-2}P_{k-1}.$$

Therefore, induction on k down to 2 gives

$$P_kQ_{k-1}-P_{k-1}Q_k = (-1)^{k-2}(P_2Q_1-Q_2P_1) = (-1)^n.$$

Then we get $\frac{(-1)^n}{Q_kQ_{k-1}}$, as desired.

(e)

(f) We need to show that the continued fraction is

(g) (f) Observe that a rational number has a unique continued fraction representation as shown by the Euclidean algorithm analogy.

(h) (g) Let the continued fraction equal x. Note that $x = 1 + \frac{1}{x}$, so $x^2 - x - 1 = 0$. Observe that the solution to this is $x = \frac{1\pm\sqrt{5}}{2}$. Since x is positive, then we must have $x = \frac{1+\sqrt{5}}{2}$, as desired.

(i) Since $u_n = u_{n-1} + u_{n-2}$, the characteristic equation is $c^2 - c - 1$, so we have roots $c_1 = \frac{1+\sqrt{5}}{2}$ and $c_2 = \frac{1-\sqrt{5}}{2}$. We must solve the equation $F_n = a_1(c_1)^n + a_2(c_2)^n$. Letting $n = 1$ and 2, the system of equations gives

$$1 = a_1\left(\frac{1+\sqrt{5}}{2}\right) + a_2\left(\frac{1-\sqrt{5}}{2}\right)$$
$$1 = a_1\left(\frac{1+\sqrt{5}}{2}\right)^2 + a_2\left(\frac{1-\sqrt{5}}{2}\right)^2 = a_1\left(\frac{3+\sqrt{5}}{2}\right) + a_2\left(\frac{3-\sqrt{5}}{2}\right).$$

Subtracting the first equation from the second gives $a_1 + a_2 = 0$, so $a_2 = -a_1$. Plugging this into the first equation gives

$$1 = a_1\left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}\right) = a_1\sqrt{5},$$

so $a_1 = \frac{1}{\sqrt{5}}$ and $a_2 = -\frac{1}{\sqrt{5}}$. This gives

$$u_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n,$$

as desired.

(j) The convergents $R_k = \frac{P_k}{Q_k}$.

Since $u_n = u_{n-1}+u_{n-2}$, the characteristic equation is c^2-c-1, so we have roots $c_1 = \frac{1+\sqrt{5}}{2}$ and $c_2 = \frac{1-\sqrt{5}}{2}$. subtracting the first equation from the second gives $a_1 + a_2 = 0$, so $a_2 = -a_1$. Plugging this into the first equation gives

$$1 = a_1\left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}\right)$$

5

1.

6 Find the limits of the sequences.

7 Show that if $a > 0$, the sequence $x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right)$ converges to the square root of a for any $x_1 > 0$.

Estimate the rate of convergence, that is, the magnitude of the absolute error $|x_n - \sqrt{a}| =$
$|$

8

(a) We prove this by induction. The first two are obvious. The base case $n = 1$ is obvious. Assume the 3rd expression holds for some n. Then

$$\begin{aligned} S_2(n+1) &= S_2(n) + (n+1)^2 \\ &= \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n + (n+1)^2 = \frac{1}{3}n^3 + \frac{3}{2}n^2 + \frac{13}{6}n + 1 \\ &= \frac{1}{3}(n+1)^3 + \frac{1}{2}(n+1)^2 + \frac{1}{6}(n+1) \end{aligned}$$

So we're done by induction. Similarly,

$$\begin{aligned} S_3(n+1) = S_3(n) + (n+1)^3 &= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\ &= (n+1)^2\left(\frac{n^2}{2} + (n+1)\right) = \frac{(n+1)^2(n+2)^2}{4} \end{aligned}$$

as desired.

(b) Observe that in general, $a_{k+1} = \frac{1}{k+1}$, so

$$\lim_{n\to\infty} = \frac{\frac{1}{k+1}n^{k+1} + a_k n^k + \cdots + a_0}{n^{k+1}}.$$

For $k > 1$, $\lim_{n\to\infty} \frac{1}{n^k} = 0$, so splitting the fraction into terms gives the desired expression.

3.2 The Limit of a Function

1

(a) Observe that any exponential function satisfies this. Take, for example, $f(x) = e^x$. Then $f(1) = e$ which is definitely positive and not equal to 1. $f(x_1) \cdot f(x_2) = e^{x_1} \cdot e^{x_2} = e^{x_1+x_2} = f(x_1+x_2)$. And since this function is continuous, $f(x) \to f(x_0)$ as $x \to x_0$, as desired.

(b) Observe that any logarithmic function satisfies this. Take, for example, $f(x) = \ln x$. Then $f(e) = \ln e = 1$, and $e > 0$ and $e \neq 1$. Then $f(x_1) + f(x_2) = \ln x_1 + \ln x_2 = \ln(x_1 \cdot x_2) = f(x_1 \cdot x_2)$ and $f(x) \to f(x_0)$ for $x \to x_0$ since ln is a continuous function, as desired.

2

(a) Observe that the function $\phi(x) = e^x$ works. Note that $\phi(x) \cdot \phi(y) = e^x \cdot e^y = e^{x+y} = \phi(x+y)$ and that the exponential function is always positive, so ϕ maps the set of real numbers to the set of positive real numbers.

(b) It suffices to prove that there does not exist a function $\varphi(x+y) = \varphi(x) \cdot \varphi(y)$ mapping $x + y \in \mathbb{R}$ to $\varphi(x) \cdot \varphi(y) \in \mathbb{R}\backslash 0$. Observe that $\mathbb{R}_0$ is a subset of $\mathbb{R}$ because there is an obvious injective mapping from $\mathbb{R}\backslash 0$ to $\mathbb{R}$, but there is no mapping in the opposite direction.

3

(a) Suppose $x < 1$, and let $x = \frac{1}{a}$. Then $x^x = \frac{1}{a^{1/a}} = \frac{1}{\sqrt[a]{a}}$. As $x \to +0$, $a \to \infty$, and it's easy to see that $\sqrt[a]{a} \to 1$ as $a \to \infty$, so the limit is 1.

(b) This is $\sqrt[x]{x}$, which approaches 1 as x approaches infinity because the exponential function grows faster than linearly.

(c)

$$\begin{aligned} \lim_{x\to 0} \frac{\log(1+x)}{x} &= \lim_{x\to 0}\left(\frac{x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots}{x}\right) \\ &= \lim_{x\to 0}\left(1 - \frac{x}{2} + \frac{x^2}{3} - \cdots\right) \\ &= 1. \end{aligned}$$

(d) Since a^x grows faster than x, it also shrinks faster, so this approaches 0 as x approaches 0.

4 Since we know that

$$\ln \frac{n+1}{n} = \ln\left(1+\frac{1}{n}\right) = \frac{1}{n} + O\left(\frac{1}{n^2}\right) \qquad \text{as } n \to \infty,$$

then

$$\lim_{n\to\infty} 1 + \frac{1}{2} + \cdots + \frac{1}{n} = \ln\left(\frac{2}{1}\right) + \ln\left(\frac{3}{2}\right) + \cdots + \ln\left(\frac{n+1}{n}\right) - O\left(\frac{1}{n^2}\right)$$
$$= \ln\left(\frac{n+1}{1}\right) = \ln n + c + o(1),$$

as desired.

5

(a) Since $a_n \sim b_n$, then there exists $\gamma(x)$ such that $a(x) = \gamma(x)b(x)$ and $\lim_{\mathcal{B}} \gamma(x) = 1$. Therefore, because

$$\sum_{n=1}^{\infty} a_n = \gamma(x) \sum_{n=1}^{\infty} b_n$$

and due to the limit of $\gamma(x)$, then

$$\sum_{n=1}^{\infty} a_n \sim \sum_{n=1}^{\infty} b_n,$$

so they must either both converge or both diverge, as desired.

(b) Observe that $\sin \frac{1}{n^p} \sim \frac{1}{n^p}$ because

$$\lim_{n\to\infty} \frac{\sin\left(\frac{1}{n^p}\right)}{\frac{1}{n^p}} = \lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Therefore $\sum_{n=1}^{\infty} \sin \frac{1}{n^p}$ converges if and only if $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges, which is only for $p > 1$.

6

(a) Observe that $a_1 \geq a_2 \geq \ldots$

7

(a) the series $\sum_{n=1}^{\infty} \ln a_n$, where $a_n > 0$, $n \in \mathbb{N}$, converges if and only if the seqence $\{\prod_n = a_1 \cdots a_n\}$ has a finite nonzero limit.

(b) the series $\sum_{n=1}^{\infty} \ln(1+\alpha_n)$, where $|\alpha_n| < 1$, converges absolutely if and only if the series $\sum_{n=1}^{\infty} \alpha_n$ converges absolutely.

8

(a) Instead of proving that if an infinite product $\prod_{n=1}^{\infty} e_n$ converges, then $e_n \to 1$, we prove that if $\lim_{n\to\infty} e_n \neq 1$, then the infinite product $\prod_{n=1}^{\infty} e_n$ cannot converge. Suppose that the limit approaches a, where $a \neq 1$. Then

$$\lim_{k\to\infty} \prod_{n=1}^{k} e_n = a^k.$$

If $a \neq 1$, then this product will not converge, so we must have $a = 1$.

(b) Note that $e_i = e^{\ln e_i}$, so since we know that $e^{x+y} = e^x \cdot e^y$ for $x, y \in \mathbb{R}$, so

$$\prod_{n=1}^{\infty} e_n = e^{\sum_{n=1}^{\infty} \ln e_n},$$

so $\prod_{n=1}^{\infty} e_n$ converges if and only if $\sum_{n=1}^{\infty} \ln e_n$ converges.

(c) We know by part (b) that $\prod_{n=1}^{\infty}(1+\alpha_n)$ converges if and only if the series $\sum_{n=1}^{\infty} \ln(1+\alpha_n)$ converges, so it suffices to show that $\ln(1+\alpha_n)$ and α_n are equivalent. By problem 3(c), we know that $\lim_{x\to 0} \frac{ln(1+x)}{x} = 1$, so they are equivalent. Therefore $\sum_{n=1}^{\infty} \ln(1+\alpha_n)$ converges if and only if $\sum_{n=1}^{\infty} \alpha_n$ converges, so we're done.

9

(a) This is asking us to find

$$\begin{aligned}(1+x)(1+x^2)(1+x^4)\cdots &= \frac{1-x^2}{1-x}(1+x^2)(1+x^4)\cdots \\ &= \frac{1-x^4}{1-x}(1+x^4)(1+x^8)\cdots \\ &= \frac{1-x^8}{1-x}(1+x^8)\cdots.\end{aligned}$$

So if we try to find

$$\prod_{n=1}^{k}(1+x^{2^{n-1}}) = \frac{1-x^{2^k}}{1-x},$$

then as $\lim_{k\to\infty} x^{2^k} = 0$, then the answer is $\frac{1}{1-x}$.

(b) Observe that

$$\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2} = 2^2\sin\frac{x}{2^2}\cos\frac{x}{2^2}\cos\frac{x}{2} = 2^3\sin\frac{x}{2^3}\cos\frac{x}{2^3}\cos\frac{x}{2^2}\cos\frac{x}{2} = \cdots.$$

In general,

$$\sin x = 2^n \sin\frac{x}{2^n}\prod_{k=1}^{n}\cos\frac{x}{2^k}.$$

Therefore

$$\prod_{k=1}^{n}\cos\frac{x}{2^k} = \frac{\frac{\sin x}{2^n}}{\sin\frac{x}{2^n}} = \frac{\sin x}{x}\cdot\frac{x/2^n}{\sin\frac{x}{2^n}}.$$

Thus because $\lim_{x\to 0}\frac{x}{\sin x} = 1$, then

$$\lim_{n\to\infty}\prod_{k=1}^{n}\cos\frac{x}{2^k} = \frac{\sin x}{x}\cdot\lim_{n\to\infty}\frac{x/2^n}{\sin\frac{x}{2^n}} = \frac{\sin x}{x}.$$

For the second part, simply observe that

$$\frac{\sin\frac{pi}{2}}{\frac{pi}{2}} = \frac{1}{\pi/2} = \cos\frac{pi}{2^2}\cos\frac{\pi}{2^3}\cos\frac{pi}{2^4}\cdots.$$

By the cos double angle formula,

$$\cos\frac{x}{2^{n+1}} = \sqrt{\frac{\cos\frac{x}{2^n}+1}{2}},$$

so since $\cos \frac{pi}{2^2} = \sqrt{\frac{1}{2}}$, this gives the $n+1$th term in the denominator of the given expression. This results in

$$\frac{1}{\pi/2} = \sqrt{1/2} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}},$$

which gives the desired expression.

(c)

$$f(x) = f(2 \cdot \frac{x}{2}) = \cos^2 \frac{x}{2} \cdot f(\frac{x}{2}) = \cos^2 \frac{x}{2} \cos^2 \frac{x}{4} \cdot f(\frac{x}{4}).$$

Continuing,

$$f(x) = \cos^2 \frac{x}{2} \cos^2 \frac{x}{4} \cdots \cos^2 \frac{x}{2^n} \cdot f(\frac{x}{2^n}),$$

so as $n \to \infty$,

$$f(x) = \left(\frac{\sin x}{x}\right)^2 f(0) = \frac{\sin^2 x}{x^2}.$$

10

(a) Note that $b_{n+1} = \frac{b_n}{1+\beta_n}$, so the limit, if it exists, is equal to

$$b = \frac{b_1}{\prod_{k=1}^{\infty}(1+\beta_k)}.$$

By problem 8, the infinite product $\prod_{k=1}^{\infty}(1+\beta_k)$ converges if and only if the infinite sum $\sum_{k=1}^{\infty} \beta_k$ converges, so we're done.

(b) Since $\frac{a_n}{\alpha_{n+1}} = 1 + \frac{p}{n} + \alpha_n$, then $a_n = \alpha_{n+1}(1 + \frac{p}{n} + \alpha_n)$.

11 Note that

$$\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x,$$

so setting $x = a_{n+1} - a_n$ gives

$$\lim_{n\to\infty} \left(1 + \frac{a_{n+1} - a_n}{n}\right)^n = e^{a_{n+1}-a_n}$$

Since $\lim_{n\to\infty} \frac{a_{n+1}-a_n}{n} = 0$, then we get $\lim_{n\to\infty} \left(\frac{1+a_{n+1}}{a_n}\right)^n \geq e$, as desired.

We now show that this estimate cannot be improved by giving a sequence satisfying $\lim_{n\to\infty} \left(\frac{1+a_{n+1}}{a_n}\right)^n = e$. Take $a_n = a_{n+1} = n$ for odd n, so $\frac{1+a_{n+1}}{a_n} = \frac{n+1}{n} = 1 + \frac{1}{n}$. Since

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e,$$

we're done.

Chapter 4

Zorich Mathematical Analysis

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4.2 Properties of Continuous Functions

1

(a) If $f \in C(A)$, then f is a continuous function defined on A. Since $B \subset A$, then f is also continuous on B, so $f|_B \in C(B)$, as desired.

(b) Consider, for example, that $E_1 = [-1, 0]$ and $E_2 = [0, 1]$. Let $f|E_1 = x$ and $f|E_2 = x + 1$. Then $f|E_1(0) = 0$ and $f|E_2(0) = 1$, which is not continuous, so F is not continuous on $E_1 \cup E_2$.

(c) Observe that the Riemann function $\mathcal{R}$ is discontinuous because there is always an irrational number between any two rational numbers. The irrational number gives 0 but the rationals are nonzero, so this is discontinuous. The Riemann function defined on only the rational $\mathbb{Q}$'s is also irrational. Take, for example, all of the rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$. $\mathcal{R}(2/5) = \mathcal{R}(3/5) = \frac{1}{5}$ but the rational numbers in between do not approach $\frac{1}{5}$. All of these points of discontinuity are removable by removing the rational numbers, leaving us with a function that outputs 0 for each input.

2 We show that $m(x) = \min_{a \le t \le x} f(t)$ is continuous, the $M(x)$ case is symmetric. Any function $f(t)$ consists of increasing and decreasing intervals. $m(x)$ cannot be discontinuous because this would require the minimum of $f(t)$ on that current interval to be discontinuous. Since the given intervals are continuous, then since we know that f is continuous on the interval $[a, b]$, then $m(x)$ is continuous.

3

(a) If this function is continuous, then $f([a, b]) = [c, d]$. Since the function is monotonic, then the function inverse is defined, and since all of the inputs are distinct, the function inverse is injective and surjective.

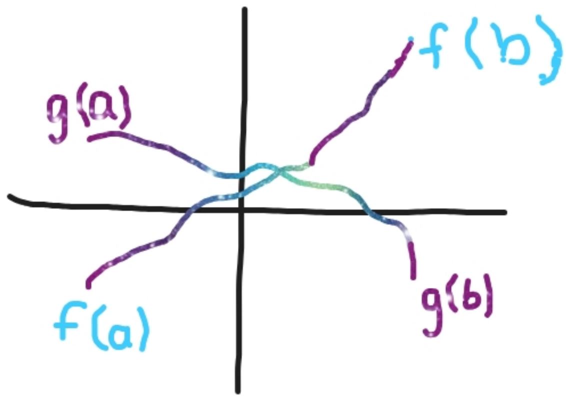
Suppose we have an input x_0 and output y_0. Since f is continuous, as $x \to x_0$, $f(x) = y \to f(x_0) = y_0$. Therefore, as $y \to y_0$, $f^{-1}(y) = x \to f^{-1}(y_0) = x_0$, so f^{-1} is continuous, as desired.

(b) Consider the function that maps $[0, 1)$ to $[0, 1)$ and $(1, 2]$ to $(1, 2]$. Then the function is monotonic but it has one discontinuity, which is 1.

(c) Observe that X and Y are not necessarily open intervals, so there may be an element $x \in X$ which is equal to the greatest lower bound or least upper bound. Then this boundary point might not be necessarily continuous.

4

(a) Consider the points graphically. The point $(a, g(a))$ is right above $(a, f(a))$, and the point $(b, g(b))$ is right below $(b, f(b))$. Therefore, there is no way for $f(a)$ to take a detour around $(a, g(a))$ or $(b, g(b))$ because otherwise f wouldn't be a function, and likewise for g. There is also no way for f to suddenly 'jump' over g because then f would be noncontinuous, and vice versa.



(b) Consider the graph of the function $g(x) = x$ in the coordinate plane. $g(0) = 0$ and $g(1) = 1$. We assume that $f(0) > 0$ and $f(1) < 1$ because otherwise we would already have a value of x that satisfies $f(x) = x$. Then $f(0) > g(0)$ and $f(1) < g(1)$, so by the previous part, f and g must intersect at a point x. Since this point lies on g, it satisfies $f(x) = x$, so we're done.

(c) Consider $f(x) = 2^x$ and $g(x) = x$ on the interval $[0, \infty)$. Then $f(x)$ has no fixed points, so they do not always have a common fixed point.

(d) Let $f(x) = x - 1$. Obviously there is no fixed point, period.

(e) Consider $f(x) = 2^x - 1$. Note that $f(0) = 0$ and $f(1) = 1$, but since this is an exponential function, $x > 2^x - 1$ for $0 < x < 1$, so this does not have a fixed point.

(f) Let $g(x) = f(f(x)) - f(x)$. Since $g(0) = g(1) = 0$ and f is continuous, then g must also be continuous. Thus it is impossible for g to be nonzero (it can't increase then decrease), so $g(x) = 0$ for all x. Therefore $f(x) = f(f(x)) = x$, as desired.

5 Suppose the function f is continuous on the interval $[a, b]$. Then for $x \in [a, b]$, $f(x)$ must be bounded because it's continuous; we cannot have $f(x) = \infty$ or $-\infty$ because otherwise $f(x)$ would not be continuous or closed. If this asymptote/limit occurs in the middle of a and b, f is not continuous. If this occurs at a or b, the domain of f would not be closed.

Furthermore, observe that the bounds of f are achieved because if they aren't, the range of f isn't closed. Note that if the minimum/maximum occurs inside the range $[a, b]$, it obviously equals the greatest lower bound/least upper bound. If this minimum/maximum occurs at the endpoints, suppose WLOG $\inf = f(a)$. Then if inf is not achieved, since f is not continuous, neither is a, so we're done.

6

(a) Let $g(x) = f^n(x) - f(x)$. Observe that $g(0) = g(1) = 0$ and f and $f^n(x)$ are continuous, so $g(x)$ is also continuous. Therefore g must equal zero because otherwise, if x increases, then $f(x) \geq x$, so $f^2(x) = f(f(x)) \geq f(x)$, etc., so

$f^n(x) > f(x)$. Therefore $g(x)$ increases, so it is impossible for $g(1) = g(0)$. Therefore $g(x) = 0$ for all x and hence $f(x) = x$ for all x.

(b) From the previous part, we already have $f^n(x) \geq f^{n-1}(x) \geq \cdots \geq f(x)$. If $f(x) = x$, then all of these are equal, x is a fixed point, and we're done. Otherwise, if $f^k(x) = f^{k-1}(x)$, we must have $f(x) = x$, so assume $f^n(x) > f^{n-1}(x) > \cdots > f(x)$. Then $f^n(x)$ increases as n increases, but since f maps from $[0, 1]$ to $[0, 1]$, $f^n(x) < 1$, so $\lim_{n\to\infty} f^n(x) - f^{n-1}(x) = 0$. Therefore since $f^{n-1}(x)$ approaches $f^n(x)$, $f^n(x)$ tends to a fixed point.

7

(a) Let $g(x) = f(x + \frac{1}{n}) - f(x)$ on the interval $[0, \frac{1}{n}]$. Then $g(0) = f\left(\frac{1}{n}\right) - f(0)$ and $g\left(\frac{1}{n}\right) = f(1) - f(1 - \frac{1}{n})$. Since $f(0) = f(1)$, then $g(x) = 0$, so g is continuous on the interval $[0, \frac{1}{n}]$ because f is continuous. Then by the intermediate value theorem, for any value between $g(0)$ and $g(1/n)$, there exists a point in the interval $[0, \frac{1}{n}]$ where $g(x)$ attains the value

$$\exists x \in [0, \frac{1}{n}] \text{ such that } g(x) = f(x + \frac{1}{n}) - f(x) = 0.$$

Therefore $f(x + \frac{1}{n}) = f(x)$, so there's a closed interval of length $\frac{1}{n}$ with both endpoints on the graph of the function, as desired.

(b) Consider the function $h(x) = \sin(2\pi x)$. Note that $h(0) = h(1) = 0$. Assume for the sake of contradiction that there does exists points x_1 and x_2 such that $h(x_1) = h(x_2)$ and $x_2 = x_1 + l$, where l is not of the form $\frac{1}{n}$. Because $h(x) = \sin(2\pi x)$, then $\sin(2\pi x_1) = \sin(2\pi x_2)$, so $2\pi l = 2\pi k$ or $2\pi l = 2\pi k + \pi$. In the first and second case, $l = k$ or $l = k + \frac{1}{2}$, so we're done.

8

(a) Note that $\omega(\delta)$ must be nonnegative because it's a supremum of nonnegative values. If we let $\delta_1 < \delta_2$, it suffices to show that $\omega(\delta_1) \leq \omega(\delta_2)$. The set of pairs of points (x_1, x_2) with $|x_1 - x_2| < \delta_1$ is the subset of the set of pairs of points (x_1, x_2) with $|x_1 - x_2| < \delta$. Thus $\sup f$ must be at least equal to the smaller set, so $\omega(\delta_1) \leq \omega(\delta_2)$, so $\omega(\delta)$ is nondecreasing.

It suffices to show that $\lim_{\delta\to 0} \omega(\delta)$ exists. $\omega(\delta)$ is nondecreasing and nonnegative, so the function is either constant or strictly increasing. The limit as δ approaches 0 exists, and is denoted by $\omega(0+)$.

(b) Let $\epsilon > 0$. Since $\omega(+0) = \lim_{\delta\to 0} \omega(\delta)$ exists, we can choose a $\delta > 0$ such that $\omega(\delta) \leq \omega(+0) + \epsilon$. Given any $x_1, x_2 \in E$ with $|x_1 - x_2| < \delta$, we have $|f(x_1) - f(x_2)| \leq \omega(\delta) \leq \omega(+0) + \epsilon$, which is the required δ for the given ϵ.

(c) Consider $\delta_1, \delta_2 > 0$. Let $x_1, x_2, x_3 \in E$ such that $|x_1 - x_3| < \delta_1 + \delta_2$, $|x_1 - x_2| < \delta_1$, and $|x_2 - x_3| < \delta_2$. Using the triangle inequality gives $|f(x_1) - f(x_3)| \leq |f(x_1) - f(x_2)| + |f(x_2) - f(x_3)|$. Since $|f(x_1) - f(x_2)| \leq \omega(\delta_1)$ and $|f(x_2) - f(x_3)| \leq \omega(\delta_2)$. Substituting this into the previous inequality gives $|f(x_1) - f(x_3)| \leq \omega(\delta_1) + \omega(\delta_2)$. Since this holds for all x_1, x_3, the inequality $\omega(\delta_1 + \delta_2) \leq \omega(\delta_1) + \omega(\delta_2)$ holds.

(d) For $f(x) = x$, we have $|f(x_1) - f(x_2)| = |x_1 - x_2|$, so the modulus of continuity for this function is $\omega(\delta) = \delta$.

For $f(x) = \sin(x^2)$, since this is a sin function, then the maximum value is 1 and the minimum is -1, so the modulus of continuity is $\omega(\delta) = 1 - (-1) = 2$.

(e) Assume that f is uniformly continuous on E. Then for every $\epsilon > 0$, there exists a $\delta > 0$ such that $|x_1 - x_2| < \delta$ implies that $|f(x_1) - f(x_2)| < \epsilon$. Since $\omega(\delta)$ is a supremum, then there exists $\delta > 0$ such that $\omega(\delta) < \epsilon$. Then $\lim_{\delta\to 0}\omega(\delta) = \omega(+0) = 0$.

Assume that $\omega(+0) = 0$. Then $\lim_{\delta\to 0}\omega(\delta) = 0$, so for any $\epsilon > 0$, there exists $\delta > 0$ such that $\omega(\delta) = \epsilon$. If $|x_1 - x_2| < \delta$, we have $|f(x_1) - f(x-2)| \leq \omega(\delta) < \epsilon$. Thus f is uniformly continuous on E, so a function f is uniformly continuous on E if and only if $\omega(+0) = 0$.

9 Note that since $f(x)$ is continuous over an interval $[a, b]$, by the extreme value theorem, the maximum and minimum value of $f(x)$ must exist on this interval. Similarly for $g(x)$ (and note the extreme value function only applies for continuous functions so this wouldn't work in general for arbitrary bounded functions). Therefore, the maximum value of $|f(x_0) - g(x_0)|$ occurs in the set X, because if it didn't exist, this would imply that $|f(x) - g(x)|$ has no maximum value, which would imply that $f(x)$ or $g(x)$ has no maximum/minimum value. One of $f(x)$ or $g(x)$ can't be increasing/decreasing in the middle of the interval if it's not approaching its bound, so we're done.

10

(a) Note that obviously, $f(x)$ is bounded, so $E_n(f)$ should equal half of the minimum difference between $\inf f(x)$ and $\sup f(x)$. This way, the distance to either the minimum or the maximum is minimized, so if we let $a_0 = \frac{\inf f(x)+\sup f(x)}{2}$, the constant function $P_0(x) = a_0$ is the polynomial of best approximation.

(b) Given a fixed polynomial $P_n(x)$, we want to find a polynomial of the form $Q_\lambda(x) = \lambda P_n(x)$ such that $\Delta(Q_{\lambda_0}) = \min_{\lambda\in\mathbb{R}}\Delta(Q_\lambda)$. $\Delta(Q_\lambda)$ is continuous with respect to λ, so there exists $\lambda_0 \in \mathbb{R}$ that minimizes $\Delta(Q_\lambda)$ in the interval $[a, b]$, implying the existence of a polynomial of best approximation of the form $\lambda P_n(x)$.

(c) Let $P_n(x)$ be the polynomial of best approximation of degree n, and let $\Delta_n = \Delta(P_n) = E_n(f)$. We show that there exists a polynomial of best approximation of degree $n + 1$ by adding a term of degree $n + 1$ to $P_n(x)$, giving

$$P_{n+1}(x) = P_n(x) + c(x - a)^{n+1}$$

for some constant c, so we're done.

(d) We know that there exists a polynomial of best approximation of degree 0, and we know that if there exists a polynomial of best approximation of degree n, there is also one of degree $n + 1$, so we're done by induction.

11 Prove the following statements.

(a) Observe that if a polynomial has complex roots (not real roots), they must come in conjugate pairs, so there are an even number of imaginary roots. The number of roots a polynomial (repeats count) has is the same as the degree of the polynomial, so there are an odd number of roots. Therefore there must be at least one real root.

(b) The sign function is:

$$\operatorname{sgn} x = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x = -1 \end{cases}$$

Therefore, if on some interval, $P_n(x)$ is the same sign, then $\text{sgn}P_n(x)$ will be continuous on that interval. Therefore, sgn is only not continuous when $P_n(x)$ changes sign, so basically when $P_n(x) = 0$, so when x is a root of P_n. We know there are at most n roots (since there are double roots too), so there are at most n points of discontinuity.

(c) I'm not sure how the fact that $\text{sgn}[(f(x_i) - P_n(x_i))(-1)^i]$ comes into play, but if we let $g(x) = f(x) - P_n(x)$, we get: either $g(x_0)$ is negative, $g(x_1)$ is positive, $g(x_2)$ is negative, etc. or, $g(x_0)$ is positive, $g(x_1)$ is negative, $g(x_2)$ is positive, etc. or, $g(x_0) = g(x_1) = \cdots = g(x_{n+1}) = 0$. Therefore $g(x)$ must have at least $n+1$ roots, since the sign changes $n+1$ times.

So we try to show that

$$E_n(f) = \inf_{P_n} \sup_{x\in[a,b]} |g(x_i)| \geq \min_{0\leq i\leq n+1} |g(x_i)|.$$

Since $|g(x_i)| \leq \sup_{x\in[a,b]} |g(x_i)|$, we're done.

12

(a) By Demoivre's theorem,

$$\text{cis}nx = \cos nx + i\sin nx = (\cos x + i\sin x)^n.$$

So $\cos nx$ is equal to the real part, which is

$$\cos nx = \cos^n x - \binom{n}{2}\cos^{n-2} x\sin^2 x + \binom{n}{4}\cos^{n-4} x\sin^4 x - \cdots + \ldots.$$

Obviously if the degree of $\sin x$ is odd, then it wouldn't be real, so $\sin x$ is always raised to an even power. Since $\sin^2 x = 1 - \cos^2 x$, we can express $\sin^{2k} x$ in terms of $\cos x$, so $\cos nx$ can be expressed as an algebraic polynomial in terms of $\cos x$, so we're done.

(b) We have $\cos(x) = \cos x$, so $T_1(x) = x$ obviously. We have $\cos(2x) = \cos x\cos x - \sin x\sin x = \cos^2 x - (1-\cos^2 x) = 2\cos^2 x - 1$, so $T_2(x) = 2x^2 - 1$. We know that $\sin(2x) = 2\sin x\cos x$, so

$$\cos(3x) = \cos(2x)\cos x - \sin(2x)\sin x = (2\cos^2 x - 1)\cos x - 2\sin^2 x\cos x = 4\cos^3 x - 3\cos x,$$

so $T_3(x) = 4x^3 - 3x$. Since

$$\sin(3x) = \sin(2x)\cos x + \cos(2x)\sin x = 2\sin x\cos^2 x + (1 - 2\sin^2 x)\sin x = 3\sin x - 4\sin^3 x.$$

Therefore

$$\begin{aligned}\cos(4x) &= \cos(3x)\cos x - \sin(3x)\sin x \\ &= (4\cos^3 x - 3\cos x)\cos x - (3\sin x - 4\sin^3 x)\sin x \\ &= 4\cos^4 x - 3\cos^2 x - 3\sin^2 x + 4\sin^4 x \\ &= 4\cos^4 x - 3\cos^2 x - 3(1-\cos^2 x) + 4(1-\cos^2 x)^2 \\ &= 8\cos^4 x - 8\cos^2 x + 1.\end{aligned}$$

Thus $T_4(x) = 8x^4 - 8x^2 + 1$.

(c) The trick is to not think of $T_n(x)$ as a polynomial, but rather as a cos function. $T_n(x)$ equals 0 when $n \arccos x$ equals some odd multiple of $\pi/2$. Let $n \arccos x = \frac{(2k+1)\pi}{2}$, so $\arccos x = \frac{(2k+1)\pi}{2n}$, so $x = \cos\left(\frac{(2k+1)\pi}{2n}\right)$. For all integer k until this thing repeats.

$|\cos|$ is maximized if its argument is a multiple of π, so $n \arccos x = k\pi$, so $\arccos x = \frac{k\pi}{n}$, so $x = \cos\left(\frac{k\pi}{n}\right)$ for all k until this repeats.

(d) Let $P_n(x)$ be any polynomial of degree n such that $P_n(x) = x^n + Q_{n-1}(x)$ for some polynomial $Q_{n-1}(x)$ of degree $n-1$. We need to prove that $T_n(x)$ is the polynomial of best approximation for $f = 0$. $P_n(x)T_n(-x) + P_n(-x)T_n(x) = 0$ for all $-1 \leq x \leq 1$. If $P_n(x) \neq T_n(x)$, then there is at least one point satisfying $-1 < a < 1$ such that $P_n(a) > T_n(a)$. However, $E_n(T_n) \geq \frac{P_n(a)}{T_n(a)} - 1 > 0$. Hence

$$\max_{-1\leq x\leq 1} |G(x)| \geq \max_{|x|=1} |G(x)| \geq E_n(T_n) > 0.$$

Thus there exists $\delta > 0$ such that $|G(x)| > \delta$ for all $-1 < x < 1$, so since $G(x)$ has degree $n-1$, it has at most $n-1$ roots, so there must be an interval where $|G(x)| > \delta > 0$. But $P_n(x)T_n(-x) + P_n(-x)T_n(x) = 0$, so $P_n(x) = T_n(x)$, and thus $T_n(x)$ is the polynomial of best approximation.

13

(a) $f(x_i) - P_n(x_i) = (-1)^i \Delta(P_n) \cdot \alpha$ for $i = 0, \ldots, n+1$, $\Delta(P_n) = \max_{x\in[a,b]} |f(x) - P_n(x)|$. In order to show that $P_n(x)$ is the unique polynomial of best approximation, if we let $Q(x)$ be a polynomial of degree n, then it suffices to show that $\max_{x\in[a,b]} |f(x) - Q(x)| \geq \Delta(P_n)$. Let $R(x) = f(x) - Q(x)$, so it suffices to show that $\max_{x\in[a,b]} |R(x)| \geq \Delta(P_n)$. Assume there's some alternate sequence for x_i, so we have

$$R(x_i) = f(x_i) - Q(x_i) = (-1)^i \Gamma, \qquad i = 0, \ldots, n+1$$

where $\Gamma = \max_{x\in[a,b]} |f(x) - Q(x)|$. Then

$$Q(x)i - P_n(x_i) = (-1)^i (\Delta(P_n) \cdot \alpha - \Gamma)$$

for $i = 0, \ldots, n+1$. Then $Q(x_i) - P_n(x_i)$ has the same sign as $f(x_i) - P_n(x_i)$, so $\max_{x\in[a,b]} |Q(x) - P_n(x)| \geq \Delta(P_n)$, so we're done.

(b) By part (a), we already have the if part. It suffices to prove the only if part. By Problem 10, we already know that a polynomial of best approximation exists. We also know that $n+1$ alternating points are on the polynomial, and we can add another one by using the next hump, so we're done.

(c) Obviously, for discontinuous functions, there are sharp edges and disconnected edges such as the graph of sgn$x$. If we take the polynomial of degree 0 $P_n(x) = 0$, then the $n+2$ alternate points do not hold.

(d) Obviously, for a best approximation of degree 0 is $P_0(x) = 1$. A best approximation of degree 1 is of the form $P_1(x) = ax + b$. The most important points are the endpoints and corners, which is $(-1, 1)$, $(0, 0)$, $(2, 2)$. The vertical distance between P_1 and these points is $1 - (-a + b) = 1 + a - b$, b, and $2 - (2a + b) = 2 - 2a - b$, respectively. Note that the $\Delta(P_1)$ will be minimized when $1 + a - b = b = 2 - 2a - b$, so $a = \frac{1}{3}$ and $b = \frac{2}{3}$, so $P_1(x) = \frac{1}{3}x + \frac{2}{3}$. Then in this case, the distance will be $\frac{2}{3}$, and we're done.

14

(a) For any two functions $f(x)$ and $g(x)$ that is part of the germ of functions at a, we define $(f+g)(x) := f(x)+g(x)$, $(f \cdot g)(x) := f(x) \cdot g(x)$ for all x in the neighborhood $U(a)$.

(b) Note that the sum and product of any two continuous functions on a given range will be continuous on that range as well. Thus the arithmetic operation on germs of continuous functions is continuous, so it does not lead outside the class of germs.

(c) The requirements are closure, associativity, commutativity, existence of identity, and existence of inverse for addition. Observe that we just proved closure by the previous part and that associativity and commutativity are obvious because the underlying functions are continuous, associative, and commutative. The identity exists for both operations, being 0 for addition and 1 for multiplication. The inverse of addition is the negative of the function whose inverse we are looking for, however this does not necessarily exist for multiplication (consider, for example, $f(x) = 0$, which does not have an inverse).

(d) Consider a subring I consisting of all functions f such that $f(a) = 0$. Then if g is an element of the ring K, then $(f \cdot g)(a) = f(a) \cdot g(a) = 0$, which is in I, so I is an ideal, as desired

15 One maximal ideal is the set of continuous functions that equal 0 at some point in the interval. Obviously if $f(x) = 0$ for some $x \in [a, b]$, then $(f \cdot g)(x) = f(x) \cdot g(x) = 0$ is obviously an element of this ring. There is no other maximal ideal because 0 is the only thing that nullifies whatever is thrown at it (being the elements of $C[a, b]$ that are not actually part of the maximal ideal).

Chapter 5: Differential Calculus

Zorich Mathematical Analysis

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5.1 Differentiable Functions

1

(a) Differentiating both sides gives $\frac{2xx'}{a^2} + \frac{2yy'}{b^2} = 0$, so dividing by 2 and substituting $x = x_0$ and $y = y_0$ gives

$$\frac{2x_0x'}{a^2} + \frac{2yy'}{b^2} = 0 \implies \frac{y'}{x'} = \frac{dy}{dx} = -\frac{b^2x_0}{a^2y_0}.$$

Thus the slope of the tangent line is $-\frac{b^2x_0}{a^2y_0}$, so the equation of the tangent line is

$$\begin{aligned} y - y_0 &= -\frac{b^2x_0}{a^2y_0}(x - x_0) \\ \implies a^2y_0y - a^2y_0^2 &= -b^2x_0x + b^2x_0^2 \\ \implies a^2y_0^2 + b^2x_0^2 &= a^2y_0y + b^2x_0x. \end{aligned}$$

Dividing both sides by a^2b^2 gives

$$\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$$

because (x_0, y_0) lies on the ellipse and thus satisfies its equation.

(b) Let P be the point (x_0, y_0). We show that if we reflect F_1P on the tangent at P, we get F_2P. Note that it suffices to show that the angle that F_1P makes with the tangent equals the angle that F_2P makes with the tangent. Let $\overrightarrow{F_1P} = \overrightarrow{v_1}$ and $\overrightarrow{F_2P} = \overrightarrow{v_2}$. It then suffices to consider their dot products. Consider a vector $\overrightarrow{u} = \langle 0, -\frac{b^2x_0}{a^2y_0}\rangle$. Then it suffices to prove that $\frac{u\cdot v_1}{||u||||v_1||} = \frac{u\cdot v_2}{||u||||v_2||}$, or $||\frac{v_1}{||v_1||}|| = ||\frac{v_2}{||v_2||}||$, which is 1, as desired.

2 For α near 0, $\sin\alpha \approx \alpha$, so $\cos\alpha \approx \sqrt{1-\alpha^2}$.

(a)

$$\sin(\frac{\pi}{6} + \alpha) = \sin\frac{\pi}{6}\cos\alpha + \sin\alpha\cos\frac{\pi}{6} = \frac{1}{2}\sqrt{1-\alpha^2} + \frac{\sqrt{3}}{2}\alpha.$$

(b) $\alpha^\circ = \frac{\pi\alpha}{180}$ radians and 30° is $\pi/6$ radians, so we can convert this to radians. So we have

$$\sin(\frac{\pi}{6} + \frac{\pi\alpha}{180}) = \frac{1}{2}\sqrt{1 - \left(\frac{\pi\alpha}{180}\right)} + \frac{\sqrt{3}}{2}\frac{\pi\alpha}{180}.$$

(c)
$$\cos(\frac{\pi}{4}+\alpha) = \cos\frac{\pi}{4}\cos\alpha - \sin\frac{\pi}{4}\sin\alpha = \sqrt{2}2\sqrt{1-\alpha^2} - \frac{\sqrt{2}}{2}\alpha.$$

(d) As above, $\alpha = \frac{\pi\alpha}{180}$, so
$$\cos(\frac{\pi}{4}+\frac{\pi\alpha}{180}) = \sqrt{2}2\sqrt{1-\left(\frac{\pi\alpha}{180}\right)^2} - \frac{\sqrt{2}}{2}\cdot\frac{\pi\alpha}{180}.$$

3

1. Observe that since the water is rotating at constant angular velocity ω, the acceleration vector is $\langle -\omega^2 \sin\omega t, -\omega^2\cos\omega t\rangle$, so the acceleration is $\frac{\omega^2}{g}$. Thus $f'(x) = \frac{\omega^2}{g}x$.
2. Consider a function $f(x) = \frac{\omega^2}{2g}x^2$. The derivative of $f(x)$ is $\frac{\omega^2}{2g}\cdot 2x = \frac{\omega^2}{g}x$.
3. No, it does not, because if the axis is shifted by a constant, then the derivative of the constant is 0, which does not change.

4

(a) Consider the graph $f(x)$. Then $f'(x)$ is the graph of its derivative, which passes through $(x_0, f'(x_0))$. Then the slope of the tangent at this point is equal to $f''(x_0)$, so the equation of the tangent line is $y - f'(x_0) = f''(x_0)(x - x_0)$.

(b) $f''(x) = 2$, so the horizontal component of the acceleration is the largest when $x_0 = 0$.

5 Observe that this function is obviously continuous because each of the terms, $\varphi_n(x)$, are all continuous. Also, note that because every number, even irrational, can be expressed by an infinite base 2 representation, then x can be written in terms of
$$x = \sum_{i=1}^{\infty}\frac{a_i}{2^i},$$
where a_i is either 0 or 1. Thus there is always a term that does not have a derivative in the sum, so $f(x)$ does not have a derivative at every point.

5.2 The Basic Rules of Differentiation

1 Let $P_n(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$.
$$\begin{aligned}
P_n(x_0) &= a_0 + a_1x_0 + a_2x_0^2 + \cdots + a_nx_0^n = \alpha_0 \\
P_n^{(1)}(x_0) &= a_1 + 2a_2x_0 + 3a_3x_0^2 + \cdots + na_nx_0^{n-1} = \alpha_1 \\
P_n^{(2)}(x_0) &= 2a_2 + 3\cdot 2a_3x_0 + 4\cdot 3a_4x_0^2 + \cdots + n(n-1)a_nx_0^{n-2} = \alpha_2 \\
&\vdots \\
P_n^{(n-1)}(x_0) &= (n-1)\cdots 1\cdot a_{n-1} + n(n-1)\cdots 2a_nx_0 = \alpha_{n-1} \\
P_n^{(n)}(x_0) &= n(n-1)\cdots 2\cdot 1a_n = \alpha_n.
\end{aligned}$$
So working our way from the bottom to the top, we get $a_n = \frac{\alpha_n}{n!}$, then we can use this to solve for a_{n-1} to get $a_{n-1} = \frac{\alpha_{n-1}-\alpha_nx_0}{(n-1)!}$, etc. Solving for all of the a_i gives our desired polynomial.

2

(a) Assuming this means $f(x) = e^{-\frac{1}{x^2}}$ if $x \neq 0$. Then by chain rule,

$$f'(x) = \left(-\frac{1}{x^2}\right)' e^{-1/x^2} = 2x^{-3}e^{-1/x^2}$$

(b) We need to find the derivative of $x^2 \sin \frac{1}{x}$. By the multiplication rule, this is $2x \cdot \sin \frac{1}{x} + x^2 \cdot (\sin \frac{1}{x})'$. By chain rule, $(\sin \frac{1}{x})' = -x^{-2} \cos \frac{1}{x}$, so we get $2x \sin \frac{1}{x} - \cos \frac{1}{x}$.

(c) We already have $f'(x)$ is differentiable. Then we can take the derivative of $f^{(2)}(x)$, etc. Since $f(x)$ consists of a smooth curve, then $f'(x)$, $f''(x)$, etc., will also be smooth, and hence it will be differentiable. Note that because we have an x^2, $f(x)$ is an even function, so $f'(x) = 0$, so $f''(x) = 0$, etc., so $f^{(n)}(x) = 0$.

(d) Note that we have $2x \sin \frac{1}{x} - \cos \frac{1}{x}$ if $x \neq 0$. This is defined on $\mathbb{R}\backslash 0$, and note that $f'(0) = 0$, so it's defined on all $\mathbb{R}$. As x approaches 0, $\frac{1}{x}$ approaches ∞, but since sin and cos is periodic, then this function will keep wavering, so it's not continuous.

(e) Observe that

$$e^{-\frac{1}{(1+x)^2} - \frac{1}{(1-x)^2}} = e^{-\frac{1}{(1+x)^2}} \cdot e^{-\frac{1}{(1-x)^2}}.$$

So since $f(x)$ is a product of functions that are infinitely differentiable on $\mathbb{R}$ they are $f(1+x)$ and $f(1-x)$ from part (a), then this function must be infinitely differentiable as well.

3 Let $f \in C^{(\infty)}$. The first derivative of $x^{n-1} f\left(\frac{1}{x}\right)$ is

$$\begin{aligned} x^{n-1}\left(f\left(\frac{1}{x}\right)\right)' + (x^{n-1})' f\left(\frac{1}{x}\right) &= x^{n-1} \cdot \frac{-1}{x^2} f'\left(\frac{1}{x}\right) + (n-1)x^{n-2} f\left(\frac{1}{x}\right) \\ &= (n-1)x^{n-2} f\left(\frac{1}{x}\right) - x^{n-3} f'\left(\frac{1}{x}\right). \end{aligned}$$

Then the nth derivative of $x^{n-1} f\left(\frac{1}{x}\right)$ is $\frac{(-1)^n}{x^{n+1}} f^{(n)}\left(\frac{1}{x}\right)$, so we're done.

4

(a) If f is an even function, $f(x) = f(-x)$. Then taking the derivative of both sides and applying the chain rule gives $f'(x) = (-x)' f'(-x) = -f'(-x)$, so f' is an odd function.

(b) If f is an odd function, $f(x) = -f(-x)$. Then taking the derivative of both sides and applying the chain rule gives $f'(x) = -(-x)' f'(-x) = f'(-x)$, so f' is an even function.

(c) We already know that if f is even, then f' is odd. We now prove the converse. If f' is odd, then $f'(x) = -f'(-x)$. Integrating gives $f(x) + c_1 = f(-x) + c_2$. Letting $x = 0$ gives $f(0) + c_1 = f(0) + c_2$, so $c_1 = c_2$. Thus $f(x) = f(-x)$, and f is even, as desired.

5

(a) If $f(x)$ is differentiable, then there exists a function $f'(x)$ such that $f(x) - f(x_0) = f'(x)(x - x_0)$, so in this case we can let $f'(x) = \varphi(x)$. If there exists a function $\varphi(x)$ such that $f(x) - f(x_0) = \varphi(x)(x - x_0)$, then $\varphi(x) = \frac{f(x) - f(x_0)}{x - x_0}$, which is the slope of the line between the points $(x, f(x))$ and $(x_0, f(x_0))$. Since the definition of derivative is the slope of the line tangent to a curve at the point, then as $x \to x_0$, $\varphi(x)$ approaches this slope. Hence $\varphi(x)$ is the derivative of f, and f is differentiable at x_0.

(b) Since we know that $\varphi(x)$ is the first-order derivative of $f(x)$, then if $\varphi(x)$ is $n-1$-differentiable, so is $f'(x)$. This means that $f(x)$ is n-differentiable, so it has a derivative of order n at x_0.

6 We need to construct a function f that is continuous at x_0 has an inverse f^{-1}, and f^{-1} is not continuous at y_0 Consider $f(x) = x$ if $x \le 0$ and $f(x) = x^2$ if $x \ge 0$ at $x = 0$. Then f is continuous at x_0 but not continuous at y_0, and the theorem doesn't apply.

7

(a) $m_1a_1 = F = \frac{Gm_1m_2}{r^2}$, $a_1 = \frac{Gm_2}{r^2}$ and $m_2a_2 = F$, so $a_2 = \frac{Gm_1}{r^2}$. Then v_1^2 increases by $2\alpha_1\Delta r$ and v_2^2 increases by $2\alpha_2\Delta r$, so we get $\frac{(2Gm_1+2Gm_2)\Delta r}{r^2} - \frac{Gm_1m_2}{r} = 0$, so E is constant.

(b) E is the total energy of the system. K is the kinetic energy, or energy in motion, and U is the potential energy, which is kind of like energy waiting to be converted to kinetic energy.

(c) The total energy of n bodies is constant if it is under no external force.

5.3 The Basic Theorems of Differential Calculus

1 Observe that

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots .$$

We also have

$$\begin{aligned}\frac{1+ax^2}{1+bx^2} &= (1+ax^2)(1-bx^2+b^2x^4-b^3x^6+\cdots)\\ &= 1+ax^2-bx^2-abx^4+b^2x^4+ab^2x^6-b^3x^6+\cdots\\ &= 1+(a-b)x^2-(b^2-ab)x^4+(ab^2-b^3)x^6+\cdots\end{aligned}$$

To maximize the order, we want at least the first 3 coefficients of each series to match up, so $a-b=-\frac{1}{2}$ and $b(b-a)=\frac{1}{24}$. Hence, solving gives $b=\frac{1}{12}$ and $a=-\frac{5}{12}$.

2 Observe that

$$x\left[\frac{1}{e} - \left(\frac{x}{x+1}\right)^x\right] = \frac{\frac{1}{e} - \left(\frac{x}{x+1}\right)^x}{1/x}.$$

Observe that since both numerator and denominator approach 0 as x approaches ∞, then we can use L'Hopitals rule. This gives

$$\begin{aligned}\lim_{x\to\infty}\frac{\frac{1}{e}-\left(\frac{x}{x+1}\right)^x}{1/x} &= \lim_{x\to\infty}\frac{\left(\frac{1}{e}-\left(\frac{x}{x+1}\right)^x\right)}{(1/x)'}\\ &= \lim_{x\to\infty}\frac{-\left(\left(\frac{x}{x+1}\right)^x\right)'}{-\frac{1}{x^2}}\\ &= \lim_{x\to\infty} x^2\left(\left(\frac{x}{x+1}\right)^x\right)'.\end{aligned}$$

We already know that $(k^x)' = k^x \cdot \ln x$, so using chain rule gives

$$\left(\left(\frac{x}{x+1}\right)^x\right)' = \left(\frac{x}{x+1}\right)' \cdot \left(\frac{x}{x+1}\right)^x \cdot \ln\frac{x}{x+1}.$$

Since $\frac{x}{x+1} = 1 - \frac{1}{x+1}$, then $\left(\frac{x}{x+1}\right)' = -\left(\frac{1}{x+1}\right)' = \frac{1}{(x+1)^2}$. Therefore, the limit is

$$\lim_{x\to\infty} \frac{x^2}{x+1}^2 \cdot \ln\left(\frac{x}{x+1}\right) \cdot \left(\frac{x}{x+1}\right)^e = 1.$$

3 By Lagrange's form of the remainder,

$$r_n(x_0; x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi)(x - x_0)^{n+1} = \frac{1}{(n+1)!} \cdot e^{\xi} \cdot x^{n+1} < 10^{-3}.$$

Since x is at most 2, it suffices to find the smallest value of n such that $\frac{1}{(n+1)!} \cdot 2^{n+1} \leq 10^{-3}$, which is $n = 10$. Thus the taylor formula is

$$1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \cdots + \frac{1}{10!}x^{10}.$$

4 We know that the taylor series at 0 of a function is

$$1 + \frac{1}{1!}f'(0) + \frac{1}{2!}f''(0) + \frac{1}{3!}f^{(3)}(0) + \cdots$$

(a) If f is even, then f' is odd, so f'' is even, so $f^{(3)}$ is odd, etc. Thus all odd-ordered derivatives of f are odd and even-ordered derivatives of f are even. Thus $f^{(2k+1)}(x) = -f^{(2k+1)}(-x)$, and plugging in $x = 0$ gives $f^{(2k+1)}(0) = -f^{(2k+1)}(0)$, so $f^{(2k+1)}(0) = 0$. Therefore all odd powers of the Taylor series are 0, and hence the Taylor series consists only of even terms.

(b) If f is odd, then all odd-ordered derivatives of f are even and all even-ordered derivatives of f are odd. Thus $f^{(2k)}(0) = 0$ using a similar argument as above, so the Taylor series consists only of odd terms.

5 Since we know that $f^{(n)}(0) = 0$ for all nonnegative n, then the Taylor series at 0 is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = 0.$$

Also, note that the remaining terms are of the form

$$\left|\frac{f^{(n)}(x)}{n!}x^n\right| \leq \frac{\sup\limits_{-1\leq x\leq 1} |f^{(n)}(x)|}{n!} n!|x|^n \leq \frac{n!C}{n!}|x|^n = C|x|^n.$$

Since $|x| < 1$, then this is a geometric series, which converges, so $f \equiv 0$ on $[-1, 1]$.

6

(a) Observe that since the derivative of a minimum or maximum point equals 0, if there is an $x \in I_1$ satisfying $|f^{(k-1)}(x)|$ is minimized, the derivative of $f^{(k-1)}$, which is $f^{(k)}$, satisfies $f^{(k)}(x) = 0$, unless x is a boundary point. Obviously $m_k(I) = 0$ if any of x are not boundary points, and it will be less than $\frac{1}{\mu}(m_{k-1}(I_1) + m_{k-1}(I_3))$. Otherwise, if all of x are boundary points, then the solution to $m_k(I)$ is also a boundary point, and it must equal one of $m_{k-1}(I_1)$ or $m_{k-1}(I_3)$, so we are also done.

(b) We have base case $k = 0$; $m_1(I) \leq 1$. Assume that for some k, we have that

$$m_k(I) \leq \frac{2^{k(k+1)/2}k^k}{\lambda^k}.$$

Then we have

$$m_{k+1}(I) \leq \frac{1}{\mu}(m_k(I_1) + m_k(I_3))$$

by part (a), so

$$m_{k+1}(I) \leq \frac{1}{\mu}\left(\frac{2^{k(k+1)/2}k^k}{\lambda_1^k} + \frac{2^{k(k+1)/2}k^k}{\lambda_3^k}\right) = \frac{2^{k(k+1)/2}k^k}{\mu}\left(\frac{1}{\lambda_1^k} + \frac{1}{\lambda_3^k}\right).$$

We wish to prove that

$$m_{k+1}(I) \leq \frac{2^{(k+1)(k+2)/2}(k+1)^{k+1}}{\lambda^{k+1}}.$$

$\lambda = \lambda_1 + \lambda_2 + \mu$ and $2^{(k+1)(k+2)/2} = (k+1)2^{k(k+1)/2}$. Therefore, we are done.

(c) It suffices to show, by induction, that there exists k numbers $x_{k_1} < x_{k_2} < \cdots < x_{k_k}$ such that $f^{(k)}(x_{k_i})f^{(k)}(x_{k_{i+1}}) < 0$, so they are of opposite sign, which will show that there is a root between x_{k_i} and $x_{k_{i+1}}$. We have base case $n = 1$. If $|f'(0)| > a_1$, then $f'(x) = 0$ has at least 0 distinct roots. Suppose this works for some value of k. We can start by using the values $x_{k_1}, \ldots, x_{k_k}$ for $x_{(k+1)_1}, \ldots, x_{(k+1)_k}$. Then we can add $x_{(k+1)_{k+1}}$ satisfying $f^{(k+1)}(x_{(k+1)_{k+1}})$ is of opposite sign from $f^{(k+1)}(x_{k_k})$, so we're done.

7

(a) Consider the auxiliary function $h(x) = f(x) - r(x)$, where $r(x) = f'(a)(x-a)$. Then $h'(x) = f'(x) - f'(a)$ and $h(a) = h(b) = 0$, so there exists ξ such that $h'(\xi) = 0$. Then $f'(\xi) = f'(a)$, and we can replace $f'(a)$ with $f'(\xi)$ for any ξ so that $f'(x)$ attains all values between $f'(a)$ and $f'(b)$.

(b) Lagrange's infinite-increment theorem states that if a function $f : [a, b] \to \mathbb{R}$ is continuous on a closed interval $[a, b]$ and differentiable on the open interval, $]a, b[$, there exists a point $\xi \in]a, b[$ such that $f(b) - f(a) = f'(\xi)(b - a)$. Because we know that f'' exists, then f' must be differentiable on $]a, b[$, so Lagrange's theorem holds for f'. Hence there exists $\xi \in]a, b[$ such that $f'(b) - f'(a) = f''(\xi)(b - a)$, as desired.

8

(a) Suppose for the sake of contradiction that $f'(x)$ has a discontinuity of the first kind. Suppose at some point $x = a$, $f'(a)$ has a discontinuity. This means that $f'(x)$ has a limit at a from both the left side and the right side, and these limits are equal, but $f'(a)$ is not equal to this limit. But because $f'(a)$ exists, this is a contradiction, so $f'(x)$ has no removable discontinuities, so it can only have discontinuities of the second kind.

(b) The error in the proof lies in the assumption that $\lim_{x\to x_0} f'(\xi) = \lim_{\xi\to x_0} f'(\xi)$. It should have been $\lim_{x\to x_0} f'(\xi(x))$ where $\xi(x)$ is a function that lies between x and x_0. Note that the limit of $f'(\xi(x))$ as $x \to x_0$ does not necessarily equal $f'(x_0)$ since $\xi(x)$ is a function of x and does not necessarily to approach x_0 in the same way as x. The "proof" assumes that the limit equals $f'(x_0)$, so it is incorrect.

9

(a) Let $x \in [-a, a]$. Then by the mean value theorem, there exists a point $c \in [-a, x]$ if $x > 0$ or $c \in [x, a]$ if $x < 0$ such that $f'(x) - f'(0) = f''(c) \cdot x$. Since $|f'(x)| \leq |f'(0)| + |f''(c)||x|$, since $c \in [-a, a]$, then $|f''(c)| \leq M_2$ and $|x| \leq a$. Hence we have $|f'(x)| \leq |f'(0)| + M_2|x|$.

Note that for some $d \in [-a, a]$, $f'(0) - f(0) = f'(d) \cdot 0$. We know that $|f'(0)| \leq |f(0)| + |f'(d)|$. Since $d \in [-a, a]$, then $|f'(d)| \leq M_1$ and $|f(0)| \leq M_0$, so $|f'(0)| \leq M_0 + M_1$. Then $|f'(x)| \leq M_0 + M_1 + M_2|x|$, so $M_1 \leq \frac{M_0}{a} + \frac{a}{2}M_2$. Since the coefficient of M_2, $\frac{x^2+a^2}{2a}$ is at least equal to $\frac{a}{2}$, we are done.

(b) Observe that when x is at its maximum, $x = a$, so from part (a), $M_1 \leq \frac{M_0}{a} + aM_2$. If the length of I, which equals $2a$, is not less than $2\sqrt{M_0/M_2}$, then $a \geq \sqrt{M_0/M_2}$. By AM-GM, $\frac{M_0}{a} + aM_2 \geq 2\sqrt{M_0M_2}$, which proves the first case.

For the second case, if $I = \mathbb{R}$, then we can't have $x = a$, and a approaches infinity. The minimum value of $\frac{M_0}{a} + \frac{a}{2}M_2$ is $\sqrt{2M_0M_2}$ by AM-GM.

(c) If we have equality cases, then we cannot have smaller numbers. For the number 2, we have $a = \sqrt{M_0/M_2}$, and for the number $\sqrt{2}$, we have $a = \sqrt{2M_0/M_2}$.

(d) We prove this by induction on p. We have the base case $p = 2$ from part 9(b), and assume this works for some p. We prove that it also works for $p + 1$. First note that by AM-GM,

$$\left(1 - \frac{k}{p}\right) M_0 + \frac{k}{p} M_p = \frac{\underbrace{M_p + M_p + \cdots + M_p}_{\text{summed } k \text{ times}} + \underbrace{M_0 + M_0 + \cdots + M_0}_{\text{summed } p-k \text{ times}}}{p}$$
$$\geq \sqrt[p]{M_p^k M_0^{p-k}} = M_0^{1-k/p} \cdot M_p^{k/p}.$$

Thus it suffices to prove that

$$M_k \leq 2^{k(p-k)/2} \left((1 - \frac{k}{p}) M_0 + \frac{k}{p} M_p \right)$$

which is obvious, so we are done.

10 By Lagrange's form of remainder, we get

$$r_n(x_0; x) = \frac{1}{n!} f^{(n+1)}(\xi)(x - x_0)^n,$$

so it suffices to prove that $\xi = x_0 + \theta(x - x_0)$. By the mean value theorem, there exists $\xi \in [0, 1]$ such that $f(x) - f(x_0) = f'(\xi)(x - x_0)$, so if we replace $f'(\xi)(x - x_0)$ with $\theta(x - x_0)$ and replace $f(x)$ with ξ, we get $\xi - x_0 = \theta(x - x_0)$.

11

(a) The Taylor expansion of f at 0 is

$$f(x) = 1 + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n.$$

Since f has $n + 1$ roots, it has order $n + 1$, so $f^{(n)}$ has order 1, which means that it has a root.

(b) Consider the Taylor expansion of f at 0, which is

$$f(x) = 1 + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n.$$

Observe that a polynomial is determined by any n values, and this will determine a polynomial of degree $n-1$, which is perfect for $L(x)$. We can use the method of undetermined coefficients to solve for $L(x)$. Note that because $f(x)$ can also be expressed in polynomial form to a highly accurate degree, then $f(x) - L(x) = 0$ for $x = x_1, \ldots, x_n$. So since $f(x) - L(x)$ is a polynomial, it must be a scalar multiple of $(x - x_1)\cdots(x - x_n)$, since $f(x) - L(x)$ has degree n. It suffices to prove that the remaining coefficient equals $\frac{f^{(n)}(\xi)}{n!}$ for some $\xi \in I$, which follows from the mean value theorem.

(c) Since $n_1 + n_2 + \cdots + n_p = n$, then the sum of the orders, and the number of roots of f, f', f'', etc is $n_1 + n_2 + \cdots + n_p = n$. Therefore, once you take the derivative $n-1$ times to get $f^{(n-1)}$, the remainging order is 1, so it has one root, so there exists ξ such that $f^{(n-1)}(\xi) = 0$.

(d) Let $g(x) = f(x) - H(x)$, so note that $g^{(k)}(x) = f^{(k)}(x) - H^{(k)}(x)$. Then for x_i, $g(x_i) = g'(x_i) = \cdots = g^{(n_i-1)}(x_i) = 0$, and we want to prove that

$$g(x) = \frac{(x - x_1)^{n_1}\cdots(x - x_p)^{n_p}}{n!} f^{(n)}(\xi).$$

If we look at the taylor polynomial forms of $g(x)$, $g'(x)$, $g''(x)$, and so on, we see that for x_i, we have that $g(x), g'(x), g''(x), \ldots, g^{(n_i-1)}(x)$ are multiples of $(x - x_i)$. But in order for $g^{(n_i-1)}(x)$ to be a multiple of $(x - x_i)$, $g(x)$ needs to be a multiple of $(x - x_i)^{n_i}$, since the $n_i - 1$th derivative of $(x - x_i)^{n_i}$ is $n_i! \cdot (x - x_i)$. So over all i, $g(x)$ is a multiple of $(x - x_1)^{n_1}\cdots(x - x_p)^{n_p}$, and the there must exist ξ such that the coefficient is $\frac{f^{(n)}(\xi)}{n!}$.

12

(a) Consider 2 roots of $P(x)$ and call them x_1 and x_2, so $P(x_1) = P(x_2)$. By the mean value theorem, there exists $\xi \in [x_1, x_2]$ such that $P(x_2) - P(x_1) = 0 = P'(\xi)(x_2 - x_1)$, so $P'(\xi) = 0$.

(b) Suppose $P(x)$ has degree n and that the multiplicity of a root a is k, so let $P(x) = (x-a)^k Q(x)$ for some polynomial Q. Observe that Q does not have a factor of $x - a$. Then by the chain rule,

$$\begin{aligned} P'(x) &= ((x-a)^k)'Q(x) + (x-a)^k Q'(x) \\ &= k(x-a)^{k-1}Q(x) + (x-a)^k Q'(x) \\ &= (x-a)^{k-1}(kQ(x) + (x-a)Q'(x)). \end{aligned}$$

Obviously $P'(x)$ is also has a as a root, but its multiplicity is $k-1$.

(c) Suppose the roots of $P(x)$ are $a_1, \ldots, a_k$ with multiplicities $n_1, \ldots, n_k$, so that

$$P(x) = c(x - a_1)^{n_1}(x - a_2)^{n_2}\cdots(x - a_k)^{n_k},$$

where c is a constant. Then by the previous part, we know that $P(x)$ has the roots $a_1, \ldots, a_k$ with multiplicities $n_1 - 1, \ldots, n_k - 1$, so $P'(x)$ is a scalar multiple of

$$(x - a_1)^{n_1-1}(x - a_2)^{n_2-1}\cdots(x - a_k)^{n_k-1},$$

and thus $Q(x)$ is also a scalar multiple of

$$(x-a_1)^{n_1-1}(x-a_2)^{n_2-1}\cdots(x-a_k)^{n_k-1}.$$

Hence dividing $P(x)$ by $Q(x)$ leaves us with a scalar multiple of $(x-a_1)(x-a_2)\cdots(x-a_k)$, so $\frac{P(x)}{Q(x)}$ has $a_1,\ldots,a_k$ as its roots, but all of them have multiplicity 1.

13

(a) Take a polynomial $Q(x) = c_0 + c_1x + \cdots + c_nx^n$. Then obviously if we take $P(x) = Q(x-x_0)$, then $P(x) = c_0 + c_1(x-x_0) + \cdots + c_n(x-x_0)^n$. So in order to express $P(x)$ in terms of $x - x_0$, we first find $P(x+x_0) = Q(x)$, and find the coefficients of $Q(x)$. Then we plug $x - x_0$ into $Q(x)$ to get $P(x)$ in terms of $x - x_0$.

(b) Consider the Taylor expansion of $f(x)$ with respect to x_0, which looks like

$$f(x) = 1 + \frac{f'(x_0)}{1!}x + \cdots + \frac{f^{(n)}(x_0)}{n!}x^n + o((x-x_0)^n).$$

Then if we let $P(x)$ be the first $n+1$ terms, we're done.

14

(a) By the mean value theorem, there exists a value ξ_{1_1} such that $f(x_0+h_1) - f(x_0) = f'(\xi_1)\cdot h_1$ and a value ξ_{1_2} such that $f(x_0+h_1+h_2) - f(x_0+h_2) = f'(\xi_2)\cdot h_1$, so there exists a value ξ_2 such that

$$\begin{aligned}(f(x_0+h_1+h_2) &- f(x_0+h_2)) - (f(x_0+h_1) - f(x_0))\\ &= f'(\xi_{1_2})\cdot h_1 - f'(\xi_{1_1})\cdot h_1\\ &= (f'(\xi_{1_2}) - f'(\xi_{1_1}))\cdot h_1 = f''(\xi_2)h_1h_2.\end{aligned}$$

As you can see, we have the base case. Assume there exists a value ξ_{k_1} such that

$$\Delta^k f(x_0; h_1, \ldots, h_{k-1}) = f^{(k)}(\xi_{k_1})h_1\cdots h_k$$

and

$$\Delta^k f(x_0; h_1, \ldots, h_k) = f^{(k)}(\xi_{k_2})h_1\cdots h_k.$$

Then

$$\begin{aligned}\Delta^{k+1} f(x_0; h_1, \ldots, h_{k+1}) &= \Delta^k f(x_0; h_1, \ldots, h_k) - \Delta^k f(x_0; h_1, \ldots, h_{k-1})\\ &= (f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1}))\cdot h_1\cdots h_k.\end{aligned}$$

By the mean value theorem, there exists some ξ such that

$$f^{(n+1)}(\xi)\cdot h_{k+1} = f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1}),$$

so substituting this in gives $f^{(k+1)}(\xi)h_1\ldots h_{k+1}$, so we are done by induction.

(b) Using our result from part (a), we know that

$$\begin{aligned}|\Delta^n f(x_0; &h_1, \ldots, h_n) - f^{(n)}(x_0)h_1\cdots h_n|\\ &= |f^{(n)}(\xi)h_1\cdots h_n - f^{(n)}(x_0)h_1\cdots h_n|\\ &= |f^{(n)}(\xi) - f^{(n)}(x_0)|\cdot|h_1|\cdots|h_n|\\ &\le \sup_{x\in]a,b[}\left|f^n(x) - f^{(n)}(x_0)\right|\cdot|h_1|\cdots|h_n|,\end{aligned}$$

as desired

(c) From part (a), we know that there exists ξ such that

$$\Delta^n f(x_0; h^n) = f^{(n)}(\xi)h^n \implies f^{(n)}(\xi) = \frac{\Delta^n f(x_0; h^n)}{h^n}.$$

As $h \to 0$, $\xi \to x_0$, so

$$f^{(n)}(x_0) = \lim_{h\to 0} \frac{\Delta^n f(x_0; h^n)}{h^n}.$$

(d) For example, take $f(x) = \frac{1}{x}$ so that $f^{(n)}(x)$ is a multiple of $\frac{1}{x^n}$. Then the limit equals 1.

15

(a) Consider the function $f(x) = \frac{1}{x^\alpha} = x^{-\alpha}$. Then $f'(x) = -\alpha x^{-\alpha-1}$, so there exists $\xi \in]n-1, n[$ such that

$$\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha} = f(n-1) - f(n) = f'(\xi) \cdot ((n-1) - n) = -f'(\xi) = \alpha \xi^{-\alpha-1}$$

by Lagrange's theorem. Hence

$$\frac{1}{\xi^{\alpha+1}} = \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right).$$

Since $\xi < n$, then $\frac{1}{\xi^{\alpha+1}} > \frac{1}{n^{\alpha+1}}$, so

$$\frac{1}{n^{\alpha+1}} < \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right),$$

as desired.

(b) It suffices to prove that the series $\sum_{n=1}^{\infty} \frac{1}{n^{1+\alpha}}$ converges for $\alpha > 0$.

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{1}{n^{1+\alpha}} &< \sum_{n=1}^{\infty} \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right) \\ &= \frac{1}{\alpha}\sum_{n=1}^{\infty}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right) \\ &= \frac{1}{\alpha},\end{aligned}$$

which converges, so $\sum_{n=1}^{\infty} \frac{1}{n^\sigma}$ also converges.

5.4 Differential Calculus Used to Study Functions

1

(a) Observe that

$$\ln M_t(x, \alpha) = \ln \left(\sum_{i=1}^{n} \alpha_i x_i^t\right)^{1/t} = \frac{1}{t} \ln \left(\sum_{i=1}^{n} \alpha_i x_i^t\right).$$

Now we can take the limit of this as $t \to 0$, and since when $t = 0$, both numerator and denominator equal 0, we use L'Hopitals rule. The derivative of the denominator is 1 and by chain rule, the derivative of the numerator is

$$\frac{\left(\sum_{i=1}^{n} \alpha_i \cdot \ln x_i \cdot x_i^t\right)}{\sum_{i=1}^{n} \alpha_i x_i^t} = \frac{\sum_{i=1}^{n} \alpha_i \cdot \ln x_i}{\sum_{i=1}^{n} \alpha_i}.$$

Because $\sum_{i=1}^{n} \alpha_i = 1$, then we get

$$\lim_{t \to 0} \ln M_t(x, \alpha) = \sum_{i=1}^{n} \alpha_i \ln x_i,$$

so

$$\lim_{t \to 0} M_t(x, \alpha) = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}.$$

(b) As $t \to +\infty$, note that the largest value of x_i will begin to dominate $M_t(x, \alpha)$, so $\lim_{t \to +\infty} M_t(x, \alpha) = \max_{1 \le i \le n} x_i$.

(c) As $t \to -\infty$, note that the smallest value of x_i will begin to dominate $M_t(x, \alpha)$, so $\lim_{t \to -\infty} M_t(x, \alpha) = \min_{1 \le i \le n} x_i$.

(d) It suffices to prove that $(M_t(x, \alpha))'$ is positive. By using the chain rule twice, we get that the derivative equals

$$\begin{aligned}
&\left(\sum_{i=1}^{n} \alpha_i x_i^t\right)^{1/t} \cdot \left(\frac{1}{t}\right)' \cdot \left(\sum_{i=1}^{n} \alpha_i x_i^t\right)' \cdot \ln\left(\sum_{i=1}^{n} \alpha_i x_i^t\right) \\
&= \left(\sum_{i=1}^{n} \alpha_i x_i^t\right)^{1/t} \cdot \left(-\frac{1}{t^2}\right) \cdot \left(\sum_{i=1}^{n} \alpha_i \ln x_i \cdot x_i^t\right) \cdot \ln\left(\sum_{i=1}^{n} \alpha_i x_i^t\right)
\end{aligned}$$

Since the first term is positive, the second is negative, the third is negative, and the fourth is positive, we're done.

2 Observe that by the binomial theorem,

$$|(1+x)|^p = |(1+x)^p| = |1 + px + \binom{p}{2} x^2 + \cdots + x^p|.$$

Obviously, if we let $c_p = 0$ (not exactly sure what this means), then it suffices to show that this sum is greater than or equal to $1 + px$. If $x > 0$, this is obvious. If $x < 0$, as long as $x \ge -1$, this is also obvious. However, if $x < -1$, then the absolute value is

$$-(1+x)^p = -1 - px - \cdots - x^p \ge 1 + px$$

because x is negative.

As you can see, we have the base case. Assume there exists a value ξ_{k_1} such that

$$\Delta^k f(x_0; h_1, \ldots, h_{k-1}) = f^{(k)}(\xi_{k_1}) h_1 \cdots h_k$$

and

$$\Delta^k f(x_0; h_1, \ldots, h_k) = f^{(k)}(\xi_{k_2}) h_1 \cdots h_k.$$

Then

$$\begin{aligned}\Delta^{k+1} f(x_0; h_1, \ldots, h_{k+1}) &= \Delta^k f(x_0; h_1, \ldots, h_k) - \Delta^k f(x_0; h_1, \ldots, h_{k-1}) \\ &= (f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1})) \cdot h_1 \cdots h_k.\end{aligned}$$

By the mean value theorem, there exists some ξ such that

$$f^{(n+1)}(\xi) \cdot h_{k+1} = f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1}),$$

so substituting this in gives $f^{(k+1)}(\xi)h_1 \ldots h_{k+1}$, so we are done by induction.

3 Observe that it suffices to show that $\left(\frac{\sin x}{x}\right)^3 - \cos x > 0$. It also suffices to show that this increases as x increases. To do this, we can show that the derivative is greater than 0. The derivative is

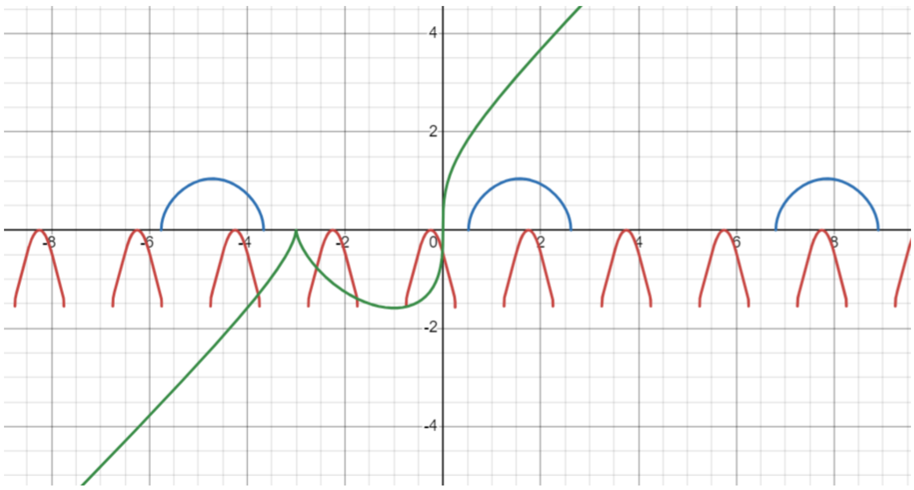
$$\left(\left(\frac{\sin x}{x}\right)^3\right)' + \sin x,$$

and by chain rule, the derivative of $\left(\frac{\sin x}{x}\right)^3$ is

$$3\left(\frac{\sin x}{x}\right)^2 \cdot \left(\frac{\sin x}{x}\right)' = \frac{3\sin^2 x}{X^2} \cdot \frac{x\cos x - \sin x}{x^2}$$

by product rule as well. Clearly this is greater than 0, as desired.

4



(a) Observe that if $f(x) = \arctan \log_2 \cos(\pi x + \frac{\pi}{4})$, we have the red graph.

(b) Observe that if $f(x) = \arccos(\frac{3}{2} - \sin x)$, we have the blue graph.

(c) Observe that if $f(x) = \sqrt[3]{x(x+3)^2}$, we have the green graph.

(d) Observe that if $\theta = \frac{r}{r^2+1}$ because I prefer using r and θ notation, then $r^2\theta + \theta = r$, so $r^2\theta - r + \theta = 0$. Then by the quadratic formula, $\theta = \frac{1 \pm \sqrt{1-4\theta^2}}{2\theta}$, so the graph starts at 0 and it goes in the positive x and y directions, it is concave down, and it has an asymptote at $y = 1$.

(e) Observe that if we have the graph of $y = f(x)$, then the graph of $y = f(x) + B$ is the graph of $y = f(x)$ shifted upwards by B units. The graph of $y = Af(x)$ is the graph of $y = f(x)$ stretched upwards by a factor of A. The graph of $y = f(x + b)$

is the graph of $y = f(x)$ shifted b units to the left. The graph of $y = f(ax)$ is the graph of $y = f(x)$ stretched horizontally by a factor of a. The graph of $-f(x)$ is the reflection of the graph of $y = f(x)$ over the x-axis. The graph of $y = f(-x)$ is the reflection of the graph of $y = f(x)$ over the y-axis.

5 If we consider two points $(x_1, f(x_1))$ and $(x_2, f(x_2))$, then the point $\left(\frac{x_1+x_2}{2}, f\left(\frac{x_1+x_2}{2}\right)\right)$ is the point on the function in between x_1 and x_2. Then $\frac{f(x_1)+f(x_2)}{2}$ is the midpoint of the chord between x_1 and x_2, so its directly above the previous point by the given inequality. Therefore the chord lies above the graph, so this is convex, as desired.

6

(a) In the given interval, note that a convex function, because it's convex, can only have one hump or drop if you want to put it that way. If it wanted to have two humps, it would need to be convex at one place and concave at another. Therefore, a convex function that's not flat would be unbounded in the positive y-direction, so if it's bounded, then it must be horizontal.

(b) Observe that in order for the limit to approach 0, we need L'Hopitals rule. The limit equals $\frac{f'(x)}{1} = 0$, so $f'(x) = 0$, so $f(x)$ is constant, as desired.

(c) Observe that for any limit, either 1. it doesn't exist. 2. it exists and is infinite, and 3. it exists and is finite. The first case doesn't hold because since f is convex, it's also continuous, so we're done.

7

(a) $f'_-(x)$ is the derivative of f at x from the left side and $f'_+(x)$ is the derivative of f at x from the right side. Observe that as f is convex, it is also continuous, and if we say that it is not differentiable at a point, for example, there is a sharp corner there (that would be the only way for it to not be differentiable at that point), then there would be a left derivative and a right derivative.

Also, since the function is convex, then at that point, despite it not being differentiable, $f''(x) > 0$, so the derivative on the right is at least than the derivative on the left, as desired.

(b) Since $x_1 < x_2$ and the function is convex, then $f''(x) > 0$, so the smallest value of $f'(x_2)$ is larger than the greatest value of $f'(x_1)$, so $f'_+(x_1) \leq f'_-(x_2)$ for $x_1, x_2 \in]a, b[$.

(c) The set of cusps in the graph is the set of sharp corners. If it's finite, it's at most countable. It must be countable, however because we can map each cusp to a natural number, so we're done.

8

(a) There are 3 cases to consider. 1, if $f(x)$ is not bounded from below for any $x \in I$, then I^* will be empty because $f^*(t)$ is infinite for any t. 2, if $f(x)$ is bounded from below for some $x \in I$, then I^* will consist of a single point. 3, if $f(x)$ is bounded from below for all $x \in I$, and the supremum is achieved at some point, then $f^*(T)$ is a real number for all t. Then I^* is an interval. It suffices to show that $f^*(t)$ is convex on I^*.

If $t_1, t_2 \in I^*$, then for any $\lambda \in (0, 1)$,

$$f^*(\lambda t_1 + (1-\lambda)t_2) = \sup_{x \in I}((\lambda t_1 + (1-\lambda)t_2)x - f(x)).$$

Suppose x_1 and x_2 achieve the supremum for t_1 and t_2, so $f^*(t_1) = t_1x_1 - f(x_1)$ and $f^*(t_2) = t_2x_2 - f(x_2)$. Consider a point $x \in I$. Then

$$(\lambda t_1 + (1-\lambda)t_2)x - f(x) \leq \lambda(t_1x_1 - f(x_1)) + (1-\lambda)(t_2x_2 - f(x_2)).$$

Now taking $\sup_{x\in I}$ on the left side,

$$f^*(\lambda t_1 + (1-\lambda)t_2) \leq \lambda f^*(t_1) + (1-\lambda)f^*(t_2),$$

so $f^*(t)$ is convex on I^*.

(b) If f is convex, then $I^* \neq \varnothing$, so for $f^* \in C(I^*)$, $(f^*)^* = \sup_{t\in I^*}(xt - f^*(t)) = f(x)$ for any $x \in I$, so the Legendre transform is involutive.

(c) Since $f^*(t) = \sup_{x\in I}(tx - f(x))$, then $tx - f(x) \leq f^*(t)$, so rearranging gives $xt \leq f(x) + f^*(t)$, as desired.

(d) If f is a convex differentiable function, then $f^*(t) = tx_t - f(x_t)$, where you can find x_t from $t = f'(x)$. $t = f'(x)$ means that the tangent to the graph of f at x has slope t, so the Legendre transform is a function defined on the set of tangents to f.

(e) We know that

$$f^*(t) = \frac{1}{\beta}t^\beta = \sup_{x\in I}(tx - f(x)) \implies tx - f(x) \leq \frac{1}{\beta}t^\beta \implies tx \leq \frac{1}{\alpha}x^\alpha + \frac{1}{\beta}x^\beta,$$

(f) Since $f^*(x)$ is a sup, then $f^*(x) \geq xt - e^x$, so $xt \leq e^x + t\ln\frac{t}{e}$.

9

(a) The angle between $\mathbf{a}$ and $\mathbf{v}$ is $\cos\theta = \frac{\mathbf{a}\cdot\mathbf{v}}{||\mathbf{a}||^2}$, so $||\mathbf{a}_t|| = ||\mathbf{a}||\cos\theta = \frac{\mathbf{a}\cdot\mathbf{v}}{||\mathbf{v}||}$. Then $\mathbf{a}_t = ||\mathbf{a}_t||\mathbf{v} = \left(\frac{\mathbf{a}\cdot\mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v}$, and $\mathbf{a}_n = \mathbf{a} - \mathbf{a}_t = \mathbf{a} - \left(\frac{\mathbf{a}\cdot\mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v}$.

(b) If we let $x = \cos t$, $y = r\sin t$, then $x' = -r\sin t$, $y' = r\cos t$, $x'' = -r\cos t$, $y' = -r\sin t$. Since $\mathbf{a}_n$ is perpendicular to $\mathbf{v}$, then $\mathbf{a}_n = \mathbf{a}$. $\frac{|\mathbf{v}(t)|}{|\mathbf{a}_n(t)|} = r$, as desired.

(c) Since

$$a_n = \frac{||v\times a||}{||v||} = \frac{||(x'y'' - x''y')k||}{\sqrt{(x')^2 + (y')^2}} = \frac{x'y'' - x''y'}{\sqrt{(x')^2 + (y')^2}}.$$

Then

$$r(t) = \frac{|v(t)|}{|a_n(t)|} = \frac{((x')^2 + (y')^2)^{3/2}}{|x'y'' - x''y'|}.$$

(d) Observe that the larger the radius, the smaller the curvature, so $k(t) = \frac{1}{r(t)}$. The curvature is how much a curve turns at a given point. If $k(t)$ is positive, then the curve is inward, but if $k(t)$ is negative, then the curve is outward.

(e) Setting $x(t) = x$ gives $x' = 1$ and $x'' = 0$. Then

$$\frac{1\cdot y'' - 0\cdot y'}{(1^2 + (y')^2)^{3/2}} = \frac{y''}{[1 + (y')^2]^{3/2}}$$

(f) Observe that with we let $R = r(t)$, then obviously the osculating circle would have the highest possible order of contact, so if $x(t)$ and $y(t)$ are twice differentiable and the osculating circle is tangent to the curve at that point, then it's center is (a, b).

(g) We know that $v^2 = 2gh$, so $(x')^2 + (y')^2 = 2gh = 2gy$. $x = 1 - y^2$, so $x' = 1 - 2yy'$.

5.5 Complex Numbers and Elementary Functions

1

(a) We know that $|z_1|$ denotes the length of the line segment from the origin to z_1, etc. So we're basically asked to prove the triangle inequality, which is obvious.

(b) Let $z = a + bi$. Then $z - 1 = (a - 1) + bi$ and $z + 1 = (a + 1) + bi$. We're in for a bash.

$$\begin{aligned} |z-1| + |z+1| &= |(a-1)+bi| + |(a+1)+bi| \\ &= \sqrt{(a-1)^2+b^2} + \sqrt{(a+1)^2+b^2} \\ &\le 3. \end{aligned}$$

Thus

$$\begin{aligned} \sqrt{(a+1)^2+b^2} &\le 3 - \sqrt{(a-1)^2+b^2} \\ (a+1)^2+b^2 &\le 9 - 6\sqrt{(a-1)^2+b^2} + (a-1)^2+b^2 \\ 4a &\le 9 - 6\sqrt{(a-1)^2+b^2} \\ 6\sqrt{(a-1)^2+b^2} &\le 9 - 4a \\ 36(a-1)^2 + 36b^2 &\le 81 - 72a + 16a^2 \\ 20a^2 + 36b^2 &\le 45. \end{aligned}$$

This is the area inside an ellipse with semimajor axis $\frac{3}{2}$ and semiminor axis $\frac{\sqrt{5}}{2}$.

(c) The nth roots of unity are of the form $e^{2\pi ik/n}$, where k is an integer from 0 to $n-1$. Then we need

$$1 + e^{2\pi i/n} + e^{4\pi i/n} + \cdots + e^{2\pi i(n-1)/n} = \frac{1-e^{2\pi}}{1-e^{2\pi i/n}} = 0.$$

(d) It flips z over the x-axis.

2

(a) Note that $1 - q^{n+1} = (1-q)(1+q+\cdots+q^n)$, so

$$1 + q + \cdots + q^n = \frac{1-q^{n+1}}{1-q}.$$

(b) Let $S = 1 + q + \cdots + q^n + \cdots$. Then $Sq = q + q^2 + \cdots$, so $S - Sq = 1$. Then $S = \frac{1}{1-q}$, so

$$1 + q + \cdots + q^n + \cdots = \frac{1}{1-q}.$$

(c) Note that

$$1 + e^{i\varphi} + \cdots + e^{in\varphi} = \frac{1-e^{i(n+1)\varphi}}{1-e^{i\varphi}}.$$

(d) This is basically $\frac{1-r^{n+1}e^{i(n+1)\varphi}}{1-re^{i\varphi}}$.

(e) This is basically $\frac{1}{1-re^{i\varphi}}$.

(f) This is the real part of $1 + re^{i\varphi} + \cdots + r^n e^{in\varphi}$, which equals

$$\begin{aligned}\frac{1 - r^{n+1}e^{(n+1)\varphi i}}{1 - re^{\varphi i}} &= \frac{(1 - r^{n+1}e^{(n+1)\varphi i})(1 - re^{-\varphi i})}{(1 - r\cos\varphi)^2 + r^2\sin^2\varphi} \\ &= \frac{1 + r^{n+2}e^{n\varphi i} - r^{n+1}e^{(n+1)\varphi i} - re^{-\varphi i}}{1 - 2r\cos\varphi + r^2}.\end{aligned}$$

The real part of this is

$$\frac{1 + r^{n+2}\cos n\varphi - r^{n+1}\cos(n+1)\varphi - r\cos\varphi}{1 - 2r\cos\varphi + r^2}.$$

(g) This is the real part of $1 + re^{i\varphi} + \cdots + r^n e^{in\varphi} + \cdots$ which equals

$$\frac{1}{1 - re^{i\varphi}} = \frac{1 - re^{-i\varphi}}{(1 - r\cos\varphi)^2 + (r\sin\varphi)^2} = \frac{1 - re^{-i\varphi}}{1 - 2r\cos\varphi + r^2}.$$

The real part of this is

$$\frac{1 - r\cos\varphi}{1 - 2r\cos\varphi + r^2}.$$

(h) This is the imaginary part of $1 + re^{i\varphi} + \cdots + r^n e^{in\varphi}$ plus 1, which is

$$1 + \frac{r^{n+2}\sin n\varphi - r^{n+1}\sin(n+1)\varphi + r\sin\varphi}{1 - 2r\cos\varphi + r^2}.$$

(i) This is the imaginary part of $1 + re^{i\varphi} + \cdots + r^n e^{in\varphi} + \cdots$ plus 1, which equals

$$1 + \frac{r\sin\varphi}{1 - 2r\cos\varphi + r^2}.$$

3 We already know that by definition, $e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n$, so

$$\lim_{n\to\infty}\left(1 + \frac{z}{n}\right)^n = \left(\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n\right)^z = e^z.$$

4

(a) Observe that

$$e^w = e^{\ln|z| + i\mathrm{Arg}z} = e^{\ln|z|}\cdot e^{i\mathrm{Arg}z} = |z|\cdot\frac{z}{|z|} = z,$$

since Arg$z$ is the angle of z, so if you take e to the i times this angle, then we get a unit complex number in the direction of z, which equals $\frac{z}{|z|}$.

(b) We find that $\mathrm{Ln}z = \ln|z| + i\mathrm{Arg}z$. Thus $\mathrm{Ln}1 = \ln|1| + i\mathrm{Arg}1 = 0 + i\cdot 0 = 0$. Then $\mathrm{Ln}i = \ln|i| + i\mathrm{Arg}i = 0 + i\cdot\frac{\pi}{2} = \frac{\pi}{2}i$.

(c) $1^\pi = e^{\pi\mathrm{Ln}1} = e^{\pi\cdot 0} = 1$ and $i^i = e^{i\mathrm{Ln}i} = e^{i\cdot\frac{pi}{2}i} = e^{-\frac{pi}{2}}$.

(d) $z = \arcsin w = \arcsin\left(\frac{1}{2i}(e^{iz} - e^{-iz})\right)$.

(e) This is because we know that $\sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$, so $|\sin z| = \frac{1}{2}|e^{iz} - e^{-iz}|$, so we would need $|e^{iz} - e^{-iz}| = 4$. Since $|e^{iz}| = |e^{-iz}|$, then we need $e^{iz} = -e^{-iz}$, so z is a real number, but $|e^{iz}| = 1$ if z is a real number, so this is impossible.

5

(a) $f(z) = \frac{1}{1+z^2}$ is continuous. If $z = a + bi$, then $f(z) = \frac{1}{1+(a+bi)^2} = \frac{1}{(1+a^2-b^2)+2abi}$, which is continuous.

(b) Note that
$$\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} = \sum_{n=0}^{\infty}(-z^2)^n = \sum_{n=0}^{\infty}(-1)^n z^{2n},$$
which has radius of convergence 1.

(c) Note that
$$\frac{1}{1+\lambda^2 z^2} = \frac{1}{1-(-\lambda^2 z^2)} = \sum_{n=0}^{\infty}(-\lambda^2 z^2)^n = \sum_{n=0}^{\infty}(-1)\lambda^2 n z^{2n},$$
so the radius of convergence is $\frac{1}{\lambda}$.

The radius of convergence is probably determined by something like $\frac{1}{d}$, where d is the scalar factor. It also makes since on the real line, for example in $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$.

6

(a) Yes, it is continuous. As $z \to 0$, $-1/z^2 \to -\infty$, so $\lim_{z\to} f(z) = 0$, which equals $f(0)$.

(b) Yes, it is. Just apply the same reasoning as in part (a) except with real numbers.

(c) We know that
$$e^z = 1 + \frac{1}{1!}z + \frac{1}{2!}z^2 + \cdots,$$
so
$$e^{-1/z^2} = 1 - \frac{1}{1!z^2} + \frac{1}{2!z^4} - \cdots.$$
Nope, it doesn't exist at $z_0 = 0$.

(d) In that case, the radius of convergence would equal 0, so we would need $\lim_{n\to 0} \sqrt[n]{|c_n|} = \infty$. If we let $c_n = 2^{n^2}$, then this works.

(e) $c_n = 2^{n^2}$ gives $\sqrt[n]{|c_n|} = 2^n$, which definitely doesn't converge.

7

(a) We show that $m(x) = \min_{a\le t\le x} f(t)$ is continuous, the $M(x)$ case is symmetric. Any function $f(t)$ consists of increasing and decreasing intervals. $m(x)$ cannot be discontinuous because this would require the minimum of $f(t)$ on that current interval to be discontinuous. Since the given intervals are continuous, then since we know that f is continuous on the interval $[a, b]$, then $m(x)$ is continuous.

(b) If this function is continuous, then $f([a, b]) = [c, d]$. Since the function is monotonic, then the function inverse is defined, and since all of the inputs are distinct, the function inverse is injective and surjective.

Suppose we have an input x_0 and output y_0. Since f is continuous, as $x \to x_0$, $f(x) = y \to f(x_0) = y_0$. Therefore, as $y \to y_0$, $f^{-1}(y) = x \to f^{-1}(y_0) = x_0$, so f^{-1} is continuous, as desired.

(c) Consider the function that maps $[0,1)$ to $[0,1)$ and $(1,2]$ to $(1,2]$. Then the function is monotonic but it has one discontinuity, which is 1.

Observe that X and Y are not necessarily open intervals, so there may be an element $x \in X$ which is equal to the greatest lower bound or least upper bound. Then this boundary point might not be necessarily continuous.

8

(a) Consider the graph of the function $g(x) = x$ in the coordinate plane. $g(0) = 0$ and $g(1) = 1$. We assume that $f(0) > 0$ and $f(1) < 1$ because otherwise we would already have a value of x that satisfies $f(x) = x$. Then $f(0) > g(0)$ and $f(1) < g(1)$, so by the previous part, f and g must intersect at a point x. Since this point lies on g, it satisfies $f(x) = x$, so we're done.

(b) Consider $f(x) = 2^x$ and $g(x) = x$ on the interval $[0, \infty)$. Then $f(x)$ has no fixed points, so they do not always have a common fixed point. Let $f(x) = x - 1$. Obviously there is no fixed point, period.

(c) Consider $f(x) = 2^x - 1$. Note that $f(0) = 0$ and $f(1) = 1$, but since this is an exponential function, $x > 2^x - 1$ for $0 < x < 1$, so this does not have a fixed point. Let $g(x) = f(f(x)) - f(x)$. Since $g(0) = g(1) = 0$ and f is continuous, then g must also be continuous. Thus it is impossible for g to be nonzero (it can't increase then decrease), so $g(x) = 0$ for all x. Therefore $f(x) = f(f(x)) = x$, as desired.

5.6 Examples of Differential Calculus in Natural Science

1

(a) In the case of solid fuel, $\alpha = \frac{\omega^2}{2Q} = \frac{2^2}{2000} = \frac{1}{500}$. In the case of liquid fuel, $\alpha = \frac{3^2}{5000} = \frac{9}{5000}$.

(b) We know that the efficiency equals $\frac{m_R \frac{v^2}{2}}{m_F Q}$, and from part (a), we know that $\frac{1}{2}\omega^2 = \alpha Q$, so $Q = \frac{\frac{1}{2}\omega^2}{\alpha}$. Then the efficiency is

$$\frac{m_R \frac{v^2}{2}}{m_F \left(\frac{1}{2}\omega^2/\alpha\right)} = \frac{m_R v^2 \alpha}{m_F \omega^2}.$$

(c) It's equal to $\frac{m_R \cdot 60^2}{m_F \cdot 3^2} = \frac{400 m_R}{m_F}$.

(d) Simply $\frac{m_R \cdot v^2}{9 m_F}$.

(e) Suppose the function f is continuous on the interval $[a, b]$. Then for $x \in [a, b]$, $f(x)$ must be bounded because it's continuous; we cannot have $f(x) = \infty$ or $-\infty$ because otherwise $f(x)$ would not be continuous or closed. If this asymptote/limit occurs in the middle of a and b, f is not continuous. If this occurs at a or b, the domain of f would not be closed.

(f) Observe that the bounds of f are achieved because if they aren't, the range of f isn't closed. Note that if the minimum/maximum occurs inside the range $[a, b]$, it obviously equals the greatest lower bound/least upper bound. If this minimum/maximum occurs at the endpoints, suppose WLOG $\inf = f(a)$. Then if inf is not achieved, since f is not continuous, neither is a, so we're done.

2

(a) From section 5.6.2, we already know that $p = \rho \cdot \frac{R}{M}T$ which equals $\frac{R}{V}T$, so the correction term would be $+\frac{R}{V}\Delta T$.

(b) $p = p_0 e^{-(g/\lambda)h} + \frac{R}{V}T$, so we can just plug in the respective values of T.

(c) Let $g(x) = f^n(x) - f(x)$. Observe that $g(0) = g(1) = 0$ and f and $f^n(x)$ are continuous, so $g(x)$ is also continuous. Therefore g must equal zero because otherwise, if x increases, then $f(x) \geq x$, so $f^2(x) = f(f(x)) \geq f(x)$, etc., so $f^n(x) > f(x)$. Therefore $g(x)$ increases, so it is impossible for $g(1) = g(0)$. Therefore $g(x) = 0$ for all x and hence $f(x) = x$ for all x.

(d) From the previous part, we already have $f^n(x) \geq f^{n-1}(x) \geq \cdots \geq f(x)$. If $f(x) = x$, then all of these are equal, x is a fixed point, and we're done.

(e) Otherwise, if $f^k(x) = f^{k-1}(x)$, we must have $f(x) = x$, so assume $f^n(x) > f^{n-1}(x) > \cdots > f(x)$. Then $f^n(x)$ increases as n increases, but since f maps from $[0, 1]$ to $[0, 1]$, $f^n(x) < 1$, so $\lim_{n\to\infty} f^n(x) - f^{n-1}(x) = 0$. Therefore since $f^{n-1}(x)$ approaches $f^n(x)$, $f^n(x)$ tends to a fixed point.

3

(a) We know that $T = \frac{\ln 2}{\lambda}$, so $\lambda = \frac{\ln 2}{T}$. Then $N(T) = N_0 e^{\lambda t} = N_0 e^{-\ln 2/T \cdot t} = N_0 \cdot 2^{-t/T}$, so $m = (m + r)2^{-t/T}$ and thus $-\frac{t}{T} = \log_2 \frac{m}{m+r}$, so $t = T \log_2 \left(1 + \frac{r}{m}\right)$.

(b) Assume that f is uniformly continuous on E. Then for every $\epsilon > 0$, there exists a $\delta > 0$ such that $|x_1 - x_2| < \delta$ implies that $|f(x_1) - f(x_2)| < \epsilon$. Since $\omega(\delta)$ is a supremum, then there exists $\delta > 0$ such that $\omega(\delta) < \epsilon$. Then $\lim_{\delta\to 0} \omega(\delta) = \omega(+0) = 0$.

(c) Assume that $\omega(+0) = 0$. Then $\lim_{\delta\to 0} \omega(\delta) = 0$, so for any $\epsilon > 0$, there exists $\delta > 0$ such that $\omega(\delta) = \epsilon$. If $|x_1 - x_2| < \delta$, we have $|f(x_1) - f(x - 2)| \leq \omega(\delta) < \epsilon$. Thus f is uniformly continuous on E, so a function f is uniformly continuous on E if and only if $\omega(+0) = 0$.

(d) Note that since $f(x)$ is continuous over an interval $[a, b]$, by the extreme value theorem, the maximum and minimum value of $f(x)$ must exist on this interval. Similarly for $g(x)$ (and note the extreme value function only applies for continuous functions so this wouldn't work in general for arbitrary bounded functions).

(e) Therefore, the maximum value of $|f(x_0) - g(x_0)|$ occurs in the set X, because if it didn't exist, this would imply that $|f(x) - g(x)|$ has no maximum value, which would imply that $f(x)$ or $g(x)$ has no maximum/minimum value. One of $f(x)$ or $g(x)$ can't be increasing/decreasing in the middle of the interval if it's not approaching its bound, so we're done.

4

(a) Observe that $I(l + h) - I(l) = -\lambda I(l)$, for some λ in terms of $I(l)$. Observe that $I(l + h) = I(l) \cdot I(h)$, since having one layer $l + h$ thick is the same as having two layers, which are l and h thick. So, it must be an exponential function. Then $I(l) = ce^{-kl}$ for some c and k. Plugging in $l = 0$ gives $I_0 = c$, so $I(l) = I_0 e^{-kl}$ for some k.

(b) We assume that the water is the same throughout, since it's pure, so there's no muddy water at the bottom. We also know $l = 10$ meters, so $I = I_0 e^{-10k}$. Then we can plug in k and done!

5

(a) Since $mx'' + kx = 0$, then $x'' = -\frac{k}{m}x$. We know that the total mechanical energy of an isolated system is constant, and that the kinetic energy equals $\frac{1}{2}mv^2 = \frac{m\dot{x}^2(t)}{2}$, and that the potential energy is $\frac{mx^2(t)}{2}$, so add them together to get $E = \frac{m\dot{x}^2(t)}{2} + \frac{mx^2(t)}{2}$ is constant.

(b) Since velocity equals 0, then we know that position, which is the integral of velocity, is some constant. Then $x(t) = c$, but $x(0) = 0 = c$, so $x(t) = 0$.

(c) If we already have $x(0) = x_0$ and $\dot{x}(0) = v_0$, then we automatically have $\ddot{x}(0) = a_0$. Then there exists $x = x(t)$ such that $m\ddot{x} + kx = 0$.

(d) Since we have friction, then energy is being taken away since the velocity must decrease, so E decreases.

6

(a) Observe that if a polynomial has complex roots (not real roots), they must come in conjugate pairs, so there are an even number of imaginary roots. The number of roots a polynomial (repeats count) has is the same as the degree of the polynomial, so there are an odd number of roots. Therefore there must be at least one real root.

(b) Given a fixed polynomial $P_n(x)$, we want to find a polynomial of the form $Q_\lambda(x) = \lambda P_n(x)$ such that $\Delta(Q_{\lambda_0}) = \min_{\lambda \in \mathbb{R}} \Delta(Q_\lambda)$. $\Delta(Q_\lambda)$ is continuous with respect to λ, so there exists $\lambda_0 \in \mathbb{R}$ that minimizes $\Delta(Q_\lambda)$ in the interval $[a, b]$, implying the existence of a polynomial of best approximation of the form $\lambda P_n(x)$.

(c) Let $P_n(x)$ be the polynomial of best approximation of degree n, and let $\Delta_n = \Delta(P_n) = E_n(f)$. We show that there exists a polynomial of best approximation of degree $n+1$ by adding a term of degree $n+1$ to $P_n(x)$, giving

$$P_{n+1}(x) = P_n(x) + c(x-a)^{n+1}$$

for some constant c, so we're done.

(d) We know that there exists a polynomial of best approximation of degree 0, and we know that if there exists a polynomial of best approximation of degree n, there is also one of degree $n+1$, so we're done by induction.

7

(a) The sign function is:

$$\operatorname{sgn} x = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x = -1 \end{cases}$$

Therefore, if on some interval, $P_n(x)$ is the same sign, then $\operatorname{sgn} P_n(x)$ will be continuous on that interval. Therefore, sgn is only not continuous when $P_n(x)$ changes sign, so basically when $P_n(x) = 0$, so when x is a root of P_n. We know there are at most n roots (since there are double roots too), so there are at most n points of discontinuity.

(b) I'm not sure how the fact that $\operatorname{sgn}[(f(x_i) - P_n(x_i))(-1)^i]$ comes into play, but if we let $g(x) = f(x) - P_n(x)$, we get: either $g(x_0)$ is negative, $g(x_1)$ is positive, $g(x_2)$ is negative, etc. or, $g(x_0)$ is positive, $g(x_1)$ is negative, $g(x_2)$ is positive, etc. or, $g(x_0) = g(x_1) = \cdots = g(x_{n+1}) = 0$. Therefore $g(x)$ must have at least $n+1$ roots, since the sign changes $n+1$ times.

So we try to show that

$$E_n(f) = \inf_{P_n} \sup_{x\in[a,b]} |g(x_i)| \geq \min_{0\leq i\leq n+1} |g(x_i)|.$$

Since $|g(x_i)| \leq \sup_{x\in[a,b]} |g(x_i)|$, we're done.

(c) By Demoivre's theorem,

$$\text{cis}nx = \cos nx + i \sin nx = (\cos x + i \sin x)^n.$$

So $\cos nx$ is equal to the real part, which is

$$\cos nx = \cos^n x - \binom{n}{2} \cos^{n-2} x \sin^2 x + \binom{n}{4} \cos^{n-4} x \sin^4 x - \cdots + \ldots.$$

(d) Obviously if the degree of $\sin x$ is odd, then it wouldn't be real, so $\sin x$ is always raised to an even power. Since $\sin^2 x = 1 - \cos^2 x$, we can express $\sin^{2k} x$ in terms of $\cos x$, so $\cos nx$ can be expressed as an algebraic polynomial in terms of $\cos x$, so we're done.

5.7 Primitives

1

(a) We have

$$\frac{P(x)}{Q(x)} - \frac{p(x)}{q(x)} = \frac{p(x)}{q(x)} \left(\frac{P_1(x)}{Q_1(x)} - 1\right),$$

so taking the integral gives

$$\int \frac{P(x)}{Q(x)} \mathrm{d}x - \int \frac{p(x)}{d(x)} \mathrm{d}x = \int \frac{p(x)}{q(x)} \left(\frac{P_1(x)}{Q_1(x)} - 1\right) \mathrm{d}x = \frac{P_1(x)}{Q_1(x)}.$$

(b) We have

$$\begin{aligned}
\left(\frac{P_1(x)}{Q_1(x)}\right)' &= \frac{P_1'(x)Q_1(x) - P_1(x)Q_1'(x)}{Q_1(x)^2} \\
&= \frac{\left(\frac{P(x)}{p(x)}\right)' \frac{Q(x)}{q(x)} - \frac{P(x)}{p(x)} \left(\frac{Q(x)}{q(x)}\right)'}{\left(\frac{Q(x)}{q(x)}\right)^2} \\
&= \frac{\left(\frac{P'(x)p(x)-P(x)p'(x)}{p(x)^2}\right) \frac{Q(x)}{q(x)} - \frac{P(x)}{p(x)} \left(\frac{Q'(x)q(x)-Q(x)q'(x)}{q(x)^2}\right)}{\left(\frac{Q(x)}{q(x)}\right)^2} \\
&= \frac{\frac{Q(x)(P'(x)p(x)-P(x)p'(x))}{p(x)} - \frac{P(x)(Q'(x)q(x)-Q(x)q'(x))}{q(x)}}{Q(x)^2 \cdot \frac{p(x)}{q(x)}} \\
&= \frac{q(x)Q(x)(P'(x)p(x) - P(x)p'(x)) - p(x)P(x)(Q'(x)q(x) - Q(x)q'(x))}{Q(x)^2 \cdot p(x)q(x)}.
\end{aligned}$$

Adding $\frac{p(x)}{q(x)}$ to this mess gives $\frac{P(x)}{Q(x)}$, as desired.

(c) Observe that if r is a double root of $Q(x)$, then $Q'(r) = 0$, but the multiplicity of r for Q' will be one less than that for Q. Then we can use the Euclidean algorithm to find the greatest common divisor between Q' and Q, which equals $Q_1(x)$. Then we can find $q(x)$ by $\frac{Q(x)}{Q_1(x)}$.

(d) We have

$$Q'(x) = 7x^6 + 18x^5 + 25x^4 + 28x^3 + 21x^2 + 10x + 3,$$

so by the euclidean algorithm we obtain $Q_1(x) = x^4 + 2x^3 + 2x^2 + 2x + 1$, and similarly we can obtain $P_1(x)$.

2

(a) Since $R(-u, v) = R(u, v)$, then R is an even function at least for u, so every u term must be an even function. Therefore, since this is a rational function, then it's made up of the sum of the polynomials, so they must be of the form $u^{2a}v^b$. Thus we can express this in terms of $(u^2)^a v^b$, so this can be expressed in the form $R_1(u^2, v)$. $f(x_i) - P_n(x_i) = (-1)^i \Delta(P_n) \cdot \alpha$ for $i = 0, \ldots, n+1$, $\Delta(P_n) = \max_{x\in[a,b]} |f(x) - P_n(x)|$. In order to show that $P_n(x)$ is the unique polynomial of best approximation, if we let $Q(x)$ be a polynomial of degree n, then it suffices to show that $\max_{x\in[a,b]} |f(x) - Q(x)| \geq \Delta(P_n)$.

(b) Let $R(x) = f(x) - Q(x)$, so it suffices to show that $\max_{x\in[a,b]} |R(x)| \geq \Delta(P_n)$. Assume there's some alternate sequence for x_i, so we have

$$R(x_i) = f(x_i) - Q(x_i) = (-1)^i \Gamma, \qquad i = 0, \ldots, n+1$$

where $\Gamma = \max_{x\in[a,b]} |f(x) - Q(x)|$. Then

$$Q(x)i - P_n(x_i) = (-1)^i (\Delta(P_n) \cdot \alpha - \Gamma)$$

for $i = 0, \ldots, n+1$. Then $Q(x_i) - P_n(x_i)$ has the same sign as $f(x_i) - P_n(x_i)$, so $\max_{x\in[a,b]} |Q(x) - P_n(x)| \geq \Delta(P_n)$, so we're done.

(c) By part (a), we already have the if part. It suffices to prove the only if part. By Problem 10, we already know that a polynomial of best approximation exists. We also know that $n+1$ alternating points are on the polynomial, and we can add another one by using the next hump, so we're done.

(d) Obviously, for discontinuous functions, there are sharp edges and disconnected edges such as the graph of sgn$x$. If we take the polynomial of degree 0 $P_n(x) = 0$, then the $n+2$ alternate points do not hold.

4

(a) Obviously, for a best approximation of degree 0 is $P_0(x) = 1$. A best approximation of degree 1 is of the form $P_1(x) = ax + b$. The most important points are the endpoints and corners, which is $(-1, 1)$, $(0, 0)$, $(2, 2)$. The vertical distance between P_1 and these points is $1 - (-a + b) = 1 + a - b$, b, and $2 - (2a + b) = 2 - 2a - b$, respectively. Note that the $\Delta(P_1)$ will be minimized when $1 + a - b = b = 2 - 2a - b$, so $a = \frac{1}{3}$ and $b = \frac{2}{3}$, so $P_1(x) = \frac{1}{3}x + \frac{2}{3}$. Then in this case, the distance will be $\frac{2}{3}$, and we're done.

(b) The requirements are closure, associativity, commutativity, existence of identity, and existence of inverse for addition. Observe that we just proved closure by the previous part and that associativity and commutativity are obvious because the underlying functions are continuous, associative, and commutative. The identity exists for both operations, being 0 for addition and 1 for multiplication. The inverse of addition is the negative of the function whose inverse we are looking for, however this does not necessarily exist for multiplication (consider, for example, $f(x) = 0$, which does not have an inverse).

5

(a) Obviously, since its degree is odd, it has to have a real root x_0, so we can let the polynomial $p(x)$ be equal to

$$p(x) = (x - x_0)(ax^2 + bx + c)$$

for real coefficients a, b, and c. Plugging in t^2 for $x - x_0$ and $t^2 + x_0$ for x, we get

$$p(x) = t^2(a(t^2 + x_0)^2 + b(t^2 + x_0) + c) = t^2(at^4 + (2ax_0 + b)t^2 + (ax_0^2 + bx_0 + c)),$$

as desired.

(b) $R(u, v)$ is some combination of u and v that includes products, then summing them all together. $R(x, \sqrt{P(x)})$ is some combination of $x = x_0 + t^2$ and $\sqrt{P(x)} = t\sqrt{at^4 + bt^3 + ct^2 + dt + e}$. A combination of $x_0 + t^2$ and $t\sqrt{at^4 + bt^3 + ct^2 + dt + e}$, say for $u^a v^b$ gives

$$(x_0 + t^2)^a \left[t\sqrt{at^4 + bt^3 + ct^2 + dt + e}\right]^b (x_0 + t^2)^a t^b \left[\sqrt{at^4 + bt^3 + ct^2 + dt + e}\right]^b.$$

Use distributive, then the terms on the left side are in terms of t, and the ones on the right are in terms of $\sqrt{at^4 + bt^3 + ct^2 + dt + e}$, as desired.

(c) Since imaginary roots must come in conjugates, then obviously they can come together to form quadratics, so a fourth degree polynomial must be able to be represented as the product of two quadratics, so it must equal $a(x^2 + p_1x + q_1)(x^2 + p_2x + q_2)$. Then plugging in $x = \frac{\alpha t + \beta}{\gamma t + 1}$ and simplifying gives an equation of the form $\frac{(M_1 + N_1t^2)(M_2 + N_2t^2)}{(\gamma t + 1)^2}$, as desired.

(d) Since we know that $ax^4 + bx^3 + \cdots + e$ can be expressed in the form

$$\frac{(M_1 + N_1t^2)(M_2 + N_2t^2)}{(\gamma t + 1)^2} = \frac{M_1M_2(1 + \frac{N_1}{M_1}t^2)(1 + \frac{N_2}{M_2}t^2)}{(\gamma t + 1)^2}.$$

Thus the square root of this is of the form $\frac{\sqrt{A(1 + m_1t^2)(1 + m_2t^2)}}{\gamma t + 1}$, and when combined with x which is also a combination of t, we get a combination of t and $\sqrt{A(1 + m_1t^2)(1 + m_2t^2)}$, as desired.

(e) $R(x, \sqrt{y})$ will consist of terms in the form $kx^a(\sqrt{y})^b$. b is either even or odd, so the terms will be of 2 forms: $kx^a y^{b/2}$ is b is even, or $kx^a y^{\frac{b-1}{2}}\sqrt{y}$ is b is odd. If we let $R_1(x, y)$ be all of the terms with b even, and $R_2(x, y)$ be all of the terms with b odd times $\sqrt{y}$, so that it would be a valid rational function, then we're done.

(f) Consider a subring I consisting of all functions f such that $f(a) = 0$. Then if g is an element of the ring K, then $(f \cdot g)(a) = f(a) \cdot g(a) = 0$, which is in I, so I is an ideal, as desired.

(g) One maximal ideal is the set of continuous functions that equal 0 at some point in the interval. Obviously if $f(x) = 0$ for some $x \in [a, b]$, then $(f \cdot g)(x) = f(x) \cdot g(x) = 0$ is obviously an element of this ring. There is no other maximal ideal because 0 is the only thing that nullifies whatever is thrown at it (being the elements of $C[a, b]$ that are not actually part of the maximal ideal).

(h) Consider the term that is of the form $cx^a(\sqrt{y})^b$. We have two cases. If b is even, then we have a purely integral term for all of the exponents. Let the sum of all these terms be $R_1(x, y)$. If b is odd, then we get a leftover $\sqrt{y}$ product. In this latter case, we have another two cases. If a is even, then we have the $\frac{R_2(x^2,y)}{\sqrt{y}}$, and if a is odd, we have the $\frac{R_3(x^2,y)}{\sqrt{y}}x$ term. Putting these terms altogether, we get

$$R(x, \sqrt{y}) = R_1(x, y) + \frac{R_2(x^2, y)}{\sqrt{y}} + \frac{R_3(x^2, y)}{\sqrt{y}}x.$$

(i) We already know that $R(x, \sqrt{P(x)})$ can be reduced to the form

$$R_1\left(t, \sqrt{A_1(1 + m_1t^2)(1 + m_2t^2)}\right)$$

by part (d). This expression can also be reduced to the form

$$\frac{r(t^2)}{\sqrt{A_2(1 + m_1t^2)(1 + m_2t^2)}}$$

since its a rational function and we have the same base terms (like for $R(u, v)$, the base terms are u and v). Put an integral around both of them and we're done.

(j) Consider, for example, that $E_1 = [-1, 0]$ and $E_2 = [0, 1]$. Let $f|E_1 = x$ and $f|E_2 = x + 1$. Then $f|E_1(0) = 0$ and $f|E_2(0) = 1$, which is not continuous, so F is not continuous on $E_1 \cup E_2$. If $f \in C(A)$, then f is a continuous function defined on A. Since $B \subset A$, then f is also continuous on B, so $f|_B \in C(B)$, as desired.

(k) Observe that the Riemann function $\mathcal{R}$ is discontinuous because there is always an irrational number between any two rational numbers. The irrational number gives 0 but the rationals are nonzero, so this is discontinuous. The Riemann function defined on only the rational $\mathbb{Q}$'s is also irrational. Take, for example, all of the rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$. $\mathcal{R}(2/5) = \mathcal{R}(3/5) = \frac{1}{5}$ but the rational numbers in between do not approach $\frac{1}{5}$. All of these points of discontinuity are removable by removing the rational numbers, leaving us with a function that outputs 0 for each input.

(l) We already know that any elliptic integral of the form $\int R(x, \sqrt{P(x)})\mathrm{d}x$ can be expressed in the form

$$\int \frac{r(t^2)\mathrm{d}t}{\sqrt{A(1 + m_1t^2)(1 + m_2t^2)}}.$$

Since the standard forms are also of this form, we're done.

(m) Suppose a number $x \in \mathbb{R}$'s base q representation has periodically repeating digits. Separate the repeating part from the non-repeating part. For example, if we have $x = 0.1232323\ldots$, we can separate this into $a = 0.1$ and $y = 0.0\overline{23}$. Then we can multiply x by q^n to get $x \cdot q^n - (x - a)$ is equal to a rational number. Therefore we can solve for x, which is a rational number, and then convert the numerators and denominators into standard base.

(n) The heights are h, qh, q^2h, etc. Note that if the initial or final velocity is v, then the final or initial velocity is 0, so $v^2 = 2gy$. This means that $v = \sqrt{2gy}$, so $t = \sqrt{\frac{2y}{g}}$. Thus the time in total is

$$\sqrt{\frac{2h}{g}} + \sqrt{\frac{2qh}{g}} + \sqrt{\frac{2q^2h}{g}} + \cdots = \sqrt{\frac{2h}{g}}(1 + \sqrt{q} + \sqrt{q^2} + \cdots) = \frac{\sqrt{\frac{2h}{g}}}{1 - \sqrt{q}}.$$

The total distance traveled is $h + qh + q^2h + \cdots = \frac{h}{1-q}$.

6

(a) We mark all the points on a cicle obtained from a fixed point by rotations of the circle through angles of n radians, where $n \in \mathbb{Z}$ ranges over all integers. This way, every single point on the circle will eventually be marked because we need to find the possible values of $n \pmod{2\pi}$ span the entirety of the range $[0, 2\pi)$.

(b) This gives

$$\frac{m}{n} = q_1 + \frac{1}{q_2 + \frac{1}{r_1/r_2}}.$$

Then $r_1/r_2 = q_3 + \frac{r_3}{r_2} = q_3 + \frac{1}{r_2/r_3}$, so we get

$$\frac{m}{n} = q_1 + \frac{1}{q_2 + \frac{1}{q_3 + \frac{1}{r_2/r_3}}}.$$

Because the m and n are relatively prime, then there will be n equations, so there will be n of those nested fractions.

(c) Observe that a rational number has a unique continued fraction representation as shown by the Euclidean algorithm analogy. Let the continued fraction equal x. Note that $x = 1 + \frac{1}{x}$, so $x^2 - x - 1 = 0$. Observe that the solution to this is $x = \frac{1 \pm \sqrt{5}}{2}$. Since x is positive, then we must have $x = \frac{1+\sqrt{5}}{2}$, as desired.

(d) Since $u_n = u_{n-1} + u_{n-2}$, the characteristic equation is $c^2 - c - 1$, so we have roots $c_1 = \frac{1+\sqrt{5}}{2}$ and $c_2 = \frac{1-\sqrt{5}}{2}$. We must solve the equation $F_n = a_1(c_1)^n + a_2(c_2)^n$. Letting $n = 1$ and 2, the system of equations gives

$$1 = a_1\left(\frac{1+\sqrt{5}}{2}\right) + a_2\left(\frac{1-\sqrt{5}}{2}\right)$$
$$1 = a_1\left(\frac{1+\sqrt{5}}{2}\right)^2 + a_2\left(\frac{1-\sqrt{5}}{2}\right)^2 = a_1\left(\frac{3+\sqrt{5}}{2}\right) + a_2\left(\frac{3-\sqrt{5}}{2}\right).$$

(e) Subtracting the first equation from the second gives $a_1 + a_2 = 0$, so $a_2 = -a_1$. Plugging this into the first equation gives

$$1 = a_1 \left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} \right) = a_1\sqrt{5},$$

so $a_1 = \frac{1}{\sqrt{5}}$ and $a_2 = -\frac{1}{\sqrt{5}}$. This gives

$$u_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n,$$

as desired.

(f) We know that

$$\begin{aligned}
\int e^{ax} \cos bx \mathrm{d}x &= \frac{1}{a} \int \cos bx \mathrm{d}e^{ax} \\
&= \frac{1}{a} e^{ax} \cos bx - \frac{1}{a} \int e^{ax} \mathrm{d} \cos bx \\
&= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} \int e^{ax} \sin bx \mathrm{d}x \\
&= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} \int \sin bx \mathrm{d}e^{ax} \\
&= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} e^{ax} \sin bx - \frac{b}{a^2} \int e^{ax} \mathrm{d} \sin bx \\
&= \frac{a \cos bx + b \sin bx}{a^2} - \frac{b^2}{a^2} \int e^{ax} \cos bx \mathrm{d}x.
\end{aligned}$$

Thus we obtain

$$\int e^{ax} \cos bx \mathrm{d}x = \frac{a \cos b + b \sin bx}{a^2 + b^2} e^{ax}.$$

(g) Since $u_n = u_{n-1} + u_{n-2}$, the characteristic equation is $c^2 - c - 1$, so we have roots $c_1 = \frac{1+\sqrt{5}}{2}$ and $c_2 = \frac{1-\sqrt{5}}{2}$. subtracting the first equation from the second gives $a_1 + a_2 = 0$, so $a_2 = -a_1$. Plugging this into the first equation gives

$$1 = a_1 \left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} \right)$$

(h) Then

$$\begin{aligned}
S_2(n+1) &= S_2(n) + (n+1)^2 \\
&= \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n + (n+1)^2 = \frac{1}{3}n^3 + \frac{3}{2}n^2 + \frac{13}{6}n + 1 \\
&= \frac{1}{3}(n+1)^3 + \frac{1}{2}(n+1)^2 + \frac{1}{6}(n+1)
\end{aligned}$$

So we're done by induction. Similarly,

$$\begin{aligned}
S_3(n+1) = S_3(n) + (n+1)^3 &= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\
&= (n+1)^2 \left(\frac{n^2}{2} + (n+1) \right) = \frac{(n+1)^2(n+2)^2}{4}
\end{aligned}$$

as desired.

(i) Observe that any exponential function satisfies this. Take, for example, $f(x) = e^x$. Then $f(1) = e$ which is definitely positive and not equal to 1. $f(x_1) \cdot f(x_2) = e^{x_1} \cdot e^{x_2} = e^{x_1+x_2} = f(x_1+x_2)$. And since this function is continuous, $f(x) \to f(x_0)$ as $x \to x_0$.

Also, observe that any logarithmic function satisfies this. Take, for example, $f(x) = \ln x$. Then $f(e) = \ln e = 1$, and $e > 0$ and $e \neq 1$. Then $f(x_1) + f(x_2) = \ln x_1 + \ln x_2 = \ln(x_1 \cdot x_2) = f(x_1 \cdot x_2)$ and $f(x) \to f(x_0)$ for $x \to x_0$ since ln is a continuous function, as desired.

7

(a) It suffices to prove that $\frac{e^x}{x} = \frac{1}{\ln x}$. Then $x = \ln e^x$, so $\frac{e^x}{\ln e^x} = \frac{1}{\ln x}$, so it suffices to prove that

$$\int \frac{\mathrm{d}e^x}{\ln e^x} = \int \frac{\mathrm{d}x}{\ln x},$$

which is true by substitution.

(b) It suffices to prove that

$$\int \frac{\cosh x}{x}\mathrm{d}x = \frac{\int \frac{e^x}{x} + \int \frac{e^{-x}}{-x}}{2} = \int \frac{\frac{e^x - e^{-x}}{2}}{x}\mathrm{d}x.$$

So it suffices to show that $\cosh x = \frac{e^x - e^{-x}}{2}$, and we already know that, so we're done.

(c) It suffices to prove

$$\int \frac{\sinh x}{x}\mathrm{d}x = \frac{\int \frac{e^x}{x}\mathrm{d}x - \int \frac{e^{-x}}{-x}\mathrm{d}x}{2} = \int \frac{e^x + e^{-x}}{2x}\mathrm{d}x.$$

So it suffices to prove that $\sinh x = \frac{e^x + e^{-x}}{2}$, so we're done.

(d) We have

$$\mathrm{Ei}(ix) = \int \frac{e^{ix}}{ix}\mathrm{d}(ix) = \int \frac{e^{ix}}{x}\mathrm{d}x$$

equals

$$\mathrm{Ci}(x) + i\mathrm{Si}(x) = \int \frac{\cos x}{x}\mathrm{d}x + i\int \frac{\sin x}{x}\mathrm{d}x = \int \frac{\cos x + i\sin x}{x}\mathrm{d}x.$$

Thus it suffices to show that $e^{ix} = \cos x + i\sin x$, so we're done.

(e) Observe that

$$\begin{aligned} e^{i\pi/4}\Phi(xe^{-i\pi/4}) &= e^{i\pi/4}\int e^{-(xe^{-i\pi/4})^2}\mathrm{d}(xe^{-i\pi/4}) \\ &= e^{i\pi/4}\int e^{-x^2e^{-i\pi/2}} \cdot e^{-i\pi/4} \cdot \mathrm{d}x \\ &= \int e^{x^2i}\mathrm{d}x \end{aligned}$$

and

$$C(x) + iS(x) = \int \cos x^2\mathrm{d}x + i\int \sin x^2\mathrm{d}x = \int (\cos x^2 + i\sin x^2)\mathrm{d}x.$$

Thus since $e^{x^2i} = \cos x^2 + i\sin x^2$, then $e^{i\pi/4}\Phi(xe^{-i\pi/4}) = C(x) + iS(x)$, as desired.

8

(a) $2x^3yy' = 2 - y^2$, so $\frac{2yy'}{2-y^2} = \frac{1}{x^3}$. Thus

$$\begin{aligned}\int \frac{2y}{2-y^2}\mathrm{d}y &= \int \frac{1}{x^3}\mathrm{d}x \\ -\int \frac{\mathrm{d}(2-y^2)}{2-y^2} &= \int x^{-3}\mathrm{d}x \\ -\ln|2-y^2| &= -\frac{x^{-2}}{2} + c \\ |2-y^2| &= cx^{x^{-2}/2} \\ y^2 = 2 - cx^{1/2x^2}.\end{aligned}$$

(b) We get $yy' = \frac{\sqrt{1+x^2}}{x}$, so

$$\int y\mathrm{d}y = \int \frac{\sqrt{1+x^2}}{x}\mathrm{d}x.$$

Then

$$\int \frac{1}{y}\mathrm{d}y = \int x\sqrt{1+x^2}\mathrm{d}x,$$

so $\ln y = \frac{1}{3}(1+x^2)^{3/2} + c$, so $y = ce^{\frac{1}{3}(1+x^2)^{3/2}}$.

(c) Let $u = y + x$, so $y' = \cos u$. Then

$$\int 1\mathrm{d}y = \int \cos u\mathrm{d}u = \sin u,$$

so $y = \sin u = \sin(y + x)$. $y + x = \arcsin y$, so $x = \arcsin y - y$.

(d) We have $x^2y' = 1 + \cos 2y$, so $\frac{y'}{1+\cos 2y} = \frac{1}{x^2}$. Then

$$\int \frac{\mathrm{d}y}{1+\cos 2y} = \int \frac{1}{x^2}\mathrm{d}x.$$

Since $1 + \cos 2y = 1 + (2\cos^2 y - 1) = 2\cos^2 y$, then

$$\int \frac{\mathrm{d}y}{2\cos^2 y} = \frac{1}{2}\tan y = -x + c.$$

Then $y = \arctan(-2x + c)$. As $x \to \infty$, $\arctan(-2x) = \frac{-\pi}{2}$.

(e) Observe that the function $\phi(x) = e^x$ works. Note that $\phi(x)\cdot\phi(y) = e^x \cdot e^y = e^{x+y} = \phi(x + y)$ and that the exponential function is always positive, so ϕ maps the set of real numbers to the set of positive real numbers.

(f) It suffices to prove that there does not exist a function $\varphi(x + y) = \varphi(x)\cdot\varphi(y)$ mapping $x + y \in \mathbb{R}$ to $\varphi(x)\cdot\varphi(y) \in \mathbb{R}\backslash 0$. Observe that $\mathbb{R}_0$ is a subset of $\mathbb{R}$ because there is an obvious injective mapping from $\mathbb{R}\backslash 0$ to $\mathbb{R}$, but there is no mapping in the opposite direction.

9

(a) Suppose the force of air resistance, f_c, equals $-kv$ for some constant k, since its proportional to v. Then $F_{\text{net}} = F_g + f_c = mg - kv = ma = m\frac{\mathrm{d}v}{\mathrm{d}t}$, so

$$\frac{m \cdot \mathrm{d}v}{mg - kv} = \mathrm{d}t \implies \frac{\mathrm{d}v}{\frac{mg}{k} - v} = \frac{k}{m} \cdot \mathrm{d}t.$$

Therefore

$$\int \frac{dv}{\frac{mg}{k} - v} = \int \frac{k}{m} \cdot \mathrm{d}t,$$

so $-\ln\left|\frac{mg}{k} - v\right| = \frac{k}{m}t + c$. Then $\frac{mg}{k} - v = ce^{-\frac{k}{m}t}$ for some constant c, so $v = \frac{mg}{k} - ce^{-\frac{k}{m}t}$. Since $\lim_{t\to\infty} v(t) = 50$, then $\frac{mg}{k} = 50$. Then plugging in $t = 0$ gives $v(0) = 0$, so $c = 50$. Then we get $v(t) = 50(1 - e^{-gt/50})$. Since

$$x(t) = \int v(t)\mathrm{d}t = 50 \int (1 - e^{-gt/50})\mathrm{d}t = 50(t + \frac{50}{g}e^{-gt/50}) = 1000,$$

so $t + \frac{50}{g}e^{-gt/50} = 20$. Solving gives $t = 19.9$ seconds.

(b) Suppose $f_c = -kv^2$, so $F_{\text{net}} = F_g + f_c = mg - kv^2 = ma = m\frac{\mathrm{d}v}{\mathrm{d}t}$, so $\frac{\mathrm{d}v}{\frac{mg}{k} - v^2} = \frac{k}{m}\mathrm{d}t$. Then

$$\int \frac{\mathrm{d}v}{\frac{mg}{k} - v^2} = \int \frac{k}{m}\mathrm{d}t \implies \frac{k}{m}t + c = \frac{1}{2\sqrt{\frac{mg}{k}}} \ln\left(\frac{\sqrt{\frac{mg}{k}} + v}{\sqrt{\frac{mg}{k}} - v}\right).$$

Thus

$$\begin{aligned}
\ln\left(\frac{\sqrt{\frac{mg}{k}} + v}{\sqrt{\frac{mg}{k}} - v}\right) &= 2\sqrt{\frac{mg}{k}} \cdot \frac{k}{m}t + c = 2\sqrt{\frac{kg}{m}}t + c \\
\frac{\sqrt{\frac{mg}{k}} + v}{\sqrt{\frac{mg}{k}} - v} &= ce^{2\sqrt{\frac{kg}{m}}t} \\
\sqrt{\frac{mg}{k}} + v &= c\sqrt{\frac{mg}{k}}e^{2\sqrt{\frac{kg}{m}}t} - ce^{2\sqrt{\frac{kg}{m}}t}v \\
\left(ce^{2\sqrt{\frac{kg}{m}}t} + 1\right)v &= \sqrt{\frac{mg}{k}}\left(ce^{2\sqrt{\frac{kg}{m}}t} - 1\right) \\
v &= \sqrt{\frac{mg}{k}} \cdot \frac{ce^{2t\sqrt{\frac{kg}{m}}} - 1}{ce^{2t\sqrt{\frac{kg}{m}}} + 1}.
\end{aligned}$$

(c) As $t \to \infty$, the right factor approaches 1, so $\sqrt{\frac{mg}{k}} = 50$. If $t = 0$, then $v(t) = 0$, so $c = 1$. Then $\sqrt{\frac{kg}{m}} = \frac{g}{50}$, and

$$v(t) = 50 \cdot \frac{e^{gt/25} - 1}{e^{gt/25} + 1}.$$

Thus

$$d(t) = \int v(t)\mathrm{d}t = 50 \int_{t=0}^{x} \frac{e^{gt/25} - 1}{e^{gt/25} + 1} = 1000,$$

so solving gives $x = 23.5$ seconds.

10 Suppose $x < 1$, and let $x = \frac{1}{a}$. Then $x^x = \frac{1}{a^{1/a}} = \frac{1}{\sqrt[a]{a}}$. As $x \to +0$, $a \to \infty$, and it's easy to see that $\sqrt[a]{a} \to 1$ as $a \to \infty$, so the limit is 1. This is $\sqrt[x]{x}$, which approaches 1 as x approaches infinity because the exponential function grows faster than linearly.

$$\begin{aligned}\lim_{x\to 0}\frac{\log(1+x)}{x} &= \lim_{x\to 0}\left(\frac{x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots}{x}\right)\\ &= \lim_{x\to 0}\left(1 - \frac{x}{2} + \frac{x^2}{3} - \cdots\right)\\ &= 1.\end{aligned}$$

Since a^x grows faster than x, it also shrinks faster, so this approaches 0 as x approaches 0.

11 Since we know that

$$\ln\frac{n+1}{n} = \ln\left(1 + \frac{1}{n}\right) = \frac{1}{n} + O\left(\frac{1}{n^2}\right) \qquad \text{as } n \to \infty,$$

then

$$\begin{aligned}\lim_{n\to\infty} 1 + \frac{1}{2} + \cdots + \frac{1}{n} &= \ln\left(\frac{2}{1}\right) + \ln\left(\frac{3}{2}\right) + \cdots + \ln\left(\frac{n+1}{n}\right) - O\left(\frac{1}{n^2}\right)\\ &= \ln\left(\frac{n+1}{1}\right) = \ln n + c + o(1),\end{aligned}$$

as desired.

12 Since $a_n \sim b_n$, then there exists $\gamma(x)$ such that $a(x) = \gamma(x)b(x)$ and $\lim_{\mathcal{B}}\gamma(x) = 1$. Therefore, because

$$\sum_{n=1}^{\infty} a_n = \gamma(x)\sum_{n=1}^{\infty} b_n$$

and due to the limit of $\gamma(x)$, then

$$\sum_{n=1}^{\infty} a_n \sim \sum_{n=1}^{\infty} b_n,$$

so they must either both converge or both diverge. Observe that $\sin\frac{1}{n^p} \sim \frac{1}{n^p}$ because

$$\lim_{n\to\infty}\frac{\sin\left(\frac{1}{n^p}\right)}{\frac{1}{n^p}} = \lim_{x\to 0}\frac{\sin x}{x} = 1.$$

Therefore $\sum_{n=1}^{\infty}\sin\frac{1}{n^p}$ converges if and only if $\sum_{n=1}^{\infty}\frac{1}{n^p}$ converges, which is only for $p > 1$.

Chapter 6: Integration

Zorich Mathematical Analysis

Carissa Zeng

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6.1 Definition of the Integral

1

(a) We know that

$$s(f;P) = \sum_{i=1}^{n} m_i \Delta x_i$$

and

$$S(f;P) = \sum_{i=1}^{n} M_i \Delta x_i.$$

Since $m_i = \inf_{x \in \Delta_i} f(x) \leq \sup_{x \in \Delta_i} f(x) = M_i$, then $s(f;P) \leq S(f;P)$, as desired.

(b) We know that since for $\tilde{P}$, we have Δx, M get smaller, so $S(f;P) - S(f;P) \geq 0$. But m gets larger, so $S(f;\tilde{P}) - S(f;P) \geq 0$. However, since these values are the same for intervals other than $\Delta x_{i_1}, \dots, \Delta x_{i_k}$, then those intervals don't matter. In each interval that does matter, the max difference is $\omega(f;[a,b])$, so we're done.

(c) We need to show that the maximum value of the smallest value is less than the minimum value of the largest value. Since, by definition, the largest value is bigger than the smaller value, then no matter how small the largest value gets, it's still bigger than the smaller value.

(d) We know that $\underline{I} = \sup_P s(f;P)$. Since we know that $s(f;P) \leq s(f;\tilde{P})$ always, and that $\lambda(P) \geq \lambda(\tilde{P})$, then the smaller $\lambda(P)$ becomes, the larger $s(f;P)$ becomes. Therefore, $s(f;P)$ is at its largest when $\lambda(P) \to 0$, and thus $\underline{I} = \lim_{\lambda(P)\to 0} s(f;P)$.

Similarly, we know that $S(f;P) \geq S(f;\tilde{P})$ always, and that $\lambda(P) \geq \lambda(\tilde{P})$. Thus as $\lambda(P) \to 0$, $S(f;P)$ gets smaller, so the maximum value of $S(f;P)$ occurs when $\lambda(P) \to 0$. And since $\overline{I} = \inf_P S(f;P)$, then $\overline{I} = \lim_{\lambda(P)\to 0} S(f;P)$.

(e) In order for f to be integrable, then the integrals from both sides need to be equal, so if $f \in \mathcal{R}[a,b]$, then $\underline{I} = \overline{I}$. For the other way, if $\underline{I} = \overline{I}$, then since the upper and lower integrals exist and are equal to each other, then f is integrable.

(f) From the previous part, $f \in \mathcal{R}[a,b]$ is equivalent to $\underline{I} = \overline{I}$. $\underline{I} = \overline{I}$ is equivalent to $\lim_{\lambda(P)\to 0} s(f;P) = \lim_{\lambda(P)\to 0} S(f;P)$, so

$$\lim_{\lambda(P)\to 0} (S(f;P) - s(f;P)) = 0.$$

By the formal definition of limit, we use some sort of epsilon-delta notation. For any ϵ such that $S(f;P) - s(f;P) < \epsilon$ (since we know that this must be positive, so we don't need to worry about magnitude), there exists δ such that there exists P for which this works and $\lambda(P) < \delta$. Basically, there exists P, so we're done.

2

(a) Differentiating both sides gives $\frac{2xx'}{a^2} + \frac{2yy'}{b^2} = 0$, so dividing by 2 and substituting $x = x_0$ and $y = y_0$ gives

$$\frac{2x_0x'}{a^2} + \frac{2yy'}{b^2} = 0 \implies \frac{y'}{x'} = \frac{dy}{dx} = -\frac{b^2x_0}{a^2y_0}.$$

Thus the slope of the tangent line is $-\frac{b^2x_0}{a^2y_0}$, so the equation of the tangent line is

$$\begin{aligned} y - y_0 &= -\frac{b^2x_0}{a^2y_0}(x - x_0) \\ \implies a^2y_0y - a^2y_0^2 &= -b^2x_0x + b^2x_0^2 \\ \implies a^2y_0^2 + b^2x_0^2 &= a^2y_0y + b^2x_0x. \end{aligned}$$

Dividing both sides by a^2b^2 gives

$$\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$$

because (x_0, y_0) lies on the ellipse and thus satisfies its equation.

(b) Let P be the point (x_0, y_0). We show that if we reflect F_1P on the tangent at P, we get F_2P. Note that it suffices to show that the angle that F_1P makes with the tangent equals the angle that F_2P makes with the tangent. Let $\overrightarrow{F_1P} = \overrightarrow{v_1}$ and $\overrightarrow{F_2P} = \overrightarrow{v_2}$. It then suffices to consider their dot products. Consider a vector $\overrightarrow{u} = \langle 0, -\frac{b^2x_0}{a^2y_0}\rangle$. Then it suffices to prove that $\frac{u \cdot v_1}{||u||||v_1||} = \frac{u \cdot v_2}{||u||||v_2}$, or $||\frac{v_1}{||v_1||}|| = ||\frac{v_2}{||v_2||}||$, which is 1, as desired.

(c)

$$\sin(\frac{\pi}{6} + \alpha) = \sin\frac{\pi}{6}\cos\alpha + \sin\alpha\cos\frac{\pi}{6} = \frac{1}{2}\sqrt{1 - \alpha^2} + \frac{\sqrt{3}}{2}\alpha.$$

$\alpha^\circ = \frac{\pi\alpha}{180}$ radians and 30° is $\pi/6$ radians, so we can convert this to radians. So we have

$$\sin(\frac{\pi}{6} + \frac{\pi\alpha}{180}) = \frac{1}{2}\sqrt{1 - \left(\frac{\pi\alpha}{180}\right)} + \frac{\sqrt{3}}{2}\frac{\pi\alpha}{180}.$$

3

(a) You want me to integrate it? Well, ok. The integral of $\frac{1}{n}$ is $\ln n$ if $x = \frac{m}{n}$ is in lowest terms, and 0 if $x \in \mathbb{R}\backslash\mathbb{Q}$.

(b) We already know that a bounded function f belongs to $\mathcal{R}[a, b]$ if the limit of the sum of the intervals of the partition as $\lambda(P)$ approaches 0 is L. So it has to be between two numbers, ϵ and δ, for any two $\delta > \epsilon > 0$.

(c)

$$\cos(\frac{\pi}{4} + \alpha) = \cos\frac{\pi}{4}\cos\alpha - \sin\frac{\pi}{4}\sin\alpha = \sqrt{2}2\sqrt{1 - \alpha^2} - \frac{\sqrt{2}}{2}\alpha.$$

As above, $\alpha = \frac{\pi\alpha}{180}$, so

$$\cos(\frac{\pi}{4} + \frac{\pi\alpha}{180}) = \sqrt{2}2\sqrt{1 - \left(\frac{\pi\alpha}{180}\right)^2} - \frac{\sqrt{2}}{2} \cdot \frac{\pi\alpha}{180}.$$

(d) Observe that since the water is rotating at constant angular velocity ω, the acceleration vector is $\langle -\omega^2 \sin \omega t, -\omega^2 \cos \omega t\rangle$, so the acceleration is $\frac{\omega^2}{g}$. Thus $f'(x) = \frac{\omega^2}{g}x$. Consider a function $f(x) = \frac{\omega^2}{2g}x^2$. The derivative of $f(x)$ is $\frac{\omega^2}{2g} \cdot 2x = \frac{\omega^2}{g}x$. If the axis is shifted by a constant, then the derivative of the constant is 0, which does not change.

4 Consider the graph $f(x)$. Then $f'(x)$ is the graph of its derivative, which passes through $(x_0, f'(x_0))$. Then the slope of the tangent at this point is equal to $f''(x_0)$, so the equation of the tangent line is $y - f'(x_0) = f''(x_0)(x - x_0)$. $f''(x) = 2$, so the horizontal component of the acceleration is the largest when $x_0 = 0$.

5

(a) Observe that this function is obviously continuous because each of the terms, $\varphi_n(x)$, are all continuous. Also, note that because every number, even irrational, can be expressed by an infinite base 2 representation, then x can be written in terms of

$$x = \sum_{i=1}^{\infty} \frac{a_i}{2^i},$$

where a_i is either 0 or 1. Thus there is always a term that does not have a derivative in the sum, so $f(x)$ does not have a derivative at every point.

(b) Let $P_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$.

$$\begin{aligned}
P_n(x_0) &= a_0 + a_1 x_0 + a_2 x_0^2 + \cdots + a_n x_0^n = \alpha_0 \\
P_n^{(1)}(x_0) &= a_1 + 2a_2 x_0 + 3a_3 x_0^2 + \cdots + n a_n x_0^{n-1} = \alpha_1 \\
P_n^{(2)}(x_0) &= 2a_2 + 3 \cdot 2a_3 x_0 + 4 \cdot 3a_4 x_0^2 + \cdots + n(n-1)a_n x_0^{n-2} = \alpha_2 \\
&\vdots \\
P_n^{(n-1)}(x_0) &= (n-1)\cdots 1 \cdot a_{n-1} + n(n-1)\cdots 2 a_n x_0 = \alpha_{n-1} \\
P_n^{(n)}(x_0) &= n(n-1)\cdots 2 \cdot 1 a_n = \alpha_n.
\end{aligned}$$

So working our way from the bottom to the top, we get $a_n = \frac{\alpha_n}{n!}$, then we can use this to solve for a_{n-1} to get $a_{n-1} = \frac{\alpha_{n-1} - \alpha_n x_0}{(n-1)!}$, etc. Solving for all of the a_i gives our desired polynomial.

6

(a) First, find the integral of $\mathbf{v}(t)$, equal to $\mathbf{V}(t) = \int \mathbf{v}(t)\mathrm{d}t$. Plug 0 into this to get $\mathbf{r}_0 = \mathbf{V}(0) + c$, so $c = \mathbf{r}_0 - \mathbf{V}(0)$. Thus $\mathbf{r}(t) = \mathbf{V}(t) + \mathbf{V}(0) - \mathbf{r}_0$.

(b) Yep. A vector is basically a list of real-valued functions, so the integral of a vector is the list of the integral of each part. So it's the same as integrating real valued functions.

(c) Yes, provided that it applies to all the components of the vector, since integrating a vector-valued function is the same as integrating many real-valued functions.

(d) Yes, provided that it applies to all the components of the vector, since integrating a vector-valued function is the same as integrating many real-valued functions.

(e) All of them?

6.2 Linearity, Additivity, and Monotonicity of the Integral

1

(a) We already know that $f(x) \geq 0$, so $f(x)$ is above the x-axis. Furthermore, since there is a continuous point x_0 such that $f(x_0) > 0$, then $f(x)$ is above the x-axis around this point (the fact that this part is continous is very important because we will know that f is not just a straight line with a couple of dots). Since $\int f(x)\mathrm{d}x$ equals the area under the graph but above the x-axis, then it must be positive.

(b) If $\int_a^b f(x)\mathrm{d}x = 0$, then the net area is 0, but $f(x) \geq 0$, so f is basically a flat line (because how else would it have 0 area?). However, points of discontinuity would be exceptions, but they are countable, so it's 'almost all' points.

2

(a) Since $m = \inf_{]a,b[} f(x)$, then $f(x) \geq m$ for all $x \in]a,b[$, and since $M = \inf_{]a,b[} f(x)$, so $f(x) \leq M$ for all $x \in]a,b[$. Since $\int_a^b f(x)\mathrm{d}x$ is the area under f, then it is at least $m(b-a)$ and at most $M(b-a)$. Thus, if it equals $\mu(b-a)$, then $m \leq \mu \leq M$.

(b) This is an application of *drum roll* the intermediate value theorem, I think. We're trying to prove that there exists a value $\xi \in]a,b[$ such that $f(\xi) = \mu$. Since μ is between m and M, which are the infinum and supremum of f, then this must happen because f is continuous.

3 We already have $f'(x)$ is differentiable. Then we can take the derivative of $f^{(2)}(x)$, etc. Since $f(x)$ consists of a smooth curve, then $f'(x)$, $f''(x)$, etc., will also be smooth, and hence it will be differentiable. Note that because we have an x^2, $f(x)$ is an even function, so $f'(x) = 0$, so $f''(x) = 0$, etc., so $f^{(n)}(x) = 0$. Note that we have $2x \sin \frac{1}{x} - \cos \frac{1}{x}$ if $x \neq 0$. This is defined on $\mathbb{R} \backslash 0$, and note that $f'(0) = 0$, so it's defined on all $\mathbb{R}$. As x approaches 0, $\frac{1}{x}$ approaches ∞, but since sin and cos is periodic, then this function will keep wavering, so it's not continuous.

4

(a) Lebesgue's criterion states that if f is integrable on an interval, then it must be bounded and continuous at almost every point on the interval. If f is integrable and satisfies this criterion, then so must f^p, so we're done.

(b) Note that

$$\int_a^b (f \cdot g)(x)\mathrm{d}x = \lim_{n\to\infty} \left(\sum_{i=0}^{n-1} \frac{b-a}{n} \cdot f\left(a + i\left(\frac{b-a}{n}\right)\right)\right).$$

Therefore, if we look at Holder's Inequality for sums, which is

$$\sum_{i=1}^{n} a_i b_i \leq \left(\sum_{i=1}^{n} a_i^p\right)^{1/p} \left(\sum_{i=1}^{n} b_i^q\right)^{1/q}.$$

If we let $f(a + i(\frac{b-a}{n})) = a_i$ and $g(a + i(\frac{b-a}{n})) = b_i$, then multiplying both sides by $\left(\frac{b-a}{n}\right)^2$ gives

$$\left|\int_a^b (f \cdot g)(x)\mathrm{d}x\right| \leq \left(\int_a^b |f|^p(x)\mathrm{d}x\right)^{1/p} \cdot \left(\int_a^b |g|^p(x)\mathrm{d}x\right)^{1/q}.$$

(c) Observe that

$$e^{-\frac{1}{(1+x)^2}-\frac{1}{(1-x)^2}} = e^{-\frac{1}{(1+x)^2}} \cdot e^{-\frac{1}{(1-x)^2}}.$$

So since $f(x)$ is a product of functions that are infinitely differentiable on $\mathbb{R}$ they are $f(1+x)$ and $f(1-x)$ from part (a), then this function must be infinitely differentiable as well. Let $f \in C^{(\infty)}$. The first derivative of $x^{n-1} f\left(\frac{1}{x}\right)$ is

$$\begin{aligned} x^{n-1}\left(f\left(\frac{1}{x}\right)\right)' + (x^{n-1})' f\left(\frac{1}{x}\right) &= x^{n-1} \cdot \frac{-1}{x^2} f'\left(\frac{1}{x}\right) + (n-1)x^{n-2} f\left(\frac{1}{x}\right) \\ &= (n-1)x^{n-2} f\left(\frac{1}{x}\right) - x^{n-3} f'\left(\frac{1}{x}\right). \end{aligned}$$

Then the nth derivative of $x^{n-1} f\left(\frac{1}{x}\right)$ is $\frac{(-1)^n}{x^{n+1}} f^{(n)}\left(\frac{1}{x}\right)$, so we're done.

(d) If f is an even function, $f(x) = f(-x)$. Then taking the derivative of both sides and applying the chain rule gives $f'(x) = (-x)' f'(-x) = -f'(-x)$, so f' is an odd function. If f is an odd function, $f(x) = -f(-x)$. Then taking the derivative of both sides and applying the chain rule gives $f'(x) = -(-x)' f'(-x) = f'(-x)$, so f' is an even function. We already know that if f is even, then f' is odd. We now prove the converse. If f' is odd, then $f'(x) = -f'(-x)$. Integrating gives $f(x) + c_1 = f(-x) + c_2$. Letting $x = 0$ gives $f(0) + c_1 = f(0) + c_2$, so $c_1 = c_2$. Thus $f(x) = f(-x)$, and f is even, as desired.

6.3 The Integral and the Derivative

1

(a) Let $f(x) = \frac{n}{x^2}$, so

$$\frac{n}{(n+1)^2} + \cdots + \frac{n}{(2n)^2} = \int_{n+1}^{2n} \frac{n}{x^2} \mathrm{d}x = \frac{n-1}{n+1}.$$

As $n \to \infty$, the limit equals 1.

(b) Let $f(x) = \frac{x^\alpha}{n^{\alpha+1}}$, so

$$\frac{1^\alpha + 2^\alpha + \cdots + n^\alpha}{n^{\alpha+1}} = \int_1^\alpha \frac{f(x)}{n^{\alpha+1}} \mathrm{d}x = \frac{1}{n^{\alpha+1}} \cdot \left.\frac{x^{\alpha+1}}{\alpha+1}\right|_1^n = \frac{1}{\alpha+1} \cdot \frac{n^{\alpha+1}-1}{n^{\alpha+1}}.$$

As $n \to \infty$, this approaches $\frac{1}{\alpha+1}$.

2

(a) Since the function is continuous on that open interval, then by Lebesgue's criterion, the function is integrable, so it has a primitive.

(b) Since f and its derivative is continuous, then f is a smooth curve. Construct a function that equals $-f(x)$ when f is decreasing and 0 when f is increasing, and another function that equals 0 with f is decreasing and $f(x)$ when f is increasing. Then their difference is $f(x)$.

3 We want to prove that

$$f(\xi)\int_a^b g(x)\mathrm{d}x = g(a)\int_a^\xi f(x)\mathrm{d}x + g(b)\int_\xi^b f(x)\mathrm{d}x.$$

Taking the derivative of both sides gives

$$f(\xi)(g(b)-g(a)) = g(a)(f(\xi)-f(a)) + g(b)(f(b)-f(\xi)),$$

or $2f(\xi)(g(b)-g(a)) = g(b)f(b) - g(a)f(a)$, which is true by the other mean-value theorem.

4 If $f(x)$ is differentiable, then there exists a function $f'(x)$ such that $f(x)-f(x_0) = f'(x)(x-x_0)$, so in this case we can let $f'(x) = \varphi(x)$. If there exists a function $\varphi(x)$ such that $f(x)-f(x_0) = \varphi(x)(x-x_0)$, then $\varphi(x) = \frac{f(x)-f(x_0)}{x-x_0}$, which is the slope of the line between the points $(x, f(x))$ and $(x_0, f(x_0))$. Since the definition of derivative is the slope of the line tangent to a curve at the point, then as $x \to x_0$, $\varphi(x)$ approaches this slope. Hence $\varphi(x)$ is the derivative of f, and f is differentiable at x_0.

5 Observe that

$$\left(\frac{e^t}{t}\right)' = \frac{t\cdot e^t - e^t\cdot 1}{t^2} = \frac{t-1}{t}\cdot\frac{1}{t}\cdot e^t \sim \frac{1}{t}\cdot e^t.$$

Thus $\int \frac{1}{t}\cdot e^t\mathrm{d}t \sim \frac{e^t}{t}$, so $\int_1^{x^2}\frac{e^t}{t}\mathrm{d}t \sim \frac{1}{x^2}e^{x^2}$.

6

(a) Note that

$$\left(\frac{\cos(t^2)}{2t}\right)' = \frac{2t\cdot 2t(-\sin(t^2)) - \cos(t^2)\cdot 2}{(2t)^2} = -\sin(t^2) - \frac{\cos(t^2)}{2t^2}.$$

As $t\to\infty$, $\left(-\frac{\cos(t^2)}{2t}\right)' \sim \sin(t^2)$. Thus

$$\int_x^{x+1}\sin(t^2)\mathrm{d}t \sim \frac{\cos(x^2)}{2x} - \frac{\cos(x+1)^2}{2(x+1)}.$$

(b)

$$xf(x) = \frac{\cos(x^2)}{2} - \frac{x}{x+1}\cdot\frac{\cos(x+1)^2}{2}.$$

As $x\to\infty$, this approaches

$$\frac{\cos(x^2)-\cos(x+1)^2}{2} = 0.$$

7 Since we know that $\varphi(x)$ is the first-order derivative of $f(x)$, then if $\varphi(x)$ is $n-1$-differentiable, so is $f'(x)$. This means that $f(x)$ is n-differentiable, so it has a derivative of order n at x_0. We need to construct a function f that is continuous at x_0 has an inverse f^{-1}, and f^{-1} is not continuous at y_0 Consider $f(x) = x$ if $x \le 0$ and $f(x) = x^2$ if $x \ge 0$ at $x = 0$. Then f is continuous at x_0 but not continuous at y_0, and the theorem doesn't apply.

8

(a) $m_1a_1 = F = \frac{Gm_1m_2}{r^2}$, $a_1 = \frac{Gm_2}{r^2}$ and $m_2a_2 = F$, so $a_2 = \frac{Gm_1}{r^2}$. Then v_1^2 increases by $2\alpha_1\Delta r$ and v_2^2 increases by $2\alpha_2\Delta r$, so we get $\frac{(2Gm_1+2Gm_2)\Delta r}{r^2} - \frac{Gm_1m_2}{r} = 0$, so E is constant.

(b) E is the total energy of the system. K is the kinetic energy, or energy in motion, and U is the potential energy, which is kind of like energy waiting to be converted to kinetic energy. The total energy of n bodies is constant if it is under no external force.

9

(a) Observe that

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots.$$

We also have

$$\begin{aligned}\frac{1+ax^2}{1+bx^2} &= (1+ax^2)(1-bx^2+b^2x^4-b^3x^6+\cdots)\\ &= 1+ax^2-bx^2-abx^4+b^2x^4+ab^2x^6-b^3x^6+\cdots\\ &= 1+(a-b)x^2-(b^2-ab)x^4+(ab^2-b^3)x^6+\cdots\end{aligned}$$

To maximize the order, we want at least the first 3 coefficients of each series to match up, so $a-b=-\frac{1}{2}$ and $b(b-a)=\frac{1}{24}$. Hence, solving gives $b=\frac{1}{12}$ and $a=-\frac{5}{12}$.

(b) Observe that

$$x\left[\frac{1}{e}-\left(\frac{x}{x+1}\right)^x\right] = \frac{\frac{1}{e}-\left(\frac{x}{x+1}\right)^x}{1/x}.$$

Observe that since both numerator and denominator approach 0 as x approaches ∞, then we can use L'Hopitals rule. This gives

$$\begin{aligned}\lim_{x\to\infty}\frac{\frac{1}{e}-\left(\frac{x}{x+1}\right)^x}{1/x} &= \lim_{x\to\infty}\frac{\left(\frac{1}{e}-\left(\frac{x}{x+1}\right)^x\right)}{(1/x)'}\\ &= \lim_{x\to\infty}\frac{-\left(\left(\frac{x}{x+1}\right)^x\right)'}{-\frac{1}{x^2}}\\ &= \lim_{x\to\infty}x^2\left(\left(\frac{x}{x+1}\right)^x\right)'.\end{aligned}$$

(c) We already know that $(k^x)' = k^x\cdot\ln x$, so using chain rule gives

$$\left(\left(\frac{x}{x+1}\right)^x\right)' = \left(\frac{x}{x+1}\right)'\cdot\left(\frac{x}{x+1}\right)^x\cdot\ln\frac{x}{x+1}.$$

Since $\frac{x}{x+1}=1-\frac{1}{x+1}$, then $\left(\frac{x}{x+1}\right)'=-\left(\frac{1}{x+1}\right)'=\frac{1}{(x+1)^2}$. Therefore, the limit is

$$\lim_{x\to\infty}\frac{x^2}{x+1}^2\cdot\ln\left(\frac{x}{x+1}\right)\cdot\left(\frac{x}{x+1}\right)^e = 1.$$

(d) By Lagrange's form of the remainder,

$$r_n(x_0;x) = \frac{1}{(n+1)!}f^{(n+1)}(\xi)(x-x_0)^{n+1} = \frac{1}{(n+1)!}\cdot e^{\xi}\cdot x^{n+1} < 10^{-3}.$$

Since x is at most 2, it suffices to find the smallest value of n such that $\frac{1}{(n+1)!}\cdot 2^{n+1}\leq 10^{-3}$, which is $n=10$. Thus the taylor formula is

$$1+\frac{1}{1!}x+\frac{1}{2!}x^2+\cdots+\frac{1}{10!}x^{10}.$$

(e) We know that the taylor series at 0 of a function is

$$1 + \frac{1}{1!}f'(0) + \frac{1}{2!}f''(0) + \frac{1}{3!}f^{(3)}(0) + \cdots$$

If f is even, then f' is odd, so f'' is even, so $f^{(3)}$ is odd, etc. Thus all odd-ordered derivatives of f are odd and even-ordered derivatives of f are even. Thus $f^{(2k+1)}(x) = -f^{(2k+1)}(-x)$, and plugging in $x = 0$ gives $f^{(2k+1)}(0) = -f^{(2k+1)}(0)$, so $f^{(2k+1)}(0) = 0$. Therefore all odd powers of the Taylor series are 0, and hence the Taylor series consists only of even terms.

If f is odd, then all odd-ordered derivatives of f are even and all even-ordered derivatives of f are odd. Thus $f^{(2k)}(0) = 0$ using a similar argument as above, so the Taylor series consists only of odd terms.

(f) Since we know that $f^{(n)}(0) = 0$ for all nonnegative n, then the Taylor series at 0 is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = 0.$$

Also, note that the remaining terms are of the form

$$\left|\frac{f^{(n)}(x)}{n!}x^n\right| \leq \frac{\sup\limits_{-1\leq x\leq 1} |f^{(n)}(x)|}{n!}n!|x|^n \leq \frac{n!C}{n!}|x|^n = C|x|^n.$$

Since $|x| < 1$, then this is a geometric series, which converges, so $f \equiv 0$ on $[-1, 1]$.

(g) Observe that since the derivative of a minimum or maximum point equals 0, if there is an $x \in I_1$ satisfying $|f^{(k-1)}(x)|$ is minimized, the derivative of $f^{(k-1)}$, which is $f^{(k)}$, satisfies $f^{(k)}(x) = 0$, unless x is a boundary point. Obviously $m_k(I) = 0$ if any of x are not boundary points, and it will be less than $\frac{1}{\mu}(m_{k-1}(I_1) + m_{k-1}(I_3))$. Otherwise, if all of x are boundary points, then the solution to $m_k(I)$ is also a boundary point, and it must equal one of $m_{k-1}(I_1)$ or $m_{k-1}(I_3)$, so we are also done.

(h) We have base case $k = 0$; $m_1(I) \leq 1$. Assume that for some k, we have that

$$m_k(I) \leq \frac{2^{k(k+1)/2}k^k}{\lambda^k}.$$

Then we have

$$m_{k+1}(I) \leq \frac{1}{\mu}(m_k(I_1) + m_k(I_3))$$

by part (a), so

$$m_{k+1}(I) \leq \frac{1}{\mu}\left(\frac{2^{k(k+1)/2}k^k}{\lambda_1^k} + \frac{2^{k(k+1)/2}k^k}{\lambda_3^k}\right) = \frac{2^{k(k+1)/2}k^k}{\mu}\left(\frac{1}{\lambda_1^k} + \frac{1}{\lambda_3^k}\right).$$

We wish to prove that

$$m_{k+1}(I) \leq \frac{2^{(k+1)(k+2)/2}(k+1)^{k+1}}{\lambda^{k+1}}.$$

$\lambda = \lambda_1 + \lambda_2 + \mu$ and $2^{(k+1)(k+2)/2} = (k+1)2^{k(k+1)/2}$. Therefore, we are done.

10

(a) It suffices to show, by induction, that there exists k numbers $x_{k_1} < x_{k_2} < \cdots < x_{k_k}$ such that $f^{(k)}(x_{k_i})f^{(k)}(x_{k_{i+1}}) < 0$, so they are of opposite sign, which will show that there is a root between x_{k_i} and $x_{k_{i+1}}$. We have base case $n = 1$. If $|f'(0)| > a_1$, then $f'(x) = 0$ has at least 0 distinct roots. Suppose this works for some value of k. We can start by using the values $x_{k_1}, \ldots, x_{k_k}$ for $x_{(k+1)_1}, \ldots, x_{(k+1)_k}$. Then we can add $x_{(k+1)_{k+1}}$ satisfying $f^{(k+1)}(x_{(k+1)_{k+1}})$ is of opposite sign from $f^{(k+1)}(x_{k_k})$, so we're done.

(b) Consider the auxiliary function $h(x) = f(x) - r(x)$, where $r(x) = f'(a)(x-a)$. Then $h'(x) = f'(x) - f'(a)$ and $h(a) = h(b) = 0$, so there exists ξ such that $h'(\xi) = 0$. Then $f'(\xi) = f'(a)$, and we can replace $f'(a)$ with $f'(\xi)$ for any ξ so that $f'(x)$ attains all values between $f'(a)$ and $f'(b)$.

11 Lagrange's infinite-increment theorem states that if a function $f : [a, b] \to \mathbb{R}$ is continuous on a closed interval $[a, b]$ and differentiable on the open interval, $]a, b[$, there exists a point $\xi \in]a, b[$ such that $f(b) - f(a) = f'(\xi)(b - a)$. Because we know that f'' exists, then f' must be differentiable on $]a, b[$, so Lagrange's theorem holds for f'. Hence there exists $\xi \in]a, b[$ such that $f'(b) - f'(a) = f''(\xi)(b - a)$, as desired.

6.4 Some Applications of Integration

1

(a) Suppose for the sake of contradiction that $f'(x)$ has a discontinuity of the first kind. Suppose at some point $x = a$, $f'(a)$ has a discontinuity. This means that $f'(x)$ has a limit at a from both the left side and the right side, and these limits are equal, but $f'(a)$ is not equal to this limit. But because $f'(a)$ exists, this is a contradiction, so $f'(x)$ has no removable discontinuities, so it can only have discontinuities of the second kind.

(b) The error in the proof lies in the assumption that $\lim_{x\to x_0} f'(\xi) = \lim_{\xi \to x_0} f'(\xi)$. It should have been $\lim_{x\to x_0} f'(\xi(x))$ where $\xi(x)$ is a function that lies between x and x_0. Note that the limit of $f'(\xi(x))$ as $x \to x_0$ does not necessarily equal $f'(x_0)$ since $\xi(x)$ is a function of x and does not necessarily to approach x_0 in the same way as x. The "proof" assumes that the limit equals $f'(x_0)$, so it is incorrect.

(c) Let $x \in [-a, a]$. Then by the mean value theorem, there exists a point $c \in [-a, x]$ if $x > 0$ or $c \in [x, a]$ if $x < 0$ such that $f'(x) - f'(0) = f''(c) \cdot x$. Since $|f'(x)| \le |f'(0)| + |f''(c)||x|$, since $c \in [-a, a]$, then $|f''(c)| \le M_2$ and $|x| \le a$. Hence we have $|f'(x)| \le |f'(0)| + M_2|x|$.

Note that for some $d \in [-a, a]$, $f'(0) - f(0) = f'(d) \cdot 0$. We know that $|f'(0)| \le |f(0)| + |f'(d)|$. Since $d \in [-a, a]$, then $|f'(d)| \le M_1$ and $|f(0)| \le M_0$, so $|f'(0)| \le M_0 + M_1$. Then $|f'(x)| \le M_0 + M_1 + M_2|x|$, so $M_1 \le \frac{M_0}{a} + \frac{a}{2}M_2$. Since the coefficient of M_2, $\frac{x^2+a^2}{2a}$ is at least equal to $\frac{a}{2}$, we are done.

2 Observe that when x is at its maximum, $x = a$, so from part (a), $M_1 \le \frac{M_0}{a} + aM_2$. If the length of I, which equals $2a$, is not less than $2\sqrt{M_0/M_2}$, then $a \ge \sqrt{M_0/M_2}$. By AM-GM, $\frac{M_0}{a} + aM_2 \ge 2\sqrt{M_0 M_2}$, which proves the first case.

For the second case, if $I = \mathbb{R}$, then we can't have $x = a$, and a approaches infinity. The minimum value of $\frac{M_0}{a} + \frac{a}{2}M_2$ is $\sqrt{2M_0M_2}$ by AM-GM.

3

(a) By Lagrange's form of remainder, we get

$$r_n(x_0; x) = \frac{1}{n!} f^{(n+1)}(\xi)(x - x_0)^n,$$

so it suffices to prove that $\xi = x_0 + \theta(x - x_0)$. By the mean value theorem, there exists $\xi \in [0, 1]$ such that $f(x) - f(x_0) = f'(\xi)(x - x_0)$, so if we replace $f'(\xi)(x - x_0)$ with $\theta(x - x_0)$ and replace $f(x)$ with ξ, we get $\xi - x_0 = \theta(x - x_0)$. If we have equality cases, then we cannot have smaller numbers. For the number 2, we have $a = \sqrt{M_0/M_2}$, and for the number $\sqrt{2}$, we have $a = \sqrt{2M_0/M_2}$.

(b) We prove this by induction on p. We have the base case $p = 2$ from part 9(b), and assume this works for some p. We prove that it also works for $p + 1$. First note that by AM-GM,

$$\begin{aligned}\left(1 - \frac{k}{p}\right) M_0 + \frac{k}{p} M_p &= \frac{\underbrace{M_p + M_p + \cdots + M_p}_{\text{summed } k \text{ times}} + \underbrace{M_0 + M_0 + \cdots + M_0}_{\text{summed } p-k \text{ times}}}{p} \\ &\geq \sqrt[p]{M_p^k M_0^{p-k}} = M_0^{1-k/p} \cdot M_p^{k/p}.\end{aligned}$$

Thus it suffices to prove that

$$M_k \leq 2^{k(p-k)/2} \left((1 - \frac{k}{p}) M_0 + \frac{k}{p} M_p \right)$$

which is obvious, so we are done.

(c) Consider the Taylor expansion of f at 0, which is

$$f(x) = 1 + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n.$$

Observe that a polynomial is determined by any n values, and this will determine a polynomial of degree $n - 1$, which is perfect for $L(x)$. We can use the method of undetermined coefficients to solve for $L(x)$. Note that because $f(x)$ can also be expressed in polynomial form to a highly accurate degree, then $f(x) - L(x) = 0$ for $x = x_1, \ldots, x_n$. So since $f(x) - L(x)$ is a polynomial, it must be a scalar multiple of $(x - x_1) \cdots (x - x_n)$, since $f(x) - L(x)$ has degree n. It suffices to prove that the remaining coefficient equals $\frac{f^{(n)}(\xi)}{n!}$ for some $\xi \in I$, which follows from the mean value theorem.

4

(a) Since $n_1 + n_2 + \cdots + n_p = n$, then the sum of the orders, and the number of roots of f, f', f'', etc is $n_1 + n_2 + \cdots + n_p = n$. Therefore, once you take the derivative $n - 1$ times to get $f^{(n-1)}$, the remainging order is 1, so it has one root, so there exists ξ such that $f^{(n-1)}(\xi) = 0$. The Taylor expansion of f at 0 is

$$f(x) = 1 + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n.$$

Since f has $n + 1$ roots, it has order $n + 1$, so $f^{(n)}$ has order 1, which means that it has a root.

(b) Let $g(x) = f(x) - H(x)$, so note that $g^{(k)}(x) = f^{(k)}(x) - H^{(k)}(x)$. Then for x_i, $g(x_i) = g'(x_i) = \cdots = g^{(n_i-1)}(x_i) = 0$, and we want to prove that

$$g(x) = \frac{(x-x_1)^{n_1}\cdots(x-x_p)^{n_p}}{n!} f^{(n)}(\xi).$$

If we look at the taylor polynomial forms of $g(x)$, $g'(x)$, $g''(x)$, and so on, we see that for x_i, we have that $g(x), g'(x), g''(x), \ldots, g^{(n_i-1)}(x)$ are multiples of $(x - x_i)$. But in order for $g^{(n_i-1)}(x)$ to be a multiple of $(x - x_i)$, $g(x)$ needs to be a multiple of $(x - x_i)^{n_i}$, since the $n_i - 1$th derivative of $(x - x_i)^{n_i}$ is $n_i! \cdot (x - x_i)$. So over all i, $g(x)$ is a multiple of $(x - x_1)^{n_1} \cdots (x - x_p)^{n_p}$, and the there must exist ξ such that the coefficient is $\frac{f^{(n)}(\xi)}{n!}$.

(c) Consider 2 roots of $P(x)$ and call them x_1 and x_2, so $P(x_1) = P(x_2)$. By the mean value theorem, there exists $\xi \in [x_1, x_2]$ such that $P(x_2)-P(x_1) = 0 = P'(\xi)(x_2-x_1)$, so $P'(\xi) = 0$. Suppose $P(x)$ has degree n and that the multiplicity of a root a is k, so let $P(x) = (x - a)^k Q(x)$ for some polynomial Q. Observe that Q does not have a factor of $x - a$. Then by the chain rule,

$$\begin{aligned} P'(x) &= ((x-a)^k)'Q(x) + (x-a)^k Q'(x) \\ &= k(x-a)^{k-1}Q(x) + (x-a)^k Q'(x) \\ &= (x-a)^{k-1}(kQ(x) + (x-a)Q'(x)). \end{aligned}$$

Obviously $P'(x)$ is also has a as a root, but its multiplicity is $k - 1$.

(d) Suppose the roots of $P(x)$ are $a_1, \ldots, a_k$ with multiplicities $n_1, \ldots, n_k$, so that

$$P(x) = c(x-a_1)^{n_1}(x-a_2)^{n_2}\cdots(x-a_k)^{n_k},$$

where c is a constant. Then by the previous part, we know that $P(x)$ has the roots $a_1, \ldots, a_k$ with multiplicities $n_1 - 1, \ldots, n_k - 1$, so $P'(x)$ is a scalar multiple of

$$(x-a_1)^{n_1-1}(x-a_2)^{n_2-1}\cdots(x-a_k)^{n_k-1},$$

and thus $Q(x)$ is also a scalar multiple of

$$(x-a_1)^{n_1-1}(x-a_2)^{n_2-1}\cdots(x-a_k)^{n_k-1}.$$

Hence dividing $P(x)$ by $Q(x)$ leaves us with a scalar multiple of $(x - a_1)(x - a_2)\cdots(x - a_k)$, so $\frac{P(x)}{Q(x)}$ has $a_1, \ldots, a_k$ as its roots, but all of them have multiplicity 1.

(e) Take a polynomial $Q(x) = c_0 + c_1x + \cdots + c_nx^n$. Then obviously if we take $P(x) = Q(x - x_0)$, then $P(x) = c_0 + c_1(x - x_0) + \cdots + c_n(x - x_0)^n$. So in order to express $P(x)$ in terms of $x - x_0$, we first find $P(x + x_0) = Q(x)$, and find the coefficients of $Q(x)$. Then we plug $x - x_0$ into $Q(x)$ to get $P(x)$ in terms of $x - x_0$. Consider the Taylor expansion of $f(x)$ with respect to x_0, which looks like

$$f(x) = 1 + \frac{f'(x_0)}{1!}x + \cdots + \frac{f^{(n)}(x_0)}{n!}x^n + o((x-x_0)^n).$$

Then if we let $P(x)$ be the first $n + 1$ terms, we're done.

5

(a) xy is the area of that rectangle. $\int_0^x f(t)\mathrm{d}t$ is the area from 0 to x under the curve of $f(t)$. Suppose WLOG $f(x) \leq y$ for the time being. $\int_0^y g(t)\mathrm{d}t$ is the area from 0 to y to the left of the curve of $g(t)$. Since $f(x) \leq y$, then $g(y) \geq x$, so there's an extra section (because we need to extend the graph of $y = f(x)$ so that it intersects the horizontal line). Thus $xy \leq \int_0^x f(t)\mathrm{d}t + \int_0^y g(t)\mathrm{d}t$.

(b) Let $f(t) = t^{p-1}$ and $g(t) = t^{q-1}$, so $f(g(t)) = g(f(t)) = t^{(p-1)(q-1)} = t$, so $(p-1)(q-1) = pq - p - q + 1 = 1$, so $pq = p + q$, so $\frac{1}{p} + \frac{1}{q} = 1$. Now by part (a),

$$xy \leq \int_0^x t^{p-1}\mathrm{d}t + \int_0^y t^{q-1}\mathrm{d}t = \frac{1}{p}t^p + \frac{1}{q}t^q,$$

as desired.

(c) This means that the sum of the area under $f(t)$ and the area to the left of $g(t)$ perfectly equals the rectangle, so $f(t)$ needs to pass through the point (x, y). Thus $f(x) = y$ and thus $g(y) = x$.

6 We get $p = \frac{2}{\pi} \cdot \frac{l}{h}$. Note that $\frac{l}{h}$ is the probability that the needle falls on a line given that the needle can only fall perpendicular to the line. Then $\frac{2}{\pi}$ is the additional rotation that's combined with it, since we have $4 \cdot \frac{1}{2\pi}$ (the top is because the area is $\frac{\pi l^2}{4}$ and the bottom is because there are 2π radians in a circle).

6.5 Improper Integrals

1

(a) Since $\frac{\sin t}{t}$ is continuous on all of $\mathbb{R}$ except for 0, and $\mathrm{Si}(0) = \int_0^0 \frac{\sin t}{t}\mathrm{d}t = 0$, so since otherwise $\mathrm{Si}(x)$ is the area under the graph, so it's defined on $\mathbb{R}$. Since

$$\mathrm{Si}(-x) = \int_0^{-x} \frac{\sin t}{t}\mathrm{d}t = -\int_0^x \frac{\sin t}{t}\mathrm{d}t = -\mathrm{Si}(x),$$

so it's an odd function. As $x \to \infty$, $\frac{\sin t}{t} \to \infty$, so $\mathrm{Si}(x)$, which is the area under $\frac{\sin t}{t}$, has a limit.

(b) In a similar fashion to the previous part, we know that $\mathrm{si}(x)$ is defined on all of $\mathbb{R}$. Note that

$$\mathrm{Si}(x) - \mathrm{si}(x) = \int_0^x \frac{\sin t}{t}\mathrm{d}t + \int_x^\infty \frac{\sin t}{t}\mathrm{d}t = \int_0^\infty \frac{\sin t}{t}\mathrm{d}t$$

which is a constant, so $\mathrm{Si}(x)$ and $\mathrm{si}(x)$ differ by a constant.

(c) Since

$$\left(\frac{\sin t}{t}\right)' = \frac{t \cdot \cos t - \sin t}{t^2} = \frac{\cos t}{t} - \frac{\sin t}{t^2} \approx \frac{\cos t}{t},$$

then $\int \frac{\cos t}{t}\mathrm{d}t \approx \frac{\sin t}{t}$. Hence

$$\mathrm{Ci}(x) = -\int_x^\infty \frac{\cos t}{t}\mathrm{d}t \approx -\left(0 - \frac{\sin x}{x}\right) = \frac{\sin x}{x}$$

since $\lim_{t\to\infty} \frac{\sin t}{t} = 0$.

2

(a) By the mean value theorem, there exists a value ξ_{1_1} such that $f(x_0+h_1)-f(x_0) = f'(\xi_1)\cdot h_1$ and a value ξ_{1_2} such that $f(x_0+h_1+h_2)-f(x_0+h_2) = f'(\xi_2)\cdot h_1$, so there exists a value ξ_2 such that

$$\begin{aligned}(f(x_0+h_1+h_2)-f(x_0+h_2))-(f(x_0+h_1)-f(x_0))\\ &= f'(\xi_{1_2})\cdot h_1 - f'(\xi_{1_1})\cdot h_1\\ &= (f'(\xi_{1_2})-f'(\xi_{1_1}))\cdot h_1 = f''(\xi_2)h_1h_2.\end{aligned}$$

As you can see, we have the base case. Assume there exists a value ξ_{k_1} such that

$$\Delta^k f(x_0;h_1,\ldots,h_{k-1}) = f^{(k)}(\xi_{k_1})h_1\cdots h_k$$

and

$$\Delta^k f(x_0;h_1,\ldots,h_k) = f^{(k)}(\xi_{k_2})h_1\cdots h_k.$$

(b) Then

$$\begin{aligned}\Delta^{k+1} f(x_0;h_1,\ldots,h_{k+1}) &= \Delta^k f(x_0;h_1,\ldots,h_k) - \Delta^k f(x_0;h_1,\ldots,h_{k-1})\\ &= (f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1}))\cdot h_1\cdots h_k.\end{aligned}$$

By the mean value theorem, there exists some ξ such that

$$f^{(n+1)}(\xi)\cdot h_{k+1} = f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1}),$$

so substituting this in gives $f^{(k+1)}(\xi)h_1\ldots h_{k+1}$, so we are done by induction.

3

(a) Using our result from problem 2 part (a), we know that

$$\begin{aligned}&|\Delta^n f(x_0;h_1,\ldots,h_n) - f^{(n)}(x_0)h_1\cdots h_n|\\ &\quad= |f^{(n)}(\xi)h_1\cdots h_n - f^{(n)}(x_0)h_1\cdots h_n|\\ &\quad= |f^{(n)}(\xi) - f^{(n)}(x_0)|\cdot|h_1|\cdots|h_n|\\ &\quad\le \sup_{x\in]a,b[}\left|f^n(x) - f^{(n)}(x_0)\right|\cdot|h_1|\cdots|h_n|,\end{aligned}$$

as desired.

(b) We know that there exists ξ such that

$$\Delta^n f(x_0;h^n) = f^{(n)}(\xi)h^n \implies f^{(n)}(\xi) = \frac{\Delta^n f(x_0;h^n)}{h^n}.$$

As $h\to 0$, $\xi\to x_0$, so

$$f^{(n)}(x_0) = \lim_{h\to 0}\frac{\Delta^n f(x_0;h^n)}{h^n}.$$

For example, take $f(x)=\frac{1}{x}$ so that $f^{(n)}(x)$ is a multiple of $\frac{1}{x^n}$. Then the limit equals 1.

4

(a) Consider the function $f(x) = \frac{1}{x^\alpha} = x^{-\alpha}$. Then $f'(x) = -\alpha x^{-\alpha-1}$, so there exists $\xi \in]n-1, n[$ such that

$$\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha} = f(n-1) - f(n) = f'(\xi) \cdot ((n-1) - n) = -f'(\xi) = \alpha\xi^{-\alpha-1}$$

by Lagrange's theorem. Hence

$$\frac{1}{\xi^{\alpha+1}} = \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right).$$

Since $\xi < n$, then $\frac{1}{\xi^{\alpha+1}} > \frac{1}{n^{\alpha+1}}$, so

$$\frac{1}{n^{\alpha+1}} < \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right),$$

as desired.

(b) It suffices to prove that the series $\sum_{n=1}^\infty \frac{1}{n^{1+\alpha}}$ converges for $\alpha > 0$.

$$\begin{aligned}\sum_{n=1}^\infty \frac{1}{n^{1+\alpha}} &< \sum_{n=1}^\infty \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right) \\ &= \frac{1}{\alpha}\sum_{n=1}^\infty \left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right) \\ &= \frac{1}{\alpha},\end{aligned}$$

which converges, so $\sum_{n=1}^\infty \frac{1}{n^\sigma}$ also converges.

(c) Observe that

$$\ln M_t(x,\alpha) = \ln\left(\sum_{i=1}^n \alpha_i x_i^t\right)^{1/t} = \frac{1}{t}\ln\left(\sum_{i=1}^n \alpha_i x_i^t\right).$$

Now we can take the limit of this as $t \to 0$, and since when $t = 0$, both numerator and denominator equal 0, we use L'Hopitals rule. The derivative of the denominator is 1 and by chain rule, the derivative of the numerator is

$$\frac{\left(\sum_{i=1}^n \alpha_i \cdot \ln x_i \cdot x_i^t\right)}{\sum_{i=1}^n \alpha_i x_i^t} = \frac{\sum_{i=1}^n \alpha_i \cdot \ln x_i}{\sum_{i=1}^n \alpha_i}.$$

Because $\sum_{i=1}^n \alpha_i = 1$, then we get

$$\lim_{t\to 0} \ln M_t(x,\alpha) = \sum_{i=1}^n \alpha_i \ln x_i,$$

so

$$\lim_{t\to 0} M_t(x,\alpha) = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}.$$

(d) As $t \to +\infty$, note that the largest value of x_i will begin to dominate $M_t(x, \alpha)$, so $\lim_{t\to+\infty} M_t(x, \alpha) = \max_{1\le i\le n} x_i$. As $t \to -\infty$, note that the smallest value of x_i will begin to dominate $M_t(x, \alpha)$, so $\lim_{t\to-\infty} M_t(x, \alpha) = \min_{1\le i\le n} x_i$.

5

(a) It suffices to prove that $(M_t(x, \alpha))'$ is positive. By using the chain rule twice, we get that the derivative equals

$$\begin{aligned}
&\left(\sum_{i=1}^{n} \alpha_i x_i^t\right)^{1/t} \cdot \left(\frac{1}{t}\right)' \cdot \left(\sum_{i=1}^{n} \alpha_i x_i^t\right)' \cdot \ln\left(\sum_{i=1}^{n} \alpha_i x_i^t\right) \\
&= \left(\sum_{i=1}^{n} \alpha_i x_i^t\right)^{1/t} \cdot \left(-\frac{1}{t^2}\right) \cdot \left(\sum_{i=1}^{n} \alpha_i \ln x_i \cdot x_i^t\right) \cdot \ln\left(\sum_{i=1}^{n} \alpha_i x_i^t\right)
\end{aligned}$$

Since the first term is positive, the second is negative, the third is negative, and the fourth is positive, we're done.

(b) Observe that by the binomial theorem,

$$|(1+x)|^p = |(1+x)^p| = |1 + px + \binom{p}{2} x^2 + \cdots + x^p|.$$

Obviously, if we let $c_p = 0$ (not exactly sure what this means), then it suffices to show that this sum is greater than or equal to $1 + px$. If $x > 0$, this is obvious. If $x < 0$, as long as $x \ge -1$, this is also obvious. However, if $x < -1$, then the absolute value is

$$-(1+x)^p = -1 - px - \cdots - x^p \ge 1 + px$$

because x is negative.

As you can see, we have the base case. Assume there exists a value ξ_{k_1} such that

$$\Delta^k f(x_0; h_1, \ldots, h_{k-1}) = f^{(k)}(\xi_{k_1}) h_1 \cdots h_k$$

and

$$\Delta^k f(x_0; h_1, \ldots, h_k) = f^{(k)}(\xi_{k_2}) h_1 \cdots h_k.$$

Then

$$\begin{aligned}
\Delta^{k+1} f(x_0; h_1, \ldots, h_{k+1}) &= \Delta^k f(x_0; h_1, \ldots, h_k) - \Delta^k f(x_0; h_1, \ldots, h_{k-1}) \\
&= (f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1})) \cdot h_1 \cdots h_k.
\end{aligned}$$

By the mean value theorem, there exists some ξ such that

$$f^{(n+1)}(\xi) \cdot h_{k+1} = f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1}),$$

so substituting this in gives $f^{(k+1)}(\xi) h_1 \ldots h_{k+1}$, so we are done by induction.

6

(a) Observe that it suffices to show that $\left(\frac{\sin x}{x}\right)^3 - \cos x > 0$. It also suffices to show that this increases as x increases. To do this, we can show that the derivative is greater than 0. The derivative is

$$\left(\left(\frac{\sin x}{x}\right)^3\right)' + \sin x,$$

and by chain rule, the derivative of $\left(\frac{\sin x}{x}\right)^3$ is

$$3\left(\frac{\sin x}{x}\right)^2 \cdot \left(\frac{\sin x}{x}\right)' = \frac{3\sin^2 x}{X^2} \cdot \frac{x\cos x - \sin x}{x^2}$$

by product rule as well. Clearly this is greater than 0, as desired.

(b) Observe that if we have the graph of $y = f(x)$, then the graph of $y = f(x) + B$ is the graph of $y = f(x)$ shifted upwards by B units. The graph of $y = Af(x)$ is the graph of $y = f(x)$ stretched upwards by a factor of A. The graph of $y = f(x+b)$ is the graph of $y = f(x)$ shifted b units to the left. The graph of $y = f(ax)$ is the graph of $y = f(x)$ stretched horizontally by a factor of a. The graph of $-f(x)$ is the reflection of the graph of $y = f(x)$ over the x-axis. The graph of $y = f(-x)$ is the reflection of the graph of $y = f(x)$ over the y-axis.

Chapter 7: Functions of Severable Variables: Their Limits and Continuity

Zorich Mathematical Analysis

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7.1 The Space $\mathbb{R}^m$ and Its Subsets

1 Consider the set $E_1 = [-1,0)^n$ and $E_2 = (0,1]^n$. Then they don't have any points in common since the elements of a vector in E_1 are all negative and the elements of a vector in E_2 are all positive, but they all come infinitely close to the 0 vector, so the distance between the sets is zero.

2

(a) Differentiating both sides gives $\frac{2xx'}{a^2} + \frac{2yy'}{b^2} = 0$, so dividing by 2 and substituting $x = x_0$ and $y = y_0$ gives

$$\frac{2x_0x'}{a^2} + \frac{2yy'}{b^2} = 0 \implies \frac{y'}{x'} = \frac{dy}{dx} = -\frac{b^2x_0}{a^2y_0}.$$

Thus the slope of the tangent line is $-\frac{b^2x_0}{a^2y_0}$, so the equation of the tangent line is

$$\begin{aligned} y - y_0 &= -\frac{b^2x_0}{a^2y_0}(x - x_0) \\ \implies a^2y_0y - a^2y_0^2 &= -b^2x_0x + b^2x_0^2 \\ \implies a^2y_0^2 + b^2x_0^2 &= a^2y_0y + b^2x_0x. \end{aligned}$$

Dividing both sides by a^2b^2 gives

$$\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$$

because (x_0, y_0) lies on the ellipse and thus satisfies its equation.

(b) Let P be the point (x_0, y_0). We show that if we reflect F_1P on the tangent at P, we get F_2P. Note that it suffices to show that the angle that F_1P makes with the tangent equals the angle that F_2P makes with the tangent. Let $\overrightarrow{F_1P} = \overrightarrow{v_1}$ and $\overrightarrow{F_2P} = \overrightarrow{v_2}$. It then suffices to consider their dot products. Consider a vector $\overrightarrow{u} = \langle 0, -\frac{b^2x_0}{a^2y_0}\rangle$. Then it suffices to prove that $\frac{u\cdot v_1}{||u||||v_1||} = \frac{u\cdot v_2}{||u||||v_2}$, or $||\frac{v_1}{||v_1||}|| = ||\frac{v_2}{||v_2||}||$, which is 1, as desired.

(c) Observe that since the water is rotating at constant angular velocity ω, the acceleration vector is $\langle -\omega^2 \sin\omega t, -\omega^2\cos\omega t\rangle$, so the acceleration is $\frac{\omega^2}{g}$. Thus $f'(x) = \frac{\omega^2}{g}x$. Consider a function $f(x) = \frac{\omega^2}{2g}x^2$. The derivative of $f(x)$ is $\frac{\omega^2}{2g}\cdot 2x = \frac{\omega^2}{g}x$. If the axis is shifted by a constant, then the derivative of the constant is 0, which does not change.

3 Consider the graph $f(x)$. Then $f'(x)$ is the graph of its derivative, which passes through $(x_0, f'(x_0))$. Then the slope of the tangent at this point is equal to $f''(x_0)$, so the equation of the tangent line is $y - f'(x_0) = f''(x_0)(x - x_0)$. $f''(x) = 2$, so the horizontal component of the acceleration is the largest when $x_0 = 0$.

4

(a) Observe that this function is obviously continuous because each of the terms, $\varphi_n(x)$, are all continuous. Also, note that because every number, even irrational, can be expressed by an infinite base 2 representation, then x can be written in terms of

$$x = \sum_{i=1}^{\infty} \frac{a_i}{2^i},$$

where a_i is either 0 or 1. Thus there is always a term that does not have a derivative in the sum, so $f(x)$ does not have a derivative at every point.

(b) Let $P_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$.

$$\begin{aligned}
P_n(x_0) &= a_0 + a_1 x_0 + a_2 x_0^2 + \cdots + a_n x_0^n = \alpha_0 \\
P_n^{(1)}(x_0) &= a_1 + 2a_2 x_0 + 3a_3 x_0^2 + \cdots + n a_n x_0^{n-1} = \alpha_1 \\
P_n^{(2)}(x_0) &= 2a_2 + 3 \cdot 2a_3 x_0 + 4 \cdot 3a_4 x_0^2 + \cdots + n(n-1)a_n x_0^{n-2} = \alpha_2 \\
&\vdots \\
P_n^{(n-1)}(x_0) &= (n-1)\cdots 1 \cdot a_{n-1} + n(n-1)\cdots 2a_n x_0 = \alpha_{n-1} \\
P_n^{(n)}(x_0) &= n(n-1)\cdots 2 \cdot 1 a_n = \alpha_n.
\end{aligned}$$

So working our way from the bottom to the top, we get $a_n = \frac{\alpha_n}{n!}$, then we can use this to solve for a_{n-1} to get $a_{n-1} = \frac{\alpha_{n-1} - \alpha_n x_0}{(n-1)!}$, etc. Solving for all of the a_i gives our desired polynomial.

7.2 Limits and Continuity of Functions of Several Variables

1

(a) Assuming this means $f(x) = e^{-\frac{1}{x^2}}$ if $x \neq 0$. Then by chain rule,

$$f'(x) = \left(-\frac{1}{x^2}\right)' e^{-1/x^2} = 2x^{-3} e^{-1/x^2}$$

We need to find the derivative of $x^2 \sin \frac{1}{x}$. By the multiplication rule, this is $2x \cdot \sin \frac{1}{x} + x^2 \cdot (\sin \frac{1}{x})'$. By chain rule, $(\sin \frac{1}{x})' = -x^{-2} \cos \frac{1}{x}$, so we get $2x \sin \frac{1}{x} - \cos \frac{1}{x}$.

(b) We already have $f'(x)$ is differentiable. Then we can take the derivative of $f^{(2)}(x)$, etc. Since $f(x)$ consists of a smooth curve, then $f'(x)$, $f''(x)$, etc., will also be smooth, and hence it will be differentiable. Note that because we have an x^2, $f(x)$ is an even function, so $f'(x) = 0$, so $f''(x) = 0$, etc., so $f^{(n)}(x) = 0$.

(c) Note that we have $2x \sin \frac{1}{x} - \cos \frac{1}{x}$ if $x \neq 0$. This is defined on $\mathbb{R} \backslash 0$, and note that $f'(0) = 0$, so it's defined on all $\mathbb{R}$. As x approaches 0, $\frac{1}{x}$ approaches ∞, but since sin and cos is periodic, then this function will keep wavering, so it's not continuous.

(d) Observe that

$$e^{-\frac{1}{(1+x)^2}-\frac{1}{(1-x)^2}} = e^{-\frac{1}{(1+x)^2}} \cdot e^{-\frac{1}{(1-x)^2}}.$$

So since $f(x)$ is a product of functions that are infinitely differentiable on $\mathbb{R}$ they are $f(1+x)$ and $f(1-x)$ from part (a), then this function must be infinitely differentiable as well.

(e) Let $f \in C^{(\infty)}$. The first derivative of $x^{n-1}f\left(\frac{1}{x}\right)$ is

$$\begin{aligned} x^{n-1}\left(f\left(\frac{1}{x}\right)\right)' + (x^{n-1})'f\left(\frac{1}{x}\right) &= x^{n-1}\cdot\frac{-1}{x^2}f'\left(\frac{1}{x}\right) + (n-1)x^{n-2}f\left(\frac{1}{x}\right) \\ &= (n-1)x^{n-2}f\left(\frac{1}{x}\right) - x^{n-3}f'\left(\frac{1}{x}\right). \end{aligned}$$

Then the nth derivative of $x^{n-1}f\left(\frac{1}{x}\right)$ is $\frac{(-1)^n}{x^{n+1}}f^{(n)}\left(\frac{1}{x}\right)$, so we're done.

2 If f is an even function, $f(x) = f(-x)$. Then taking the derivative of both sides and applying the chain rule gives $f'(x) = (-x)'f'(-x) = -f'(-x)$, so f' is an odd function. If f is an odd function, $f(x) = -f(-x)$. Then taking the derivative of both sides and applying the chain rule gives $f'(x) = -(-x)'f'(-x) = f'(-x)$, so f' is an even function. We already know that if f is even, then f' is odd. We now prove the converse. If f' is odd, then $f'(x) = -f'(-x)$. Integrating gives $f(x) + c_1 = f(-x) + c_2$. Letting $x = 0$ gives $f(0) + c_1 = f(0) + c_2$, so $c_1 = c_2$. Thus $f(x) = f(-x)$, and f is even, as desired.

3

(a) If $f(x)$ is differentiable, then there exists a function $f'(x)$ such that $f(x) - f(x_0) = f'(x)(x - x_0)$, so in this case we can let $f'(x) = \varphi(x)$. If there exists a function $\varphi(x)$ such that $f(x) - f(x_0) = \varphi(x)(x - x_0)$, then $\varphi(x) = \frac{f(x)-f(x_0)}{x-x_0}$, which is the slope of the line between the points $(x, f(x))$ and $(x_0, f(x_0))$. Since the definition of derivative is the slope of the line tangent to a curve at the point, then as $x \to x_0$, $\varphi(x)$ approaches this slope. Hence $\varphi(x)$ is the derivative of f, and f is differentiable at x_0.

(b) Since we know that $\varphi(x)$ is the first-order derivative of $f(x)$, then if $\varphi(x)$ is $n-1$-differentiable, so is $f'(x)$. This means that $f(x)$ is n-differentiable, so it has a derivative of order n at x_0. We need to construct a function f that is continuous at x_0 has an inverse f^{-1}, and f^{-1} is not continuous at y_0 Consider $f(x) = x$ if $x \le 0$ and $f(x) = x^2$ if $x \ge 0$ at $x = 0$. Then f is continuous at x_0 but not continuous at y_0, and the theorem doesn't apply.

(c) $m_1a_1 = F = \frac{Gm_1m_2}{r^2}$, $a_1 = \frac{Gm_2}{r^2}$ and $m_2a_2 = F$, so $a_2 = \frac{Gm_1}{r^2}$. Then v_1^2 increases by $2\alpha_1\Delta r$ and v_2^2 increases by $2\alpha_2\Delta r$, so we get $\frac{(2Gm_1+2Gm_2)\Delta r}{r^2} - \frac{Gm_1m_2}{r} = 0$, so E is constant. E is the total energy of the system. K is the kinetic energy, or energy in motion, and U is the potential energy, which is kind of like energy waiting to be converted to kinetic energy. The total energy of n bodies is constant if it is under no external force.

(d) Observe that

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots.$$

We also have

$$\begin{aligned}\frac{1+ax^2}{1+bx^2} &= (1+ax^2)(1-bx^2+b^2x^4-b^3x^6+\cdots)\\ &= 1+ax^2-bx^2-abx^4+b^2x^4+ab^2x^6-b^3x^6+\cdots\\ &= 1+(a-b)x^2-(b^2-ab)x^4+(ab^2-b^3)x^6+\cdots\end{aligned}$$

To maximize the order, we want at least the first 3 coefficients of each series to match up, so $a-b=-\frac{1}{2}$ and $b(b-a)=\frac{1}{24}$. Hence, solving gives $b=\frac{1}{12}$ and $a=-\frac{5}{12}$.

(e) Observe that

$$x\left[\frac{1}{e}-\left(\frac{x}{x+1}\right)^x\right]=\frac{\frac{1}{e}-\left(\frac{x}{x+1}\right)^x}{1/x}.$$

Observe that since both numerator and denominator approach 0 as x approaches ∞, then we can use L'Hopitals rule. This gives

$$\begin{aligned}\lim_{x\to\infty}\frac{\frac{1}{e}-\left(\frac{x}{x+1}\right)^x}{1/x} &= \lim_{x\to\infty}\frac{\left(\frac{1}{e}-\left(\frac{x}{x+1}\right)^x\right)}{(1/x)'}\\ &= \lim_{x\to\infty}\frac{-\left(\left(\frac{x}{x+1}\right)^x\right)'}{-\frac{1}{x^2}}\\ &= \lim_{x\to\infty}x^2\left(\left(\frac{x}{x+1}\right)^x\right)'.\end{aligned}$$

(f) We already know that $(k^x)'=k^x\cdot\ln x$, so using chain rule gives

$$\left(\left(\frac{x}{x+1}\right)^x\right)'=\left(\frac{x}{x+1}\right)'\cdot\left(\frac{x}{x+1}\right)^x\cdot\ln\frac{x}{x+1}.$$

Since $\frac{x}{x+1}=1-\frac{1}{x+1}$, then $\left(\frac{x}{x+1}\right)'=-\left(\frac{1}{x+1}\right)'=\frac{1}{(x+1)^2}$. Therefore, the limit is

$$\lim_{x\to\infty}\frac{x^2}{(x+1)^2}\cdot\ln\left(\frac{x}{x+1}\right)\cdot\left(\frac{x}{x+1}\right)^x=1.$$

Chapter 8: Differential Calculus in Several Variables

Zorich Mathematical Analysis

Carissa Zeng

November 5, 2024

8.3 The Basic Laws of Differentiation

1

(a) Differentiating both sides gives $\frac{2xx'}{a^2} + \frac{2yy'}{b^2} = 0$, so dividing by 2 and substituting $x = x_0$ and $y = y_0$ gives

$$\frac{2x_0x'}{a^2} + \frac{2yy'}{b^2} = 0 \implies \frac{y'}{x'} = \frac{dy}{dx} = -\frac{b^2x_0}{a^2y_0}.$$

Thus the slope of the tangent line is $-\frac{b^2x_0}{a^2y_0}$, so the equation of the tangent line is

$$\begin{aligned} y - y_0 &= -\frac{b^2x_0}{a^2y_0}(x - x_0) \\ \implies a^2y_0y - a^2y_0^2 &= -b^2x_0x + b^2x_0^2 \\ \implies a^2y_0^2 + b^2x_0^2 &= a^2y_0y + b^2x_0x. \end{aligned}$$

Dividing both sides by a^2b^2 gives

$$\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$$

because (x_0, y_0) lies on the ellipse and thus satisfies its equation.

(b) Let P be the point (x_0, y_0). We show that if we reflect F_1P on the tangent at P, we get F_2P. Note that it suffices to show that the angle that F_1P makes with the tangent equals the angle that F_2P makes with the tangent. Let $\overrightarrow{F_1P} = \overrightarrow{v_1}$ and $\overrightarrow{F_2P} = \overrightarrow{v_2}$. It then suffices to consider their dot products. Consider a vector $\overrightarrow{u} = \langle 0, -\frac{b^2x_0}{a^2y_0} \rangle$. Then it suffices to prove that $\frac{u \cdot v_1}{||u|| ||v_1||} = \frac{u \cdot v_2}{||u|| ||v_2||}$, or $|| \frac{v_1}{||v_1||} || = || \frac{v_2}{||v_2||} ||$, which is 1, as desired.

(c)

$$\sin(\frac{\pi}{6} + \alpha) = \sin\frac{\pi}{6}\cos\alpha + \sin\alpha\cos\frac{\pi}{6} = \frac{1}{2}\sqrt{1 - \alpha^2} + \frac{\sqrt{3}}{2}\alpha.$$

$\alpha^\circ = \frac{\pi\alpha}{180}$ radians and 30° is $\pi/6$ radians, so we can convert this to radians. So we have

$$\sin(\frac{\pi}{6} + \frac{\pi\alpha}{180}) = \frac{1}{2}\sqrt{1 - \left(\frac{\pi\alpha}{180}\right)} + \frac{\sqrt{3}}{2}\frac{\pi\alpha}{180}.$$

(d)
$$\cos(\frac{\pi}{4}+\alpha) = \cos\frac{\pi}{4}\cos\alpha - \sin\frac{\pi}{4}\sin\alpha = \sqrt{2}2\sqrt{1-\alpha^2} - \frac{\sqrt{2}}{2}\alpha.$$
As above, $\alpha = \frac{\pi\alpha}{180}$, so
$$\cos(\frac{\pi}{4}+\frac{\pi\alpha}{180}) = \sqrt{2}2\sqrt{1-\left(\frac{\pi\alpha}{180}\right)^2} - \frac{\sqrt{2}}{2}\cdot\frac{\pi\alpha}{180}.$$

2

(a) I think it's an elliptic paraboloid or something. Just graph all the ellipses $x^2+4y^2 = c$ on the plane $z = c$.

(b) Observe that since the water is rotating at constant angular velocity ω, the acceleration vector is $\langle -\omega^2\sin\omega t, -\omega^2\cos\omega t\rangle$, so the acceleration is $\frac{\omega^2}{g}$. Thus $f'(x) = \frac{\omega^2}{g}x$. Consider a function $f(x) = \frac{\omega^2}{2g}x^2$. The derivative of $f(x)$ is $\frac{\omega^2}{2g}\cdot 2x = \frac{\omega^2}{g}x$. If the axis is shifted by a constant, then the derivative of the constant is 0, which does not change.

(c) The gradient is $\text{grad} f(x,y) = \langle 2x, 8y\rangle$. Consider the graph $f(x)$. Then $f'(x)$ is the graph of its derivative, which passes through $(x_0, f'(x_0))$. Then the slope of the tangent at this point is equal to $f''(x_0)$, so the equation of the tangent line is $y - f'(x_0) = f''(x_0)(x-x_0)$. $f''(x) = 2$, so the horizontal component of the acceleration is the largest when $x_0 = 0$.

(d) Observe that this function is obviously continuous because each of the terms, $\varphi_n(x)$, are all continuous. Also, note that because every number, even irrational, can be expressed by an infinite base 2 representation, then x can be written in terms of
$$x = \sum_{i=1}^{\infty}\frac{a_i}{2^i},$$
where a_i is either 0 or 1. Thus there is always a term that does not have a derivative in the sum, so $f(x)$ does not have a derivative at every point.

(e) Who needs a computer to find the minimum, it's obviously $f(0,0) = 0$. But if you wanted to, you could make the computer find the solutions to $f_x(x,y) = 0$ and $f_y(x,y) = 0$, which is basically $2x = 0$ and $8y = 0$, basically, $(0,0,0)$.

3

(a) $\text{grad} f_1(x,y) = \langle 2x, 2y\rangle$, $\text{grad} f_2(x,y) = \langle -2x, -2y\rangle$, and
$$\text{grad} f_3(x,y) = \left\langle \frac{1/y}{1+x^2}, \frac{-x/y^2}{1+(1/y)^2}\right\rangle = \left\langle \frac{1}{y(1+x^2)}, -\frac{x}{1+y^2}\right\rangle.$$
$\text{grad} f_4(x,y) = \langle y, x\rangle$.

(b) Let $P_n(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$.
$$\begin{aligned}
P_n(x_0) &= a_0 + a_1x_0 + a_2x_0^2 + \cdots + a_nx_0^n = \alpha_0\\
P_n^{(1)}(x_0) &= a_1 + 2a_2x_0 + 3a_3x_0^2 + \cdots + na_nx_0^{n-1} = \alpha_1\\
P_n^{(2)}(x_0) &= 2a_2 + 3\cdot 2a_3x_0 + 4\cdot 3a_4x_0^2 + \cdots + n(n-1)a_nx_0^{n-2} = \alpha_2\\
&\vdots\\
P_n^{(n-1)}(x_0) &= (n-1)\cdots 1\cdot a_{n-1} + n(n-1)\cdots 2a_nx_0 = \alpha_{n-1}\\
P_n^{(n)}(x_0) &= n(n-1)\cdots 2\cdot 1a_n = \alpha_n.
\end{aligned}$$

So working our way from the bottom to the top, we get $a_n = \frac{\alpha_n}{n!}$, then we can use this to solve for a_{n-1} to get $a_{n-1} = \frac{\alpha_{n-1}-\alpha_n x_0}{(n-1)!}$, etc. Solving for all of the a_i gives our desired polynomial.

(c) Assuming this means $f(x) = e^{-\frac{1}{x^2}}$ if $x \neq 0$. Then by chain rule,

$$f'(x) = \left(-\frac{1}{x^2}\right)' e^{-1/x^2} = 2x^{-3}e^{-1/x^2}$$

We need to find the derivative of $x^2 \sin\frac{1}{x}$. By the multiplication rule, this is $2x \cdot \sin\frac{1}{x} + x^2 \cdot (\sin\frac{1}{x})'$. By chain rule, $(\sin\frac{1}{x})' = -x^{-2}\cos\frac{1}{x}$, so we get $2x\sin\frac{1}{x} - \cos\frac{1}{x}$.

(d) We already have $f'(x)$ is differentiable. Then we can take the derivative of $f^{(2)}(x)$, etc. Since $f(x)$ consists of a smooth curve, then $f'(x)$, $f''(x)$, etc., will also be smooth, and hence it will be differentiable. Note that because we have an x^2, $f(x)$ is an even function, so $f'(x) = 0$, so $f''(x) = 0$, etc., so $f^{(n)}(x) = 0$.

4

(a) Note that we have $2x\sin\frac{1}{x} - \cos\frac{1}{x}$ if $x \neq 0$. This is defined on $\mathbb{R}\backslash 0$, and note that $f'(0) = 0$, so it's defined on all $\mathbb{R}$. As x approaches 0, $\frac{1}{x}$ approaches ∞, but since sin and cos is periodic, then this function will keep wavering, so it's not continuous. Observe that

$$e^{-\frac{1}{(1+x)^2}-\frac{1}{(1-x)^2}} = e^{-\frac{1}{(1+x)^2}} \cdot e^{-\frac{1}{(1-x)^2}}.$$

So since $f(x)$ is a product of functions that are infinitely differentiable on $\mathbb{R}$ they are $f(1+x)$ and $f(1-x)$ from part (a), then this function must be infinitely differentiable as well.

(b) Let $f \in C^{(\infty)}$. The first derivative of $x^{n-1}f\left(\frac{1}{x}\right)$ is

$$\begin{aligned} x^{n-1}\left(f\left(\frac{1}{x}\right)\right)' + (x^{n-1})'f\left(\frac{1}{x}\right) &= x^{n-1} \cdot \frac{-1}{x^2} f'\left(\frac{1}{x}\right) + (n-1)x^{n-2}f\left(\frac{1}{x}\right) \\ &= (n-1)x^{n-2}f\left(\frac{1}{x}\right) - x^{n-3}f'\left(\frac{1}{x}\right). \end{aligned}$$

Then the nth derivative of $x^{n-1}f\left(\frac{1}{x}\right)$ is $\frac{(-1)^n}{x^{n+1}}f^{(n)}\left(\frac{1}{x}\right)$, so we're done.

(c) If f is an even function, $f(x) = f(-x)$. Then taking the derivative of both sides and applying the chain rule gives $f'(x) = (-x)'f'(-x) = -f'(-x)$, so f' is an odd function. If f is an odd function, $f(x) = -f(-x)$. Then taking the derivative of both sides and applying the chain rule gives $f'(x) = -(-x)'f'(-x) = f'(-x)$, so f' is an even function. We already know that if f is even, then f' is odd. We now prove the converse. If f' is odd, then $f'(x) = -f'(-x)$. Integrating gives $f(x) + c_1 = f(-x) + c_2$. Letting $x = 0$ gives $f(0) + c_1 = f(0) + c_2$, so $c_1 = c_2$. Thus $f(x) = f(-x)$, and f is even, as desired.

(d) If $f(x)$ is differentiable, then there exists a function $f'(x)$ such that $f(x) - f(x_0) = f'(x)(x - x_0)$, so in this case we can let $f'(x) = \varphi(x)$. If there exists a function $\varphi(x)$ such that $f(x) - f(x_0) = \varphi(x)(x - x_0)$, then $\varphi(x) = \frac{f(x)-f(x_0)}{x-x_0}$, which is the slope of the line between the points $(x, f(x))$ and $(x_0, f(x_0))$. Since the definition of derivative is the slope of the line tangent to a curve at the point, then as $x \to x_0$, $\varphi(x)$ approaches this slope. Hence $\varphi(x)$ is the derivative of f, and f is differentiable at x_0.

(e) Since we know that $\varphi(x)$ is the first-order derivative of $f(x)$, then if $\varphi(x)$ is $n-1$-differentiable, so is $f'(x)$. This means that $f(x)$ is n-differentiable, so it has a derivative of order n at x_0. We need to construct a function f that is continuous at x_0 has an inverse f^{-1}, and f^{-1} is not continuous at y_0. Consider $f(x) = x$ if $x \leq 0$ and $f(x) = x^2$ if $x \geq 0$ at $x = 0$. Then f is continuous at x_0 but not continuous at y_0, and the theorem doesn't apply.

(f) $m_1 a_1 = F = \frac{Gm_1m_2}{r^2}$, $a_1 = \frac{Gm_2}{r^2}$ and $m_2a_2 = F$, so $a_2 = \frac{Gm_1}{r^2}$. Then v_1^2 increases by $2\alpha_1 \Delta r$ and v_2^2 increases by $2\alpha_2\Delta r$, so we get $\frac{(2Gm_1+2Gm_2)\Delta r}{r^2} - \frac{Gm_1m_2}{r} = 0$, so E is constant. E is the total energy of the system. K is the kinetic energy, or energy in motion, and U is the potential energy, which is kind of like energy waiting to be converted to kinetic energy.

(g) Observe that

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots.$$

We also have

$$\begin{aligned}\frac{1+ax^2}{1+bx^2} &= (1+ax^2)(1-bx^2+b^2x^4-b^3x^6+\cdots)\\ &= 1+ax^2-bx^2-abx^4+b^2x^4+ab^2x^6-b^3x^6+\cdots\\ &= 1+(a-b)x^2-(b^2-ab)x^4+(ab^2-b^3)x^6+\cdots\end{aligned}$$

To maximize the order, we want at least the first 3 coefficients of each series to match up, so $a - b = -\frac{1}{2}$ and $b(b-a) = \frac{1}{24}$. Hence, solving gives $b = \frac{1}{12}$ and $a = -\frac{5}{12}$.

(h) This is because tea leaves prefer places of equilibrium. If they're off to the side, then there's more force on one side than the other, so they're pushed to the center, and the water above pushes them down, so they end up in the center, at the bottom of the cup.

5

(a) Observe that

$$x\left[\frac{1}{e} - \left(\frac{x}{x+1}\right)^x\right] = \frac{\frac{1}{e} - \left(\frac{x}{x+1}\right)^x}{1/x}.$$

Observe that since both numerator and denominator approach 0 as x approaches ∞, then we can use L'Hopitals rule. This gives

$$\begin{aligned}\lim_{x\to\infty}\frac{\frac{1}{e} - \left(\frac{x}{x+1}\right)^x}{1/x} &= \lim_{x\to\infty}\frac{\left(\frac{1}{e} - \left(\frac{x}{x+1}\right)^x\right)}{(1/x)'}\\ &= \lim_{x\to\infty}\frac{-\left(\left(\frac{x}{x+1}\right)^x\right)'}{-\frac{1}{x^2}}\\ &= \lim_{x\to\infty} x^2\left(\left(\frac{x}{x+1}\right)^x\right)'.\end{aligned}$$

(b) We already know that $(k^x)' = k^x \cdot \ln x$, so using chain rule gives

$$\left(\left(\frac{x}{x+1}\right)^x\right)' = \left(\frac{x}{x+1}\right)' \cdot \left(\frac{x}{x+1}\right)^x \cdot \ln\frac{x}{x+1}.$$

Since $\frac{x}{x+1} = 1 - \frac{1}{x+1}$, then $\left(\frac{x}{x+1}\right)' = -\left(\frac{1}{x+1}\right)' = \frac{1}{(x+1)^2}$. Therefore, the limit is

$$\lim_{x\to\infty} \frac{x^2}{x+1}^2 \cdot \ln\left(\frac{x}{x+1}\right)\cdot\left(\frac{x}{x+1}\right)^e = 1.$$

6

(a) By Lagrange's form of the remainder,

$$r_n(x_0;x) = \frac{1}{(n+1)!}f^{(n+1)}(\xi)(x-x_0)^{n+1} = \frac{1}{(n+1)!}\cdot e^{\xi}\cdot x^{n+1} < 10^{-3}.$$

Since x is at most 2, it suffices to find the smallest value of n such that $\frac{1}{(n+1)!}\cdot 2^{n+1} \le 10^{-3}$, which is $n = 10$. Thus the taylor formula is

$$1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \cdots + \frac{1}{10!}x^{10}.$$

(b) If f is even, then f' is odd, so f'' is even, so $f^{(3)}$ is odd, etc. Thus all odd-ordered derivatives of f are odd and even-ordered derivatives of f are even. Thus $f^{(2k+1)}(x) = -f^{(2k+1)}(-x)$, and plugging in $x = 0$ gives $f^{(2k+1)}(0) = -f^{(2k+1)}(0)$, so $f^{(2k+1)}(0) = 0$. Therefore all odd powers of the Taylor series are 0, and hence the Taylor series consists only of even terms.

(c) Since we know that $f^{(n)}(0) = 0$ for all nonnegative n, then the Taylor series at 0 is

$$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n = 0.$$

Also, note that the remaining terms are of the form

$$\left|\frac{f^{(n)}(x)}{n!}x^n\right| \le \frac{\sup\limits_{-1\le x\le 1}|f^{(n)}(x)|}{n!}n!|x|^n \le \frac{n!C}{n!}|x|^n = C|x|^n.$$

Since $|x| < 1$, then this is a geometric series, which converges, so $f \equiv 0$ on $[-1, 1]$.

(d) Observe that since the derivative of a minimum or maximum point equals 0, if there is an $x \in I_1$ satisfying $|f^{(k-1)}(x)|$ is minimized, the derivative of $f^{(k-1)}$, which is $f^{(k)}$, satisfies $f^{(k)}(x) = 0$, unless x is a boundary point. Obviously $m_k(I) = 0$ if any of x are not boundary points, and it will be less than $\frac{1}{\mu}(m_{k-1}(I_1) + m_{k-1}(I_3))$. Otherwise, if all of x are boundary points, then the solution to $m_k(I)$ is also a boundary point, and it must equal one of $m_{k-1}(I_1)$ or $m_{k-1}(I_3)$, so we are also done.

(e) We have base case $k = 0$; $m_1(I) \le 1$. Assume that for some k, we have that

$$m_k(I) \le \frac{2^{k(k+1)/2}k^k}{\lambda^k}.$$

Then we have

$$m_{k+1}(I) \le \frac{1}{\mu}(m_k(I_1) + m_k(I_3))$$

by part (a), so

$$m_{k+1}(I) \leq \frac{1}{\mu}\left(\frac{2^{k(k+1)/2}k^k}{\lambda_1^k} + \frac{2^{k(k+1)/2}k^k}{\lambda_3^k}\right) = \frac{2^{k(k+1)/2}k^k}{\mu}\left(\frac{1}{\lambda_1^k} + \frac{1}{\lambda_3^k}\right).$$

We wish to prove that

$$m_{k+1}(I) \leq \frac{2^{(k+1)(k+2)/2}(k+1)^{k+1}}{\lambda^{k+1}}.$$

$\lambda = \lambda_1 + \lambda_2 + \mu$ and $2^{(k+1)(k+2)/2} = (k+1)2^{k(k+1)/2}$. Therefore, we are done.

7

(a) It suffices to show, by induction, that there exists k numbers $x_{k_1} < x_{k_2} < \cdots < x_{k_k}$ such that $f^{(k)}(x_{k_i})f^{(k)}(x_{k_{i+1}}) < 0$, so they are of opposite sign, which will show that there is a root between x_{k_i} and $x_{k_{i+1}}$. We have base case $n = 1$. If $|f'(0)| > a_1$, then $f'(x) = 0$ has at least 0 distinct roots. Suppose this works for some value of k. We can start by using the values $x_{k_1}, \dots, x_{k_k}$ for $x_{(k+1)_1}, \dots, x_{(k+1)_k}$. Then we can add $x_{(k+1)_{k+1}}$ satisfying $f^{(k+1)}(x_{(k+1)_{k+1}})$ is of opposite sign from $f^{(k+1)}(x_{k_k})$, so we're done.

(b) Consider the auxiliary function $h(x) = f(x) - r(x)$, where $r(x) = f'(a)(x-a)$. Then $h'(x) = f'(x) - f'(a)$ and $h(a) = h(b) = 0$, so there exists ξ such that $h'(\xi) = 0$. Then $f'(\xi) = f'(a)$, and we can replace $f'(a)$ with $f'(\xi)$ for any ξ so that $f'(x)$ attains all values between $f'(a)$ and $f'(b)$.

(c) Lagrange's infinite-increment theorem states that if a function $f : [a, b] \to \mathbb{R}$ is continuous on a closed interval $[a, b]$ and differentiable on the open interval, $]a, b[$, there exists a point $\xi \in]a, b[$ such that $f(b) - f(a) = f'(\xi)(b - a)$. Because we know that f'' exists, then f' must be differentiable on $]a, b[$, so Lagrange's theorem holds for f'. Hence there exists $\xi \in]a, b[$ such that $f'(b) - f'(a) = f''(\xi)(b - a)$, as desired.

(d) Suppose for the sake of contradiction that $f'(x)$ has a discontinuity of the first kind. Suppose at some point $x = a$, $f'(a)$ has a discontinuity. This means that $f'(x)$ has a limit at a from both the left side and the right side, and these limits are equal, but $f'(a)$ is not equal to this limit. But because $f'(a)$ exists, this is a contradiction, so $f'(x)$ has no removable discontinuities, so it can only have discontinuities of the second kind.

(e) The error in the proof lies in the assumption that $\lim_{x\to x_0} f'(\xi) = \lim_{\xi\to x_0} f'(\xi)$. It should have been $\lim_{x\to x_0} f'(\xi(x))$ where $\xi(x)$ is a function that lies between x and x_0. Note that the limit of $f'(\xi(x))$ as $x \to x_0$ does not necessarily equal $f'(x_0)$ since $\xi(x)$ is a function of x and does not necessarily to approach x_0 in the same way as x. The "proof" assumes that the limit equals $f'(x_0)$, so it is incorrect.

(f) Let $x \in [-a, a]$. Then by the mean value theorem, there exists a point $c \in [-a, x]$ if $x > 0$ or $c \in [x, a]$ if $x < 0$ such that $f'(x) - f'(0) = f''(c) \cdot x$. Since $|f'(x)| \leq |f'(0)| + |f''(c)||x|$, since $c \in [-a, a]$, then $|f''(c)| \leq M_2$ and $|x| \leq a$. Hence we have $|f'(x)| \leq |f'(0)| + M_2|x|$.

Note that for some $d \in [-a, a]$, $f'(0) - f(0) = f'(d) \cdot 0$. We know that $|f'(0)| \leq |f(0)| + |f'(d)|$. Since $d \in [-a, a]$, then $|f'(d)| \leq M_1$ and $|f(0)| \leq M_0$, so $|f'(0)| \leq M_0 + M_1$. Then $|f'(x)| \leq M_0 + M_1 + M_2|x|$, so $M_1 \leq \frac{M_0}{a} + \frac{a}{2}M_2$. Since the coefficient of M_2, $\frac{x^2+a^2}{2a}$ is at least equal to $\frac{a}{2}$, we are done.

(g) We prove this by induction on p. We have the base case $p = 2$ from part 9(b), and assume this works for some p. We prove that it also works for $p + 1$. First note that by AM-GM,

$$\left(1 - \frac{k}{p}\right) M_0 + \frac{k}{p} M_p = \frac{\underbrace{M_p + M_p + \cdots + M_p}_{\text{summed } k \text{ times}} + \underbrace{M_0 + M_0 + \cdots + M_0}_{\text{summed } p-k \text{ times}}}{p}$$
$$\geq \sqrt[p]{M_p^k M_0^{p-k}} = M_0^{1-k/p} \cdot M_p^{k/p}.$$

Thus it suffices to prove that

$$M_k \leq 2^{k(p-k)/2} \left((1 - \frac{k}{p}) M_0 + \frac{k}{p} M_p \right)$$

which is obvious, so we are done.

(h) Observe that when x is at its maximum, $x = a$, so from part (a), $M_1 \leq \frac{M_0}{a} + aM_2$. If the length of I, which equals $2a$, is not less than $2\sqrt{M_0/M_2}$, then $a \geq \sqrt{M_0/M_2}$. By AM-GM, $\frac{M_0}{a} + aM_2 \geq 2\sqrt{M_0 M_2}$, which proves the first case.

For the second case, if $I = \mathbb{R}$, then we can't have $x = a$, and a approaches infinity. The minimum value of $\frac{M_0}{a} + \frac{a}{2} M_2$ is $\sqrt{2M_0 M_2}$ by AM-GM.

8.4 Real-valued Functions of Several Variables

1

(a) Yes, because this means that y is not involved in $f(x, y)$ (if it were, then its derivative would certainly not equal) because $f(x, y)$ is literally a constant to y.

(b) If we have equality cases, then we cannot have smaller numbers. For the number 2, we have $a = \sqrt{M_0/M_2}$, and for the number $\sqrt{2}$, we have $a = \sqrt{2M_0/M_2}$.

2

(a) We prove this by induction on p. We have the base case $p = 2$ from part 9(b), and assume this works for some p. We prove that it also works for $p + 1$. First note that by AM-GM,

$$\left(1 - \frac{k}{p}\right) M_0 + \frac{k}{p} M_p = \frac{\underbrace{M_p + M_p + \cdots + M_p}_{\text{summed } k \text{ times}} + \underbrace{M_0 + M_0 + \cdots + M_0}_{\text{summed } p-k \text{ times}}}{p}$$
$$\geq \sqrt[p]{M_p^k M_0^{p-k}} = M_0^{1-k/p} \cdot M_p^{k/p}.$$

Thus it suffices to prove that

$$M_k \leq 2^{k(p-k)/2} \left((1 - \frac{k}{p}) M_0 + \frac{k}{p} M_p \right)$$

which is obvious, so we are done.

(b) By Lagrange's form of remainder, we get

$$r_n(x_0; x) = \frac{1}{n!} f^{(n+1)}(\xi)(x - x_0)^n,$$

so it suffices to prove that $\xi = x_0 + \theta(x - x_0)$. By the mean value theorem, there exists $\xi \in [0, 1]$ such that $f(x) - f(x_0) = f'(\xi)(x - x_0)$, so if we replace $f'(\xi)(x - x_0)$ with $\theta(x - x_0)$ and replace $f(x)$ with ξ, we get $\xi - x_0 = \theta(x - x_0)$.

3

(a) The Taylor expansion of f at 0 is

$$f(x) = 1 + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n.$$

Since f has $n+1$ roots, it has order $n+1$, so $f^{(n)}$ has order 1, which means that it has a root.

(b) Consider the Taylor expansion of f at 0, which is

$$f(x) = 1 + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n.$$

Observe that a polynomial is determined by any n values, and this will determine a polynomial of degree $n-1$, which is perfect for $L(x)$. We can use the method of undetermined coefficients to solve for $L(x)$. Note that because $f(x)$ can also be expressed in polynomial form to a highly accurate degree, then $f(x) - L(x) = 0$ for $x = x_1, \ldots, x_n$. So since $f(x) - L(x)$ is a polynomial, it must be a scalar multiple of $(x - x_1)\cdots(x - x_n)$, since $f(x) - L(x)$ has degree n. It suffices to prove that the remaining coefficient equals $\frac{f^{(n)}(\xi)}{n!}$ for some $\xi \in I$, which follows from the mean value theorem.

4

(a) Since $n_1 + n_2 + \cdots + n_p = n$, then the sum of the orders, and the number of roots of f, f', f'', etc is $n_1 + n_2 + \cdots + n_p = n$. Therefore, once you take the derivative $n-1$ times to get $f^{(n-1)}$, the remaining order is 1, so it has one root, so there exists ξ such that $f^{(n-1)}(\xi) = 0$.

(b) Let $g(x) = f(x) - H(x)$, so note that $g^{(k)}(x) = f^{(k)}(x) - H^{(k)}(x)$. Then for x_i, $g(x_i) = g'(x_i) = \cdots = g^{(n_i-1)}(x_i) = 0$, and we want to prove that

$$g(x) = \frac{(x - x_1)^{n_1} \cdots (x - x_p)^{n_p}}{n!} f^{(n)}(\xi).$$

If we look at the taylor polynomial forms of $g(x)$, $g'(x)$, $g''(x)$, and so on, we see that for x_i, we have that $g(x), g'(x), g''(x), \ldots, g^{(n_i-1)}(x)$ are multiples of $(x - x_i)$. But in order for $g^{(n_i-1)}(x)$ to be a multiple of $(x - x_i)$, $g(x)$ needs to be a multiple of $(x - x_i)^{n_i}$, since the $n_i - 1$th derivative of $(x - x_i)^{n_i}$ is $n_i! \cdot (x - x_i)$. So over all i, $g(x)$ is a multiple of $(x - x_1)^{n_1} \cdots (x - x_p)^{n_p}$, and the there must exist ξ such that the coefficient is $\frac{f^{(n)}(\xi)}{n!}$.

(c) Consider 2 roots of $P(x)$ and call them x_1 and x_2, so $P(x_1) = P(x_2)$. By the mean value theorem, there exists $\xi \in [x_1, x_2]$ such that $P(x_2) - P(x_1) = 0 = P'(\xi)(x_2 - x_1)$, so $P'(\xi) = 0$. Suppose $P(x)$ has degree n and that the multiplicity of a root a is k, so let $P(x) = (x-a)^k Q(x)$ for some polynomial Q. Observe that Q does not have a factor of $x - a$. Then by the chain rule,

$$\begin{aligned}
P'(x) &= ((x-a)^k)'Q(x) + (x-a)^k Q'(x) \\
&= k(x-a)^{k-1}Q(x) + (x-a)^k Q'(x) \\
&= (x-a)^{k-1}(kQ(x) + (x-a)Q'(x)).
\end{aligned}$$

Obviously $P'(x)$ is also has a as a root, but its multiplicity is $k - 1$.

(d) Suppose the roots of $P(x)$ are $a_1, \dots, a_k$ with multiplicities $n_1, \dots, n_k$, so that

$$P(x) = c(x-a_1)^{n_1}(x-a_2)^{n_2}\cdots(x-a_k)^{n_k},$$

where c is a constant. Then by the previous part, we know that $P(x)$ has the roots $a_1, \dots, a_k$ with multiplicities $n_1 - 1, \dots, n_k - 1$, so $P'(x)$ is a scalar multiple of

$$(x-a_1)^{n_1-1}(x-a_2)^{n_2-1}\cdots(x-a_k)^{n_k-1},$$

and thus $Q(x)$ is also a scalar multiple of

$$(x-a_1)^{n_1-1}(x-a_2)^{n_2-1}\cdots(x-a_k)^{n_k-1}.$$

Hence dividing $P(x)$ by $Q(x)$ leaves us with a scalar multiple of $(x-a_1)(x-a_2)\cdots(x-a_k)$, so $\frac{P(x)}{Q(x)}$ has $a_1, \dots, a_k$ as its roots, but all of them have multiplicity 1.

5

(a) Take a polynomial $Q(x) = c_0 + c_1 x + \cdots + c_n x^n$. Then obviously if we take $P(x) = Q(x - x_0)$, then $P(x) = c_0 + c_1(x-x_0) + \cdots + c_n(x-x_0)^n$. So in order to express $P(x)$ in terms of $x - x_0$, we first find $P(x + x_0) = Q(x)$, and find the coefficients of $Q(x)$. Then we plug $x - x_0$ into $Q(x)$ to get $P(x)$ in terms of $x - x_0$. Consider the Taylor expansion of $f(x)$ with respect to x_0, which looks like

$$f(x) = 1 + \frac{f'(x_0)}{1!}x + \cdots + \frac{f^{(n)}(x_0)}{n!}x^n + o((x-x_0)^n).$$

Then if we let $P(x)$ be the first $n+1$ terms, we're done.

(b) By the mean value theorem, there exists a value ξ_{1_1} such that $f(x_0 + h_1) - f(x_0) = f'(\xi_1) \cdot h_1$ and a value ξ_{1_2} such that $f(x_0 + h_1 + h_2) - f(x_0 + h_2) = f'(\xi_2) \cdot h_1$, so there exists a value ξ_2 such that

$$\begin{aligned}(f(x_0+h_1+h_2) - f(x_0+h_2)) - (f(x_0+h_1) - f(x_0))\\ = f'(\xi_{1_2}) \cdot h_1 - f'(\xi_{1_1}) \cdot h_1\\ = (f'(\xi_{1_2}) - f'(\xi_{1_1})) \cdot h_1 = f''(\xi_2)h_1h_2.\end{aligned}$$

As you can see, we have the base case. Assume there exists a value ξ_{k_1} such that

$$\Delta^k f(x_0; h_1, \dots, h_{k-1}) = f^{(k)}(\xi_{k_1})h_1 \cdots h_k$$

and

$$\Delta^k f(x_0; h_1, \dots, h_k) = f^{(k)}(\xi_{k_2})h_1 \cdots h_k.$$

(c) Then

$$\begin{aligned}\Delta^{k+1} f(x_0; h_1, \dots, h_{k+1}) &= \Delta^k f(x_0; h_1, \dots, h_k) - \Delta^k f(x_0; h_1, \dots, h_{k-1})\\ &= (f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1})) \cdot h_1 \cdots h_k.\end{aligned}$$

By the mean value theorem, there exists some ξ such that

$$f^{(n+1)}(\xi) \cdot h_{k+1} = f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1}),$$

so substituting this in gives $f^{(k+1)}(\xi)h_1 \dots h_{k+1}$, so we are done by induction.

(d) Using our result from part (a), we know that

$$\begin{aligned}
&|\Delta^n f(x_0; h_1, \dots, h_n) - f^{(n)}(x_0)h_1 \cdots h_n| \\
&\quad = |f^{(n)}(\xi)h_1 \cdots h_n - f^{(n)}(x_0)h_1 \cdots h_n| \\
&\quad = |f^{(n)}(\xi) - f^{(n)}(x_0)| \cdot |h_1| \cdots |h_n| \\
&\quad \le \sup_{x \in]a,b[} \left| f^n(x) - f^{(n)}(x_0) \right| \cdot |h_1| \cdots |h_n|,
\end{aligned}$$

as desired.

6 From part (a), we know that there exists ξ such that

$$\Delta^n f(x_0; h^n) = f^{(n)}(\xi)h^n \implies f^{(n)}(\xi) = \frac{\Delta^n f(x_0; h^n)}{h^n}.$$

As $h \to 0$, $\xi \to x_0$, so

$$f^{(n)}(x_0) = \lim_{h \to 0} \frac{\Delta^n f(x_0; h^n)}{h^n}.$$

For example, take $f(x) = \frac{1}{x}$ so that $f^{(n)}(x)$ is a multiple of $\frac{1}{x^n}$. Then the limit equals 1.

7 Consider the function $f(x) = \frac{1}{x^\alpha} = x^{-\alpha}$. Then $f'(x) = -\alpha x^{-\alpha-1}$, so there exists $\xi \in]n-1, n[$ such that

$$\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha} = f(n-1) - f(n) = f'(\xi) \cdot ((n-1) - n) = -f'(\xi) = \alpha \xi^{-\alpha-1}$$

by Lagrange's theorem. Hence

$$\frac{1}{\xi^{\alpha+1}} = \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right).$$

Since $\xi < n$, then $\frac{1}{\xi^{\alpha+1}} > \frac{1}{n^{\alpha+1}}$, so

$$\frac{1}{n^{\alpha+1}} < \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right),$$

as desired.

8

(a) It suffices to prove that the series $\sum_{n=1}^\infty \frac{1}{n^{1+\alpha}}$ converges for $\alpha > 0$.

$$\begin{aligned}
\sum_{n=1}^\infty \frac{1}{n^{1+\alpha}} &< \sum_{n=1}^\infty \frac{1}{\alpha}\left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right) \\
&= \frac{1}{\alpha}\sum_{n=1}^\infty \left(\frac{1}{(n-1)^\alpha} - \frac{1}{n^\alpha}\right) \\
&= \frac{1}{\alpha},
\end{aligned}$$

which converges, so $\sum_{n=1}^\infty \frac{1}{n^\sigma}$ also converges.

(b) Observe that

$$\ln M_t(x, \alpha) = \ln\left(\sum_{i=1}^n \alpha_i x_i^t\right)^{1/t} = \frac{1}{t}\ln\left(\sum_{i=1}^n \alpha_i x_i^t\right).$$

Now we can take the limit of this as $t \to 0$, and since when $t = 0$, both numerator and denominator equal 0, we use L'Hopitals rule. The derivative of the denominator is 1 and by chain rule, the derivative of the numerator is

$$\frac{\left(\sum_{i=1}^{n} \alpha_i \cdot \ln x_i \cdot x_i^t\right)}{\sum_{i=1}^{n} \alpha_i x_i^t} = \frac{\sum_{i=1}^{n} \alpha_i \cdot \ln x_i}{\sum_{i=1}^{n} \alpha_i}.$$

Because $\sum_{i=1}^{n} \alpha_i = 1$, then we get

$$\lim_{t\to 0} \ln M_t(x, \alpha) = \sum_{i=1}^{n} \alpha_i \ln x_i,$$

so

$$\lim_{t\to 0} M_t(x, \alpha) = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}.$$

(c) As $t \to +\infty$, note that the largest value of x_i will begin to dominate $M_t(x, \alpha)$, so $\lim_{t\to+\infty} M_t(x, \alpha) = \max_{1\le i\le n} x_i$. As $t \to -\infty$, note that the smallest value of x_i will begin to dominate $M_t(x, \alpha)$, so $\lim_{t\to-\infty} M_t(x, \alpha) = \min_{1\le i\le n} x_i$.

(d) It suffices to prove that $(M_t(x, \alpha))'$ is positive. By using the chain rule twice, we get that the derivative equals

$$\begin{aligned} &\left(\sum_{i=1}^{n} \alpha_i x_i^t\right)^{1/t} \cdot \left(\frac{1}{t}\right)' \cdot \left(\sum_{i=1}^{n} \alpha_i x_i^t\right)' \cdot \ln\left(\sum_{i=1}^{n} \alpha_i x_i^t\right) \\ &= \left(\sum_{i=1}^{n} \alpha_i x_i^t\right)^{1/t} \cdot \left(-\frac{1}{t^2}\right) \cdot \left(\sum_{i=1}^{n} \alpha_i \ln x_i \cdot x_i^t\right) \cdot \ln\left(\sum_{i=1}^{n} \alpha_i x_i^t\right) \end{aligned}$$

Since the first term is positive, the second is negative, the third is negative, and the fourth is positive, we're done.

9

(a) Observe that by the binomial theorem,

$$|(1 + x)|^p = |(1 + x)^p| = |1 + px + \binom{p}{2}x^2 + \cdots + x^p|.$$

Obviously, if we let $c_p = 0$ (not exactly sure what this means), then it suffices to show that this sum is greater than or equal to $1 + px$. If $x > 0$, this is obvious. If $x < 0$, as long as $x \ge -1$, this is also obvious. However, if $x < -1$, then the absolute value is

$$-(1 + x)^p = -1 - px - \cdots - x^p \ge 1 + px$$

because x is negative.

(b) As you can see, we have the base case. Assume there exists a value ξ_{k_1} such that

$$\Delta^k f(x_0; h_1, \ldots, h_{k-1}) = f^{(k)}(\xi_{k_1})h_1 \cdots h_k$$

and

$$\Delta^k f(x_0; h_1, \ldots, h_k) = f^{(k)}(\xi_{k_2})h_1 \cdots h_k.$$

Then

$$\begin{aligned}\Delta^{k+1} f(x_0; h_1, \ldots, h_{k+1}) &= \Delta^k f(x_0; h_1, \ldots, h_k) - \Delta^k f(x_0; h_1, \ldots, h_{k-1}) \\ &= (f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1})) \cdot h_1 \cdots h_k.\end{aligned}$$

By the mean value theorem, there exists some ξ such that

$$f^{(n+1)}(\xi) \cdot h_{k+1} = f^{(k)}(\xi_{k_2}) - f^{(k)}(\xi_{k_1}),$$

so substituting this in gives $f^{(k+1)}(\xi)h_1 \ldots h_{k+1}$, so we are done by induction.

(c) Observe that it suffices to show that $\left(\frac{\sin x}{x}\right)^3 - \cos x > 0$. It also suffices to show that this increases as x increases. To do this, we can show that the derivative is greater than 0. The derivative is

$$\left(\left(\frac{\sin x}{x}\right)^3\right)' + \sin x,$$

and by chain rule, the derivative of $\left(\frac{\sin x}{x}\right)^3$ is

$$3\left(\frac{\sin x}{x}\right)^2 \cdot \left(\frac{\sin x}{x}\right)' = \frac{3\sin^2 x}{X^2} \cdot \frac{x\cos x - \sin x}{x^2}$$

by product rule as well. Clearly this is greater than 0, as desired.

10

(a) Observe that if $f(x) = \arctan \log_2 \cos(\pi x + \frac{\pi}{4})$, we have the red graph. Observe that if $f(x) = \arccos(\frac{3}{2} - \sin x)$, we have the blue graph. Observe that if $f(x) = \sqrt[3]{x(x+3)^2}$, we have the green graph. Observe that if $\theta = \frac{r}{r^2+1}$ because I prefer using r and θ notation, then $r^2\theta + \theta = r$, so $r^2\theta - r + \theta = 0$. Then by the quadratic formula, $\theta = \frac{1 \pm \sqrt{1-4\theta^2}}{2\theta}$, so the graph starts at 0 and it goes in the positive x and y directions, it is concave down, and it has an asymptote at $y = 1$.

(b) Observe that if we have the graph of $y = f(x)$, then the graph of $y = f(x) + B$ is the graph of $y = f(x)$ shifted upwards by B units. The graph of $y = Af(x)$ is the graph of $y = f(x)$ stretched upwards by a factor of A. The graph of $y = f(x + b)$ is the graph of $y = f(x)$ shifted b units to the left. The graph of $y = f(ax)$ is the graph of $y = f(x)$ stretched horizontally by a factor of a. The graph of $-f(x)$ is the reflection of the graph of $y = f(x)$ over the x-axis. The graph of $y = f(-x)$ is the reflection of the graph of $y = f(x)$ over the y-axis.

(c) If we consider two points $(x_1, f(x_1))$ and $(x_2, f(x_2))$, then the point $\left(\frac{x_1+x_2}{2}, f\left(\frac{x_1+x_2}{2}\right)\right)$ is the point on the function in between x_1 and x_2. Then $\frac{f(x_1)+f(x_2)}{2}$ is the midpoint of the chord between x_1 and x_2, so its directly above the previous point by the given inequality. Therefore the chord lies above the graph, so this is convex, as desired.

(d) In the given interval, note that a convex function, because it's convex, can only have one hump or drop if you want to put it that way. If it wanted to have two humps, it would need to be convex at one place and concave at another. Therefore, a convex function that's not flat would be unbounded in the positive y-direction, so if it's bounded, then it must be horizontal.

(e) Observe that in order for the limit to approach 0, we need L'Hopitals rule. The limit equals $\frac{f'(x)}{1} = 0$, so $f'(x) = 0$, so $f(x)$ is constant, as desired. Observe that for any limit, either 1. it doesn't exist. 2. it exists and is infinite, and 3. it exists and is finite. The first case doesn't hold because since f is convex, it's also continuous, so we're done.

(f) In the given interval, note that a convex function, because it's convex, can only have one hump or drop if you want to put it that way. If it wanted to have two humps, it would need to be convex at one place and concave at another. Therefore, a convex function that's not flat would be unbounded in the positive y-direction, so if it's bounded, then it must be horizontal.

(g) Observe that in order for the limit to approach 0, we need L'Hopitals rule. The limit equals $\frac{f'(x)}{1} = 0$, so $f'(x) = 0$, so $f(x)$ is constant, as desired. Now note that for any limit, either 1. it doesn't exist. 2. it exists and is infinite, and 3. it exists and is finite. The first case doesn't hold because since f is convex, it's also continuous, so we're done.

11

(a) $f'_-(x)$ is the derivative of f at x from the left side and $f'_+(x)$ is the derivative of f at x from the right side. Observe that as f is convex, it is also continuous, and if we say that it is not differentiable at a point, for example, there is a sharp corner there (that would be the only way for it to not be differentiable at that point), then there would be a left derivative and a right derivative.

Also, since the function is convex, then at that point, despite it not being differentiable, $f''(x) > 0$, so the derivative on the right is at least than the derivative on the left, as desired.

(b) Since $x_1 < x_2$ and the function is convex, then $f''(x) > 0$, so the smallest value of $f'(x_2)$ is larger than the greatest value of $f'(x_1)$, so $f'_+(x_1) \leq f'_-(x_2)$ for $x_1, x_2 \in]a, b[$. The set of cusps in the graph is the set of sharp corners. If it's finite, it's at most countable. It must be countable, however because we can map each cusp to a natural number, so we're done.

(c) There are 3 cases to consider. 1, if $f(x)$ is not bounded from below for any $x \in I$, then I^* will be empty because $f^*(t)$ is infinite for any t. 2, if $f(x)$ is bounded from below for some $x \in I$, then I^* will consist of a single point. 3, if $f(x)$ is bounded from below for all $x \in I$, and the supremum is achieved at some point, then $f^*(T)$ is a real number for all t. Then I^* is an interval. It suffices to show that $f^*(t)$ is convex on I^*.

If $t_1, t_2 \in I^*$, then for any $\lambda \in (0, 1)$,

$$f^*(\lambda t_1 + (1-\lambda)t_2) = \sup_{x \in I}((\lambda t_1 + (1-\lambda)t_2)x - f(x)).$$

Suppose x_1 and x_2 achieve the supremum for t_1 and t_2, so $f^*(t_1) = t_1 x_1 - f(x_1)$ and $f^*(t_2) = t_2 x_2 - f(x_2)$. Consider a point $x \in I$. Then

$$(\lambda t_1 + (1-\lambda)t_2)x - f(x) \leq \lambda(t_1 x_1 - f(x_1)) + (1-\lambda)(t_2 x_2 - f(x_2)).$$

Now taking $\sup_{x \in I}$ on the left side,

$$f^*(\lambda t_1 + (1-\lambda)t_2) \leq \lambda f^*(t_1) + (1-\lambda) f^*(t_2),$$

so $f^*(t)$ is convex on I^*.

(d) If f is convex, then $I^* \neq \varnothing$, so for $f^* \in C(I^*)$, $(f^*)^* = \sup_{t \in I^*}(xt - f^*(t)) = f(x)$ for any $x \in I$, so the Legendre transform is involutive. Since $f^*(t) = \sup_{x \in I}(tx - f(x))$, then $tx - f(x) \leq f^*(t)$, so rearranging gives $xt \leq f(x) + f^*(t)$, as desired. If f is a convex differentiable function, then $f^*(t) = tx_t - f(x_t)$, where you can find x_t from $t = f'(x)$. $t = f'(x)$ means that the tangent to the graph of f at x has slope t, so the Legendre transform is a function defined on the set of tangents to f.

8.5 The Implicit Function Theorem

1

(a) We know that $\langle F_x(x_0, y_0), F_y(x_0, y_0)\rangle$ is the direction of the line that is tangent to F at (x_0, y_0), so the slope is $\frac{F_y(x_0,y_0)}{F_x(x_0,y_0)}$, and the equation is

$$y = \frac{F_y(x_0, y_0)}{F_x(x_0, y_0)}(x - x_0) + y_0.$$

(b) We know that

$$f^*(t) = \frac{1}{\beta}t^\beta = \sup_{x \in I}(tx - f(x)) \implies tx - f(x) \leq \frac{1}{\beta}t^\beta \implies tx \leq \frac{1}{\alpha}x^\alpha + \frac{1}{\beta}x^\beta,$$

Since $f^*(x)$ is a sup, then $f^*(x) \geq xt - e^x$, so $xt \leq e^x + t \ln \frac{t}{e}$.

(c) The angle between $\mathbf{a}$ and $\mathbf{v}$ is $\cos\theta = \frac{\mathbf{a} \cdot \mathbf{v}}{||\mathbf{a}||^2}$, so $||\mathbf{a}_t|| = ||\mathbf{a}|| \cos\theta = \frac{\mathbf{a} \cdot \mathbf{v}}{||\mathbf{v}||}$. Then $\mathbf{a}_t = ||\mathbf{a}_t||\mathbf{v} = \left(\frac{\mathbf{a} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v}$, and $\mathbf{a}_n = \mathbf{a} - \mathbf{a}_t = \mathbf{a} - \left(\frac{\mathbf{a} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v}$.

2

(a) Since

$$a_n = \frac{||v \times a||}{||v||} = \frac{||(x'y'' - x''y')k||}{\sqrt{(x')^2 + (y')^2}} = \frac{x'y'' - x''y'}{\sqrt{(x')^2 + (y')^2}}.$$

Then

$$r(t) = \frac{|v(t)|}{|a_n(t)|} = \frac{((x')^2 + (y')^2)^{3/2}}{|x'y'' - x''y'|}.$$

If we let $x = \cos t$, $y = r \sin t$, then $x' = -r \sin t$, $y' = r \cos t$, $x'' = -r \cos t$, $y' = -r \sin t$. Since $\mathbf{a}_n$ is perpendicular to $\mathbf{v}$, then $\mathbf{a}_n = \mathbf{a}$. $\frac{|\mathbf{v}(t)|}{|\mathbf{a}_n(t)|} = r$, as desired.

(b) Observe that the larger the radius, the smaller the curvature, so $k(t) = \frac{1}{r(t)}$. The curvature is how much a curve turns at a given point. If $k(t)$ is positive, then the curve is inward, but if $k(t)$ is negative, then the curve is outward. Setting $x(t) = x$ gives $x' = 1$ and $x'' = 0$. Then

$$\frac{1 \cdot y'' - 0 \cdot y'}{(1^2 + (y')^2)^{3/2}} = \frac{y''}{[1 + (y')^2]^{3/2}}$$

(c) Observe that with we let $R = r(t)$, then obviously the osculating circle would have the highest possible order of contact, so if $x(t)$ and $y(t)$ are twice differentiable and the osculating circle is tangent to the curve at that point, then it's center is (a, b). We know that $v^2 = 2gh$, so $(x')^2 + (y')^2 = 2gh = 2gy$. $x = 1 - y^2$, so $x' = 1 - 2yy'$.

(d) Let $z = a + bi$. Then $z - 1 = (a-1) + bi$ and $z + 1 = (a+1) + bi$. We're in for a bash.

$$\begin{aligned} |z-1| + |z+1| &= |(a-1) + bi| + |(a+1) + bi| \\ &= \sqrt{(a-1)^2 + b^2} + \sqrt{(a+1)^2 + b^2} \\ &\leq 3. \end{aligned}$$

Thus

$$\begin{aligned} \sqrt{(a+1)^2 + b^2} &\leq 3 - \sqrt{(a-1)^2 + b^2} \\ (a+1)^2 + b^2 &\leq 9 - 6\sqrt{(a-1)^2 + b^2} + (a-1)^2 + b^2 \\ 4a &\leq 9 - 6\sqrt{(a-1)^2 + b^2} \\ 6\sqrt{(a-1)^2 + b^2} &\leq 9 - 4a \\ 36(a-1)^2 + 36b^2 &\leq 81 - 72a + 16a^2 \\ 20a^2 + 36b^2 &\leq 45. \end{aligned}$$

(e) We know that $|z_1|$ denotes the length of the line segment from the origin to z_1, etc. So we're basically asked to prove the triangle inequality, which is obvious. This is the area inside an ellipse with semimajor axis $\frac{3}{2}$ and semiminor axis $\frac{\sqrt{5}}{2}$. The nth roots of unity are of the form $e^{2\pi ik/n}$, where k is an integer from 0 to $n-1$. Then we need

$$1 + e^{2\pi i/n} + e^{4\pi i/n} + \cdots + e^{2\pi i(n-1)/n} = \frac{1 - e^{2\pi}}{1 - e^{2\pi i/n}} = 0.$$

It flips z over the x-axis.

(f) Note that $1 - q^{n+1} = (1-q)(1 + q + \cdots + q^n)$, so

$$1 + q + \cdots + q^n = \frac{1 - q^{n+1}}{1 - q}.$$

Let $S = 1 + q + \cdots + q^n + \cdots$. Then $Sq = q + q^2 + \cdots$, so $S - Sq = 1$. Then $S = \frac{1}{1-q}$, so

$$1 + q + \cdots + q^n + \cdots = \frac{1}{1-q}.$$

Note that

$$1 + e^{i\varphi} + \cdots + e^{in\varphi} = \frac{1 - e^{i(n+1)\varphi}}{1 - e^{i\varphi}}.$$

This is basically $\frac{1 - r^{n+1}e^{i(n+1)\varphi}}{1 - re^{i\varphi}}$, which is basically $\frac{1}{1 - re^{i\varphi}}$.

(g) This is the real part of $1 + re^{i\varphi} + \cdots + r^n e^{in\varphi}$, which equals

$$\begin{aligned} \frac{1 - r^{n+1}e^{(n+1)\varphi i}}{1 - re^{\varphi i}} &= \frac{(1 - r^{n+1}e^{(n+1)\varphi i})(1 - re^{-\varphi i})}{(1 - r\cos\varphi)^2 + r^2\sin^2\varphi} \\ &= \frac{1 + r^{n+2}e^{n\varphi i} - r^{n+1}e^{(n+1)\varphi i} - re^{-\varphi i}}{1 - 2r\cos\varphi + r^2}. \end{aligned}$$

The real part of this is

$$\frac{1 + r^{n+2}\cos n\varphi - r^{n+1}\cos(n+1)\varphi - r\cos\varphi}{1 - 2r\cos\varphi + r^2}.$$

3

(a) This is the real part of $1 + re^{i\varphi} + \cdots + r^n e^{in\varphi} + \cdots$ which equals

$$\frac{1}{1 - re^{i\varphi}} = \frac{1 - re^{-i\varphi}}{(1 - r\cos\varphi)^2 + (r\sin\varphi)^2} = \frac{1 - re^{-i\varphi}}{1 - 2r\cos\varphi + r^2}.$$

The real part of this is

$$\frac{1 - r\cos\varphi}{1 - 2r\cos\varphi + r^2}.$$

(b) This is the imaginary part of $1 + re^{i\varphi} + \cdots + r^n e^{in\varphi}$ plus 1, which is

$$1 + \frac{r^{n+2}\sin n\varphi - r^{n+1}\sin(n+1)\varphi + r\sin\varphi}{1 - 2r\cos\varphi + r^2}.$$

This is the imaginary part of $1 + re^{i\varphi} + \cdots + r^n e^{in\varphi} + \cdots$ plus 1, which equals

$$1 + \frac{r\sin\varphi}{1 - 2r\cos\varphi + r^2}.$$

(c) We already know that by definition, $e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n$, so

$$\lim_{n\to\infty}\left(1 + \frac{z}{n}\right)^n = \left(\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n\right)^z = e^z.$$

Observe that

$$e^w = e^{\ln|z| + i\text{Arg}z} = e^{\ln|z|}\cdot e^{i\text{Arg}z} = |z|\cdot\frac{z}{|z|} = z,$$

since Arg$z$ is the angle of z, so if you take e to the i times this angle, then we get a unit complex number in the direction of z, which equals $\frac{z}{|z|}$.

4

(a) We find that $\text{Ln}z = \ln|z| + i\text{Arg}z$. Thus $\text{Ln}1 = \ln|1| + i\text{Arg}1 = 0 + i\cdot 0 = 0$. Then $\text{Ln}i = \ln|i| + i\text{Arg}i = 0 + i\cdot\frac{\pi}{2} = \frac{\pi}{2}i$. $1^\pi = e^{\pi\text{Ln}1} = e^{\pi\cdot 0} = 1$ and $i^i = e^{i\text{Ln}i} = e^{i\cdot\frac{pi}{2}i} = e^{-\frac{pi}{2}}$. $z = \arcsin w = \arcsin\left(\frac{1}{2i}(e^{iz} - e^{-iz})\right)$. This is because we know that $\sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$, so $|\sin z| = \frac{1}{2}|e^{iz} - e^{-iz}|$, so we would need $|e^{iz} - e^{-iz}| = 4$. Since $|e^{iz}| = |e^{-iz}|$, then we need $e^{iz} = -e^{-iz}$, so z is a real number, but $|e^{iz}| = 1$ if z is a real number, so this is impossible.

(b) $f(z) = \frac{1}{1+z^2}$ is continuous. If $z = a + bi$, then $f(z) = \frac{1}{1+(a+bi)^2} = \frac{1}{(1+a^2-b^2)+2abi}$, which is continuous. Note that

$$\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} = \sum_{n=0}^{\infty}(-z^2)^n = \sum_{n=0}^{\infty}(-1)^n z^{2n},$$

which has radius of convergence 1. Note that

$$\frac{1}{1+\lambda^2 z^2} = \frac{1}{1-(-\lambda^2 z^2)} = \sum_{n=0}^{\infty}(-\lambda^2 z^2)^n = \sum_{n=0}^{\infty}(-1)\lambda^2 n z^{2n},$$

so the radius of convergence is $\frac{1}{\lambda}$.

(c) The radius of convergence is probably determined by something like $\frac{1}{d}$, where d is the scalar factor. It also makes since on the real line, for example in $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$.

(d) Yes, it is continuous. As $z \to 0$, $-1/z^2 \to -\infty$, so $\lim_{z\to} f(z) = 0$, which equals $f(0)$. Just apply the same reasoning as in part (a) except with real numbers. We know that

$$e^z = 1 + \frac{1}{1!}z + \frac{1}{2!}z^2 + \cdots,$$

so

$$e^{-1/z^2} = 1 - \frac{1}{1!z^2} + \frac{1}{2!z^4} - \cdots.$$

Nope, it doesn't exist at $z_0 = 0$.

(e) In that case, the radius of convergence would equal 0, so we would need $\lim_{n\to 0} \sqrt[n]{|c_n|} = \infty$. If we let $c_n = 2^{n^2}$, then this works. $c_n = 2^{n^2}$ gives $\sqrt[n]{|c_n|} = 2^n$, which definitely doesn't converge.

(f) We show that $m(x) = \min_{a\le t\le x} f(t)$ is continuous, the $M(x)$ case is symmetric. Any function $f(t)$ consists of increasing and decreasing intervals. $m(x)$ cannot be discontinuous because this would require the minimum of $f(t)$ on that current interval to be discontinuous. Since the given intervals are continuous, then since we know that f is continuous on the interval $[a, b]$, then $m(x)$ is continuous.

(g) If this function is continuous, then $f([a, b]) = [c, d]$. Since the function is monotonic, then the function inverse is defined, and since all of the inputs are distinct, the function inverse is injective and surjective.

Suppose we have an input x_0 and output y_0. Since f is continuous, as $x \to x_0$, $f(x) = y \to f(x_0) = y_0$. Therefore, as $y \to y_0$, $f^{-1}(y) = x \to f^{-1}(y_0) = x_0$, so f^{-1} is continuous, as desired.

5

(a) Consider the function that maps $[0, 1)$ to $[0, 1)$ and $(1, 2]$ to $(1, 2]$. Then the function is monotonic but it has one discontinuity, which is 1. Observe that X and Y are not necessarily open intervals, so there may be an element $x \in X$ which is equal to the greatest lower bound or least upper bound. Then this boundary point might not be necessarily continuous.

(b) Consider $f(x) = 2^x$ and $g(x) = x$ on the interval $[0, \infty)$. Then $f(x)$ has no fixed points, so they do not always have a common fixed point. Let $f(x) = x - 1$. Obviously there is no fixed point, period.

(c) Consider $f(x) = 2^x - 1$. Note that $f(0) = 0$ and $f(1) = 1$, but since this is an exponential function, $x > 2^x - 1$ for $0 < x < 1$, so this does not have a fixed point. Let $g(x) = f(f(x)) - f(x)$. Since $g(0) = g(1) = 0$ and f is continuous, then g must also be continuous. Thus it is impossible for g to be nonzero (it can't increase then decrease), so $g(x) = 0$ for all x. Therefore $f(x) = f(f(x)) = x$, as desired.

6 Easy-peasy. We would basicaly be finding values of z that satisfy this equation and set them equal to some function $f(x, y)$. Obviously these functions are smooth.

8.6 Some Corollaries of the Implicit Function Theorem

1 In the case of solid fuel, $\alpha = \frac{\omega^2}{2Q} = \frac{2^2}{2000} = \frac{1}{500}$. In the case of liquid fuel, $\alpha = \frac{3^2}{5000} = \frac{9}{5000}$. We know that the efficiency equals $\frac{m_R \frac{v^2}{2}}{m_F Q}$, and from part (a), we know that $\frac{1}{2}\omega^2 = \alpha Q$, so $Q = \frac{\frac{1}{2}\omega^2}{\alpha}$. Then the efficiency is

$$\frac{m_R\frac{v^2}{2}}{m_F\left(\frac{1}{2}\omega^2/\alpha\right)} = \frac{m_R v^2 \alpha}{m_F \omega^2}.$$

2

(a) Suppose the function f is continuous on the interval $[a, b]$. Then for $x \in [a, b]$, $f(x)$ must be bounded because it's continuous; we cannot have $f(x) = \infty$ or $-\infty$ because otherwise $f(x)$ would not be continuous or closed. If this asymptote/limit occurs in the middle of a and b, f is not continuous. If this occurs at a or b, the domain of f would not be closed.

(b) Observe that the bounds of f are achieved because if they aren't, the range of f isn't closed. Note that if the minimum/maximum occurs inside the range $[a, b]$, it obviously equals the greatest lower bound/least upper bound. If this minimum/maximum occurs at the endpoints, suppose WLOG $\inf = f(a)$. Then if inf is not achieved, since f is not continuous, neither is a, so we're done.

3

(a) From section 8.6.2, we already know that $p = \rho \cdot \frac{R}{M}T$ which equals $\frac{R}{V}T$, so the correction term would be $+\frac{R}{V}\Delta T$. $p = p_0 e^{-(g/\lambda)h} + \frac{R}{V}T$, so we can just plug in the respective values of T.

(b) Let $g(x) = f^n(x) - f(x)$. Observe that $g(0) = g(1) = 0$ and f and $f^n(x)$ are continuous, so $g(x)$ is also continuous. Therefore g must equal zero because otherwise, if x increases, then $f(x) \geq x$, so $f^2(x) = f(f(x)) \geq f(x)$, etc., so $f^n(x) > f(x)$. Therefore $g(x)$ increases, so it is impossible for $g(1) = g(0)$. Therefore $g(x) = 0$ for all x and hence $f(x) = x$ for all x.

4

(a) Otherwise, if $f^k(x) = f^{k-1}(x)$, we must have $f(x) = x$, so assume $f^n(x) > f^{n-1}(x) > \cdots > f(x)$. Then $f^n(x)$ increases as n increases, but since f maps from $[0, 1]$ to $[0, 1]$, $f^n(x) < 1$, so $\lim_{n\to\infty} f^n(x) - f^{n-1}(x) = 0$. Therefore since $f^{n-1}(x)$ approaches $f^n(x)$, $f^n(x)$ tends to a fixed point.

(b) If we know that $\pi^1(x^1) = f^1(x^1)$, $\pi^2(x^2) = f^2(x^2)$, etc, until $\pi^m(x^m) = f^m(x^m)$. We know that $T = \frac{\ln 2}{\lambda}$, so $\lambda = \frac{\ln 2}{T}$. Then $N(T) = N_0 e^{\lambda t} = N_0 e^{-\ln 2/T \cdot t} = N_0 \cdot 2^{-t/T}$, so $m = (m + r)2^{-t/T}$ and thus $-\frac{t}{T} = \log_2 \frac{m}{m+r}$, so $t = T \log_2 \left(1 + \frac{r}{m}\right)$.

(c) Assume that f is uniformly continuous on E. Then for every $\epsilon > 0$, there exists a $\delta > 0$ such that $|x_1 - x_2| < \delta$ implies that $|f(x_1) - f(x_2)| < \epsilon$. Since $\omega(\delta)$ is a supremum, then there exists $\delta > 0$ such that $\omega(\delta) < \epsilon$. Then $\lim_{\delta\to 0} \omega(\delta) = \omega(+0) = 0$.

(d) Assume that $\omega(+0) = 0$. Then $\lim_{\delta\to 0} \omega(\delta) = 0$, so for any $\epsilon > 0$, there exists $\delta > 0$ such that $\omega(\delta) = \epsilon$. If $|x_1 - x_2| < \delta$, we have $|f(x_1) - f(x - 2)| \leq \omega(\delta) < \epsilon$. Thus f is uniformly continuous on E, so a function f is uniformly continuous on E if and only if $\omega(+0) = 0$.

(e) Note that since $f(x)$ is continuous over an interval $[a, b]$, by the extreme value theorem, the maximum and minimum value of $f(x)$ must exist on this interval. Similarly for $g(x)$ (and note the extreme value function only applies for continuous functions so this wouldn't work in general for arbitrary bounded functions).

5 Therefore, the maximum value of $|f(x_0) - g(x_0)|$ occurs in the set X, because if it didn't exist, this would imply that $|f(x) - g(x)|$ has no maximum value, which would imply that $f(x)$ or $g(x)$ has no maximum/minimum value. One of $f(x)$ or $g(x)$ can't be increasing/decreasing in the middle of the interval if it's not approaching its bound, so we're done.

6

(a) Observe that $I(l + h) - I(l) = -\lambda I(l)$, for some λ in terms of $I(l)$. Observe that $I(l + h) = I(l) \cdot I(h)$, since having one layer $l + h$ thick is the same as having two layers, which are l and h thick. So, it must be an exponential function. Then $I(l) = ce^{-kl}$ for some c and k. Plugging in $l = 0$ gives $I_0 = c$, so $I(l) = I_0 e^{-kl}$ for some k.

(b) We assume that the water is the same throughout, since it's pure, so there's no muddy water at the bottom. We also know $l = 10$ meters, so $I = I_0 e^{-10k}$. Then we can plug in k and done! Since $mx'' + kx = 0$, then $x'' = -\frac{k}{m}x$. We know that the total mechanical energy of an isolated system is constant, and that the kinetic energy equals $\frac{1}{2}mv^2 = \frac{m\dot{x}^2(t)}{2}$, and that the potential energy is $\frac{mx^2(t)}{2}$, so add them together to get $E = \frac{m\dot{x}^2(t)}{2} + \frac{mx^2(t)}{2}$ is constant.

(c) Since velocity equals 0, then we know that position, which is the integral of velocity, is some constant. Then $x(t) = c$, but $x(0) = 0 = c$, so $x(t) = 0$. If we already have $x(0) = x_0$ and $\dot{x}(0) = v_0$, then we automatically have $\ddot{x}(0) = a_0$. Then there exists $x = x(t)$ such that $m\ddot{x} + kx = 0$. Since we have friction, then energy is being taken away since the velocity must decrease, so E decreases.

(d) Observe that if a polynomial has complex roots (not real roots), they must come in conjugate pairs, so there are an even number of imaginary roots. The number of roots a polynomial (repeats count) has is the same as the degree of the polynomial, so there are an odd number of roots. Therefore there must be at least one real root.

8.7 Surfaces in $\mathbb{R}^n$ and Constrained Extrema

1

(a) Given a fixed polynomial $P_n(x)$, we want to find a polynomial of the form $Q_\lambda(x) = \lambda P_n(x)$ such that $\Delta(Q_{\lambda_0}) = \min_{\lambda \in \mathbb{R}} \Delta(Q_\lambda)$. $\Delta(Q_\lambda)$ is continuous with respect to λ, so there exists $\lambda_0 \in \mathbb{R}$ that minimizes $\Delta(Q_\lambda)$ in the interval $[a, b]$, implying the existence of a polynomial of best approximation of the form $\lambda P_n(x)$.

(b) Let $P_n(x)$ be the polynomial of best approximation of degree n, and let $\Delta_n = \Delta(P_n) = E_n(f)$. We show that there exists a polynomial of best approximation of degree $n + 1$ by adding a term of degree $n + 1$ to $P_n(x)$, giving

$$P_{n+1}(x) = P_n(x) + c(x - a)^{n+1}$$

for some constant c. We know that there exists a polynomial of best approximation of degree 0, and we know that if there exists a polynomial of best approximation of degree n, there is also one of degree $n + 1$, so we're done by induction.

(c) The sign function is:

$$\text{sgn} x = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x = -1 \end{cases}$$

Therefore, if on some interval, $P_n(x)$ is the same sign, then $\operatorname{sgn}P_n(x)$ will be continuous on that interval. Therefore, sgn is only not continuous when $P_n(x)$ changes sign, so basically when $P_n(x) = 0$, so when x is a root of P_n. We know there are at most n roots (since there are double roots too), so there are at most n points of discontinuity.

2

(a) I'm not sure how the fact that $\operatorname{sgn}[(f(x_i) - P_n(x_i))(-1)^i]$ comes into play, but if we let $g(x) = f(x) - P_n(x)$, we get: either $g(x_0)$ is negative, $g(x_1)$ is positive, $g(x_2)$ is negative, etc. or, $g(x_0)$ is positive, $g(x_1)$ is negative, $g(x_2)$ is positive, etc. or, $g(x_0) = g(x_1) = \cdots = g(x_{n+1}) = 0$. Therefore $g(x)$ must have at least $n+1$ roots, since the sign changes $n+1$ times.

So we try to show that

$$E_n(f) = \inf_{P_n} \sup_{x\in[a,b]} |g(x_i)| \geq \min_{0\leq i\leq n+1} |g(x_i)|.$$

Since $|g(x_i)| \leq \sup_{x\in[a,b]} |g(x_i)|$, we're done.

(b) By Demoivre's theorem,

$$\operatorname{cis} nx = \cos nx + i\sin nx = (\cos x + i\sin x)^n.$$

So $\cos nx$ is equal to the real part, which is

$$\cos nx = \cos^n x - \binom{n}{2}\cos^{n-2} x\sin^2 x + \binom{n}{4}\cos^{n-4} x\sin^4 x - \cdots + \ldots.$$

Obviously if the degree of $\sin x$ is odd, then it wouldn't be real, so $\sin x$ is always raised to an even power. Since $\sin^2 x = 1 - \cos^2 x$, we can express $\sin^{2k} x$ in terms of $\cos x$, so $\cos nx$ can be expressed as an algebraic polynomial in terms of $\cos x$, so we're done.

(c) We have

$$\frac{P(x)}{Q(x)} - \frac{p(x)}{q(x)} = \frac{p(x)}{q(x)}\left(\frac{P_1(x)}{Q_1(x)} - 1\right),$$

so taking the integral gives

$$\int \frac{P(x)}{Q(x)}\mathrm{d}x - \int \frac{p(x)}{d(x)}\mathrm{d}x = \int \frac{p(x)}{q(x)}\left(\frac{P_1(x)}{Q_1(x)} - 1\right)\mathrm{d}x = \frac{P_1(x)}{Q_1(x)}.$$

(d) We have

$$\begin{aligned}
\left(\frac{P_1(x)}{Q_1(x)}\right)' &= \frac{P_1'(x)Q_1(x) - P_1(x)Q_1'(x)}{Q_1(x)^2} \\
&= \frac{\left(\frac{P(x)}{p(x)}\right)'\frac{Q(x)}{q(x)} - \frac{P(x)}{p(x)}\left(\frac{Q(x)}{q(x)}\right)'}{\left(\frac{Q(x)}{q(x)}\right)^2} \\
&= \frac{\left(\frac{P'(x)p(x)-P(x)p'(x)}{p(x)^2}\right)\frac{Q(x)}{q(x)} - \frac{P(x)}{p(x)}\left(\frac{Q'(x)q(x)-Q(x)q'(x)}{q(x)^2}\right)}{\left(\frac{Q(x)}{q(x)}\right)^2} \\
&= \frac{\frac{Q(x)(P'(x)p(x)-P(x)p'(x))}{p(x)} - \frac{P(x)(Q'(x)q(x)-Q(x)q'(x))}{q(x)}}{Q(x)^2\cdot\frac{p(x)}{q(x)}} \\
&= \frac{q(x)Q(x)(P'(x)p(x) - P(x)p'(x)) - p(x)P(x)(Q'(x)q(x) - Q(x)q'(x))}{Q(x)^2\cdot p(x)q(x)}.
\end{aligned}$$

Adding $\frac{p(x)}{q(x)}$ to this mess gives $\frac{P(x)}{Q(x)}$, as desired.

(e) Observe that if r is a double root of $Q(x)$, then $Q'(r) = 0$, but the multiplicity of r for Q' will be one less than that for Q. Then we can use the Euclidean algorithm to find the greatest common divisor between Q' and Q, which equals $Q_1(x)$. Then we can find $q(x)$ by $\frac{Q(x)}{Q_1(x)}$. We have

$$Q'(x) = 7x^6 + 18x^5 + 25x^4 + 28x^3 + 21x^2 + 10x + 3,$$

so by the euclidean algorithm we obtain $Q_1(x) = x^4 + 2x^3 + 2x^2 + 2x + 1$, and similarly we can obtain $P_1(x)$.

(f) Since $R(-u, v) = R(u, v)$, then R is an even function at least for u, so every u term must be an even function. Therefore, since this is a rational function, then it's made up of the sum of the polynomials, so they must be of the form $u^{2a}v^b$. Thus we can express this in terms of $(u^2)^a v^b$, so this can be expressed in the form $R_1(u^2, v)$. $f(x_i) - P_n(x_i) = (-1)^i \Delta(P_n) \cdot \alpha$ for $i = 0, \ldots, n+1$, $\Delta(P_n) = \max_{x\in[a,b]} |f(x) - P_n(x)|$. In order to show that $P_n(x)$ is the unique polynomial of best approximation, if we let $Q(x)$ be a polynomial of degree n, then it suffices to show that $\max_{x\in[a,b]} |f(x) - Q(x)| \geq \Delta(P_n)$.

3

(a) Let $R(x) = f(x) - Q(x)$, so it suffices to show that $\max_{x\in[a,b]} |R(x)| \geq \Delta(P_n)$. Assume there's some alternate sequence for x_i, so we have

$$R(x_i) = f(x_i) - Q(x_i) = (-1)^i \Gamma, \qquad i = 0, \ldots, n+1$$

where $\Gamma = \max_{x\in[a,b]} |f(x) - Q(x)|$. Then

$$Q(x)i - P_n(x_i) = (-1)^i (\Delta(P_n) \cdot \alpha - \Gamma)$$

for $i = 0, \ldots, n+1$. Then $Q(x_i) - P_n(x_i)$ has the same sign as $f(x_i) - P_n(x_i)$, so $\max_{x\in[a,b]} |Q(x) - P_n(x)| \geq \Delta(P_n)$, so we're done.

(b) By part (a), we already have the if part. It suffices to prove the only if part. By Problem 10, we already know that a polynomial of best approximation exists. We also know that $n + 1$ alternating points are on the polynomial, and we can add another one by using the next hump, so we're done.

(c) Obviously, for a best approximation of degree 0 is $P_0(x) = 1$. A best approximation of degree 1 is of the form $P_1(x) = ax + b$. The most important points are the endpoints and corners, which is $(-1, 1)$, $(0, 0)$, $(2, 2)$. The vertical distance between P_1 and these points is $1 - (-a + b) = 1 + a - b$, b, and $2 - (2a + b) = 2 - 2a - b$, respectively. Note that the $\Delta(P_1)$ will be minimized when $1 + a - b = b = 2 - 2a - b$, so $a = \frac{1}{3}$ and $b = \frac{2}{3}$, so $P_1(x) = \frac{1}{3}x + \frac{2}{3}$. Then in this case, the distance will be $\frac{2}{3}$, and we're done.

(d) The requirements are closure, associativity, commutativity, existence of identity, and existence of inverse for addition. Observe that we just proved closure by the previous part and that associativity and commutativity are obvious because the underlying functions are continuous, associative, and commutative. The identity exists for both operations, being 0 for addition and 1 for multiplication. The inverse of addition is the negative of the function whose inverse we are looking for, however this does not necessarily exist for multiplication (consider, for example, $f(x) = 0$, which does not have an inverse).

(e) Obviously, since its degree is odd, it has to have a real root x_0, so we can let the polynomial $p(x)$ be equal to

$$p(x) = (x - x_0)(ax^2 + bx + c)$$

for real coefficients a, b, and c. Plugging in t^2 for $x - x_0$ and $t^2 + x_0$ for x, we get

$$p(x) = t^2(a(t^2 + x_0)^2 + b(t^2 + x_0) + c) = t^2(at^4 + (2ax_0 + b)t^2 + (ax_0^2 + bx_0 + c)),$$

as desired.

4

(a) $R(u, v)$ is some combination of u and v that includes products, then summing them all together. $R(x, \sqrt{P(x)})$ is some combination of $x = x_0 + t^2$ and $\sqrt{P(x)} = t\sqrt{at^4 + bt^3 + ct^2 + dt + e}$. A combination of $x_0 + t^2$ and $t\sqrt{at^4 + bt^3 + ct^2 + dt + e}$, say for $u^a v^b$ gives

$$(x_0+t^2)^a \left[t\sqrt{at^4 + bt^3 + ct^2 + dt + e}\right]^b = (x_0+t^2)^a t^b \left[\sqrt{at^4 + bt^3 + ct^2 + dt + e}\right]^b.$$

Use distributive, then the terms on the left side are in terms of t, and the ones on the right are in terms of $\sqrt{at^4 + bt^3 + ct^2 + dt + e}$, as desired.

(b) Since imaginary roots must come in conjugates, then obviously they can come together to form quadratics, so a fourth degree polynomial must be able to be represented as the product of two quadratics, so it must equal $a(x^2 + p_1x + q_1)(x^2 + p_2x + q_2)$. Then plugging in $x = \frac{\alpha t+\beta}{\gamma t+1}$ and simplifying gives an equation of the form $\frac{(M_1+N_1t^2)(M_2+N_2t^2)}{(\gamma t+1)^2}$, as desired.

(c) Since we know that $ax^4 + bx^3 + \cdots + e$ can be expressed in the form

$$\frac{(M_1 + N_1t^2)(M_2 + N_2t^2)}{(\gamma t + 1)^2} = \frac{M_1M_2(1 + \frac{N_1}{M_1}t^2)(1 + \frac{N_2}{M_2}t^2)}{(\gamma t + 1)^2}.$$

Thus the square root of this is of the form $\frac{\sqrt{A(1+m_1t^2)(1+m_2t^2)}}{\gamma t+1}$, and when combined with x which is also a combination of t, we get a combination of t and $\sqrt{A(1 + m_1t^2)(1 + m_2t^2)}$, as desired.

(d) Consider a subring I consisting of all functions f such that $f(a) = 0$. Then if g is an element of the ring K, then $(f \cdot g)(a) = f(a) \cdot g(a) = 0$, which is in I, so I is an ideal. $R(x, \sqrt{y})$ will consist of terms in the form $kx^a(\sqrt{y})^b$. b is either even or odd, so the terms will be of 2 forms: $kx^a y^{b/2}$ is b is even, or $kx^a y^{\frac{b-1}{2}}\sqrt{y}$ is b is odd. If we let $R_1(x, y)$ be all of the terms with b even, and $R_2(x, y)$ be all of the terms with b odd times $\sqrt{y}$, so that it would be a valid rational function, then we're done.

(e) Consider the term that is of the form $cx^a(\sqrt{y})^b$. We have two cases. If b is even, then we have a purely integral term for all of the exponents. Let the sum of all these terms be $R_1(x, y)$. If b is odd, then we get a leftover $\sqrt{y}$ product. In this latter case, we have another two cases. If a is even, then we have the $\frac{R_2(x^2,y)}{\sqrt{y}}$, and if a is odd, we have the $\frac{R_3(x^2,y)}{\sqrt{y}}x$ term. Putting these terms altogether, we get

$$R(x, \sqrt{y}) = R_1(x, y) + \frac{R_2(x^2, y)}{\sqrt{y}} + \frac{R_3(x^2, y)}{\sqrt{y}}x.$$

5

(a) At the extrema of $d_p(x) = ||p - x||$, we're at a maximum, so $d_p'(x) = 0$. Thus since $d_p(x)$ is the distance between a point on S to some random point, then this distance is minimized. $p - x$ is basically the line connecting p to this point x on S, so at the minimization, it's basically perpendicular to the tangent plane to S at x. Thus it is orthogonal to S.

(b) A degenerate critical point is a saddle point, no? Neither a minimum nor a maximum. From part (a), we know that at any critical point, $p - x$ is orthogonal to S. Since the dimension of S is k, there are at most k critical points, so there are also at most k points p such that $d_p(x)$ has a saddle point.

(c) We already know that $R(x, \sqrt{P(x)})$ can be reduced to the form

$$R_1\left(t, \sqrt{A_1(1 + m_1t^2)(1 + m_2t^2)}\right)$$

by part (d). This expression can also be reduced to the form

$$\frac{r(t^2)}{\sqrt{A_2(1 + m_1t^2)(1 + m_2t^2)}}$$

since its a rational function and we have the same base terms (like for $R(u, v)$, the base terms are u and v). Put an integral around both of them and we're done.

6 Consider, for example, that $E_1 = [-1, 0]$ and $E_2 = [0, 1]$. Let $f|E_1 = x$ and $f|E_2 = x + 1$. Then $f|E_1(0) = 0$ and $f|E_2(0) = 1$, which is not continuous, so F is not continuous on $E_1 \cup E_2$. If $f \in C(A)$, then f is a continuous function defined on A. Since $B \subset A$, then f is also continuous on B, so $f|_B \in C(B)$, as desired.

7

(a) Observe that the Riemann function $\mathcal{R}$ is discontinuous because there is always an irrational number between any two rational numbers. The irrational number gives 0 but the rationals are nonzero, so this is discontinuous. The Riemann function defined on only the rational $\mathbb{Q}$'s is also irrational. Take, for example, all of the rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$. $\mathcal{R}(2/5) = \mathcal{R}(3/5) = \frac{1}{5}$ but the rational numbers in between do not approach $\frac{1}{5}$. All of these points of discontinuity are removable by removing the rational numbers, leaving us with a function that outputs 0 for each input.

(b) We already know that any elliptic integral of the form $\int R(x, \sqrt{P(x)})\mathrm{d}x$ can be expressed in the form

$$\int \frac{r(t^2)\mathrm{d}t}{\sqrt{A(1 + m_1t^2)(1 + m_2t^2)}}.$$

Since the standard forms are also of this form, we're done.

(c) Suppose a number $x \in \mathbb{R}$'s base q representation has periodically repeating digits. Separate the repeating part from the non-repeating part. For example, if we have $x = 0.1232323\ldots$, we can separate this into $a = 0.1$ and $y = 0.0\overline{23}$. Then we can multiply x by q^n to get $x \cdot q^n - (x - a)$ is equal to a rational number. Therefore we can solve for x, which is a rational number, and then convert the numerators and denominators into standard base.

(d) The heights are h, qh, q^2h, etc. Note that if the initial or final velocity is v, then the final or initial velocity is 0, so $v^2 = 2gy$. This means that $v = \sqrt{2gy}$, so $t = \sqrt{\frac{2y}{g}}$. Thus the time in total is

$$\sqrt{\frac{2h}{g}} + \sqrt{\frac{2qh}{g}} + \sqrt{\frac{2q^2h}{g}} + \cdots = \sqrt{\frac{2h}{g}}(1 + \sqrt{q} + \sqrt{q^2} + \cdots) = \frac{\sqrt{\frac{2h}{g}}}{1-\sqrt{q}}.$$

The total distance traveled is $h + qh + q^2h + \cdots = \frac{h}{1-q}$.

(e) We mark all the points on a cicle obtained from a fixed point by rotations of the circle through angles of n radians, where $n \in \mathbb{Z}$ ranges over all integers. This way, every single point on the circle will eventually be marked because we need to find the possible values of $n \pmod{2\pi}$ span the entirety of the range $[0, 2\pi)$.

(f) This gives

$$\frac{m}{n} = q_1 + \frac{1}{q_2 + \frac{1}{r_1/r_2}}.$$

Then $r_1/r_2 = q_3 + \frac{r_3}{r_2} = q_3 + \frac{1}{r_2/r_3}$, so we get

$$\frac{m}{n} = q_1 + \frac{1}{q_2 + \frac{1}{q_3 + \frac{1}{r_2/r_3}}}.$$

Because the m and n are relatively prime, then there will be n equations, so there will be n of those nested fractions.

8 Since $u_n = u_{n-1} + u_{n-2}$, the characteristic equation is $c^2 - c - 1$, so we have roots $c_1 = \frac{1+\sqrt{5}}{2}$ and $c_2 = \frac{1-\sqrt{5}}{2}$. We must solve the equation $F_n = a_1(c_1)^n + a_2(c_2)^n$. Letting $n = 1$ and 2, the system of equations gives

$$1 = a_1\left(\frac{1+\sqrt{5}}{2}\right) + a_2\left(\frac{1-\sqrt{5}}{2}\right)$$

$$1 = a_1\left(\frac{1+\sqrt{5}}{2}\right)^2 + a_2\left(\frac{1-\sqrt{5}}{2}\right)^2 = a_1\left(\frac{3+\sqrt{5}}{2}\right) + a_2\left(\frac{3-\sqrt{5}}{2}\right).$$

Observe that a rational number has a unique continued fraction representation as shown by the Euclidean algorithm analogy. Let the continued fraction equal x. Note that $x = 1 + \frac{1}{x}$, so $x^2 - x - 1 = 0$. Observe that the solution to this is $x = \frac{1\pm\sqrt{5}}{2}$. Since x is positive, then we must have $x = \frac{1+\sqrt{5}}{2}$, as desired.

9

(a) Subtracting the first equation from the second gives $a_1 + a_2 = 0$, so $a_2 = -a_1$. Plugging this into the first equation gives

$$1 = a_1\left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}\right) = a_1\sqrt{5},$$

so $a_1 = \frac{1}{\sqrt{5}}$ and $a_2 = -\frac{1}{\sqrt{5}}$. This gives

$$u_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n,$$

as desired.

(b) We know that

$$\begin{aligned}\int e^{ax}\cos bx\mathrm{d}x &= \frac{1}{a}\int \cos bx\mathrm{d}e^{ax}\\ &= \frac{1}{a}e^{ax}\cos bx - \frac{1}{a}\int e^{ax}\mathrm{d}\cos bx\\ &= \frac{1}{a}e^{ax}\cos bx + \frac{b}{a}\int e^{ax}\sin bx\mathrm{d}x\\ &= \frac{1}{a}e^{ax}\cos bx + \frac{b}{a^2}\int \sin bx\mathrm{d}e^{ax}\\ &= \frac{1}{a}e^{ax}\cos bx + \frac{b}{a^2}e^{ax}\sin bx - \frac{b}{a^2}\int e^{ax}\mathrm{d}\sin bx\\ &= \frac{a\cos bx + b\sin bx}{a^2} - \frac{b^2}{a^2}\int e^{ax}\cos bx\mathrm{d}x.\end{aligned}$$

Thus we obtain

$$\int e^{ax}\cos bx\mathrm{d}x = \frac{a\cos b + b\sin bx}{a^2+b^2}e^{ax}.$$

(c) Then

$$\begin{aligned}S_2(n+1) &= S_2(n) + (n+1)^2\\ &= \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n + (n+1)^2 = \frac{1}{3}n^3 + \frac{3}{2}n^2 + \frac{13}{6}n + 1\\ &= \frac{1}{3}(n+1)^3 + \frac{1}{2}(n+1)^2 + \frac{1}{6}(n+1)\end{aligned}$$

So we're done by induction. Similarly,

$$\begin{aligned}S_3(n+1) = S_3(n) + (n+1)^3 &= \frac{n^2(n+1)^2}{4} + (n+1)^3\\ &= (n+1)^2\left(\frac{n^2}{2} + (n+1)\right) = \frac{(n+1)^2(n+2)^2}{4}\end{aligned}$$

as desired.

(d) Observe that any exponential function satisfies this. Take, for example, $f(x) = e^x$. Then $f(1) = e$ which is definitely positive and not equal to 1. $f(x_1)\cdot f(x_2) = e^{x_1}\cdot e^{x_2} = e^{x_1+x_2} = f(x_1+x_2)$. And since this function is continuous, $f(x) \to f(x_0)$ as $x \to x_0$.

Also, observe that any logarithmic function satisfies this. Take, for example, $f(x) = \ln x$. Then $f(e) = \ln e = 1$, and $e > 0$ and $e \neq 1$. Then $f(x_1) + f(x_2) = \ln x_1 + \ln x_2 = \ln(x_1\cdot x_2) = f(x_1\cdot x_2)$ and $f(x) \to f(x_0)$ for $x \to x_0$ since ln is a continuous function, as desired.

10

(a) It suffices to prove that $\frac{e^x}{x} = \frac{1}{\ln x}$. Then $x = \ln e^x$, so $\frac{e^x}{\ln e^x} = \frac{1}{\ln x}$, so it suffices to prove that

$$\int \frac{\mathrm{d}e^x}{\ln e^x} = \int \frac{\mathrm{d}x}{\ln x},$$

which is true by substitution. It suffices to prove that

$$\int \frac{\cosh x}{x}\mathrm{d}x = \frac{\int \frac{e^x}{x} + \int \frac{e^{-x}}{-x}}{2} = \int \frac{\frac{e^x - e^{-x}}{2}}{x}\mathrm{d}x.$$

So it suffices to show that $\cosh x = \frac{e^x - e^{-x}}{2}$, and we already know that, so we're done.

(b) It suffices to prove

$$\int \frac{\sinh x}{x}\mathrm{d}x = \frac{\int \frac{e^x}{x}\mathrm{d}x - \int \frac{e^{-x}}{-x}\mathrm{d}x}{2} = \int \frac{e^x + e^{-x}}{2x}\mathrm{d}x.$$

So it suffices to prove that $\sinh x = \frac{e^x+e^{-x}}{2}$.We have

$$\mathrm{Ei}(ix) = \int \frac{e^{ix}}{ix}\mathrm{d}(ix) = \int \frac{e^{ix}}{x}\mathrm{d}x$$

equals

$$\mathrm{Ci}(x) + i\mathrm{Si}(x) = \int \frac{\cos x}{x}\mathrm{d}x + i\int \frac{\sin x}{x}\mathrm{d}x = \int \frac{\cos x + i\sin x}{x}\mathrm{d}x.$$

Thus it suffices to show that $e^{ix} = \cos x + i\sin x$, so we're done.

11 Observe that

$$\begin{aligned} e^{i\pi/4}\Phi(xe^{-i\pi/4}) &= e^{i\pi/4}\int e^{-(xe^{-i\pi/4})^2}\mathrm{d}(xe^{-i\pi/4}) \\ &= e^{i\pi/4}\int e^{-x^2e^{-i\pi/2}} \cdot e^{-i\pi/4} \cdot \mathrm{d}x \\ &= \int e^{x^2 i}\mathrm{d}x \end{aligned}$$

and

$$C(x) + iS(x) = \int \cos x^2 \mathrm{d}x + i\int \sin x^2 \mathrm{d}x = \int (\cos x^2 + i\sin x^2)\mathrm{d}x.$$

Thus since $e^{x^2 i} = \cos x^2 + i\sin x^2$, then $e^{i\pi/4}\Phi(xe^{-i\pi/4}) = C(x) + iS(x)$, as desired.

12

(a) $2x^3yy' = 2 - y^2$, so $\frac{2yy'}{2-y^2} = \frac{1}{x^3}$. Thus

$$\begin{aligned} \int \frac{2y}{2-y^2}\mathrm{d}y &= \int \frac{1}{x^3}\mathrm{d}x \\ -\int \frac{\mathrm{d}(2-y^2)}{2-y^2} &= \int x^{-3}\mathrm{d}x \\ -\ln|2-y^2| &= -\frac{x^{-2}}{2} + c \\ |2-y^2| &= cx^{x^{-2}/2} \\ y^2 &= 2 - cx^{1/2x^2}. \end{aligned}$$

We get $yy' = \frac{\sqrt{1+x^2}}{x}$, so

$$\int y\mathrm{d}y = \int \frac{\sqrt{1+x^2}}{x}\mathrm{d}x.$$

Then

$$\int \frac{1}{y}\mathrm{d}y = \int x\sqrt{1+x^2}\mathrm{d}x,$$

so $\ln y = \frac{1}{3}(1+x^2)^{3/2} + c$, so $y = ce^{\frac{1}{3}(1+x^2)^{3/2}}$.

(b) Let $u = y + x$, so $y' = \cos u$. Then

$$\int 1\mathrm{d}y = \int \cos u \mathrm{d}u = \sin u,$$

so $y = \sin u = \sin(y + x)$. $y + x = \arcsin y$, so $x = \arcsin y - y$. We have $x^2 y' = 1 + \cos 2y$, so $\frac{y'}{1+\cos 2y} = \frac{1}{x^2}$. Then

$$\int \frac{\mathrm{d}y}{1 + \cos 2y} = \int \frac{1}{x^2}\mathrm{d}x.$$

Since $1 + \cos 2y = 1 + (2\cos^2 y - 1) = 2\cos^2 y$, then

$$\int \frac{\mathrm{d}y}{2\cos^2 y} = \frac{1}{2}\tan y = -x + c.$$

Then $y = \arctan(-2x + c)$. As $x \to \infty$, $\arctan(-2x) = \frac{-\pi}{2}$.

(c) Observe that the function $\phi(x) = e^x$ works. Note that $\phi(x) \cdot \phi(y) = e^x \cdot e^y = e^{x+y} = \phi(x + y)$ and that the exponential function is always positive, so ϕ maps the set of real numbers to the set of positive real numbers. It suffices to prove that there does not exist a function $\varphi(x+y) = \varphi(x) \cdot \varphi(y)$ mapping $x+y \in \mathbb{R}$ to $\varphi(x) \cdot \varphi(y) \in \mathbb{R}\backslash 0$. Observe that $\mathbb{R}_0$ is a subset of $\mathbb{R}$ because there is an obvious injective mapping from $\mathbb{R}\backslash 0$ to $\mathbb{R}$, but there is no mapping in the opposite direction.

(d) Sure, it does. Lagrange murderpliers is just a way of finding the maximum value of a function given a certain constraint.